Construction Scheduling with Primavera P6

Planning • Controls • Implementation

This manual provides the basic concepts of CPM scheduling combined with software techniques that provides the knowledge required to begin using Primavera P6 to update a schedule, document delays, plan a project, schedule a project, customize the schedule, establish resources & cost accounts and viewing tabular & graphic reports.



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Chapter 1
Introduction to CPM Scheduling

This chapter provides an overview of the basic concepts of CPM scheduling and offers some pointers on how to put the critical path method to work for you. We discuss the elements of an effective schedule and factors to consider when updating a schedule and documenting delays.



- ✓ *The Contractor should make sure the schedule is comprehensive and realistic.*
- ✓ *The Owner and the Owner's representatives should review the schedule as if they were performing the work themselves.*

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The Well-Planned Schedule is a Roadmap to Success

- ✓ To know where you are going and how to get there.
- ✓ To effectively allocate five limited resources:

  TIME

  MONEY

  EQUIPMENT

  PERSONNEL

  MATERIAL

- ✓ To track work progress and to evaluate time extension requests.
- ✓ To substantiate delay/interruption claims.



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Scheduling Techniques

Bar Chart or Gantt Chart

For small and simple projects – introduced by Gantt and Frederick Taylor (Father of Scientific Management) during the British Industrial Revolution at the turn of the century.

CPM (Critical Path Method)

A powerful planning tool, CPM was developed during the late 1950s in two more-or-less parallel actions: the DuPont Company's Refinery Renovation Project and the U.S. Navy's Fleet Ballistic Missile Project.

During the middle and second half of the 1960s two developments produced substantial expansion of CPM usage in the construction industry: first, the espousal and the educational endorsement of CPM by the Associated General Contractors of America; second, the decision by many construction-oriented federal government agencies to require the use of CPM on large projects.

PERT (Program Evaluation and Review Technique)

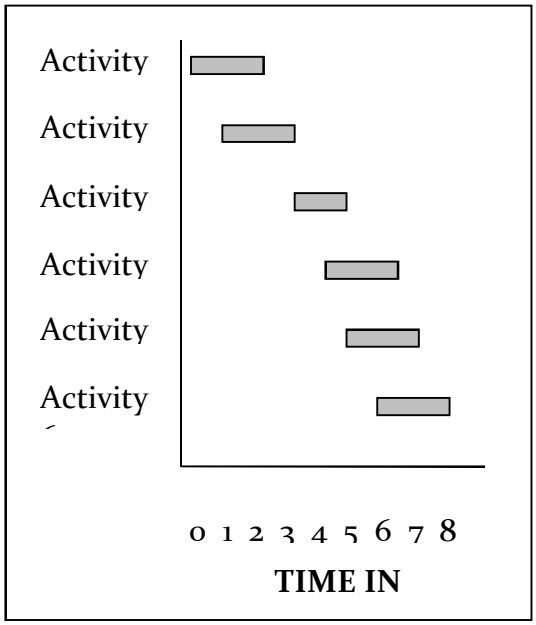
Based on the probability of outcome. Rarely used in Construction Projects.

The Bar Chart (Gantt Chart)

The bar chart represents the sequence and duration of activities in a project. Each bar stands for one activity, with the length of the bar indicating the activity's duration and the position of the bar indicating the time period when the activity is projected to take place.

Two Variables:

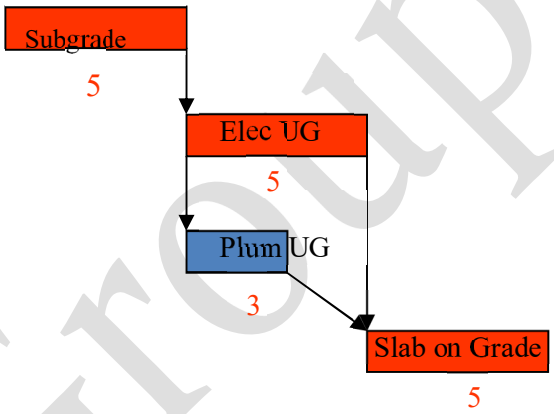
1) **Time**
2) **Scope**



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The Critical Path Method (CPM)

The critical path method (CPM) refines the bar chart representation of a project by adding relationship ties between activities, so that if anything changes, its effects on the rest of the schedule can immediately be seen.



The "critical path" is simply the chain of activities from the start to the finish of a project that has the least amount of float (takes the most time to complete), such that any delays in the critical path activities will negatively impact the finish date of the project.

The usefulness of the CPM is that it allows one to run "what if" scenarios and spot potential problems areas early, while there is still time to proactively deal with them. It also provides an effective way of documenting the impact of delays on a project.

Three Variables:

1) **Time**
2) **Scope**
3) **Logic**

The useful CPM tool is:

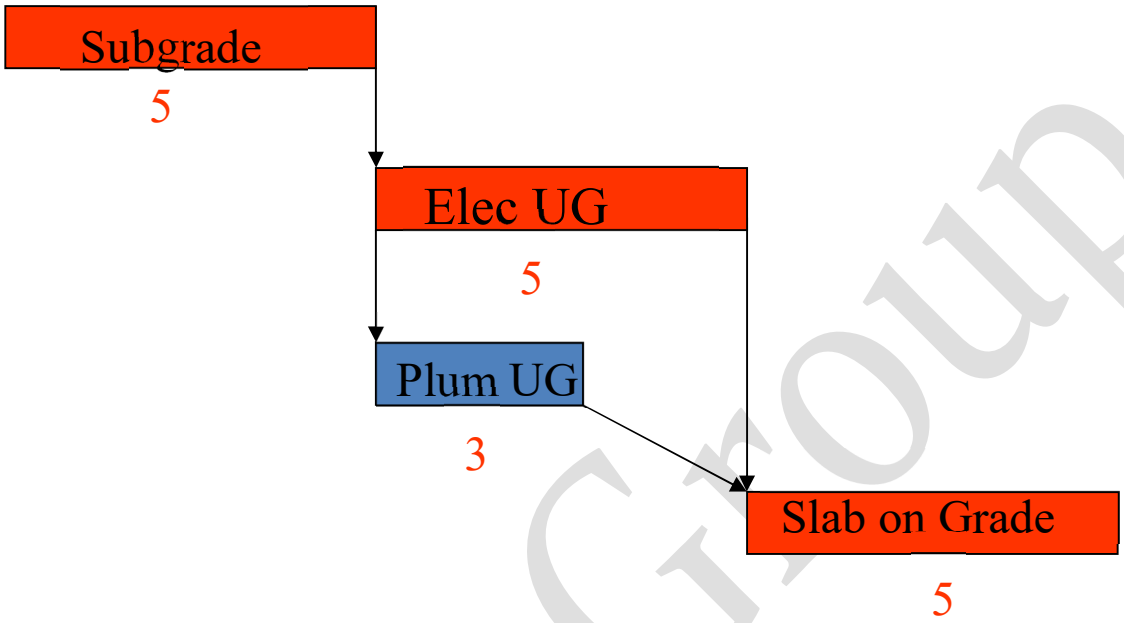
- ✓ ***Developed…*** *with the full participation of the:*
  Project Manager
  Estimator
  Purchasing Agent
  Job Superintendent
  Controller
  Subcontractors

- ✓ ***Comprehensive***
  See the checklist on the next page.

- ✓ ***Realistic***
  Allow a little extra time to absorb unexpected delays.

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Illustration of the Critical Path



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Technical Terms Used in CPM Scheduling

Activity A distinct item of work completion of which is necessary to finish the project.

Duration A realistic, best guess estimate of the amount of time necessary for accomplishing the work involved in an activity.

Early Start (ES) The earliest point in time when an activity can start, assuming all prerequisite activities take place as scheduled.

Early Finish (EF) The earliest date that an activity can finish, assuming that it began on its early start date.

Late Start (LS) The point in time by which an activity must start if there is to be no delay in subsequent activities.

Late Finish (LF) The date by which all work involved in an activity must be completed in order to avoid delaying subsequent activities.

Critical Path The sequence of activities from project start to finish that has the least amount of float, i.e., requires the longest total amount of time to complete.

Float The amount of extra time available to an activity or activities on a path when all activities start as early as possible. (Free float is uniquely available to an activity.)

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Can you Calculate the Early and Late Dates?

Positive float is the number of days an activity can be delayed without affecting the project finish (the plumber).

Zero float denotes activities that have no flexibility but must start and finish on or before their scheduled dates in order for the project to complete on time.

Negative float denotes activities scheduled to occur after their late dates. The project is delayed.

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Review Questions

1. The THREE VARIABLES in a *task* driven CPM schedule are
   - _____
   - _____
   - _____
2. True or False? Early Dates must always be met in order to avoid delaying subsequent activities.
3. Critical activities in a typical Baseline Schedule might have which of the following float types
   A. Positive Float
   B. Zero Float
   C. Negative Float
4. Complete the following formulas calculating float using *before*, *after*, or *equal to*.

   Positive Float = Early Dates are _____ the Late Dates.

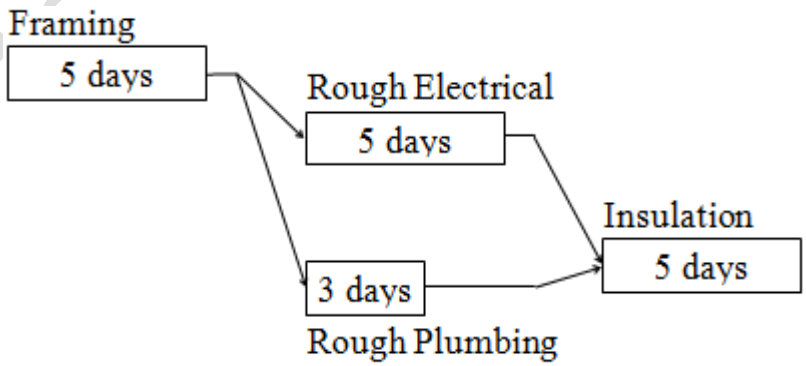
   Zero Float = Early Dates are _____ the Late Dates.

   Negative Float = Early Dates are _____ the Late Dates.

5. Rough Electrical, with a 5-day duration, and Rough Plumbing, with a 3-day duration, are scheduled to begin after Framing and before Insulation. If Framing and Insulation are both critical with zero float,

   How many days of float would Rough Electrical have? _____

   How many days of float would Rough Plumbing have? _____



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Chapter 2
Planning a Project

We'll discuss in this chapter the scope of work – the detail needed in building a schedule. We'll also look at what resources might be needed in your schedule and we'll also develop an activity code system.



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Developing a Plan

Before jumping into P6 you should take some time to think through the requirements (time, resources, milestone dates, etc.) of the project and develop a general outline of how you plan to meet those requirements given the resources available. The process of developing a good initial (target or baseline) schedule may take a while and may go through several revisions, but the time is well worth it. A good target schedule will be much easier to update and will be much more effective in documenting delays than a poor one. As you develop a plan, be sure to get input from all project management personnel, including the owner and any subcontractors, so that the target schedule is something everyone can commit to. Review the CPM checklist above as you develop your plan.

Project Duration

When considering project duration, you must take into account the amount of resources needed to complete each activity.

The majority of construction projects are *task driven* (each activity is controlled by its duration and logic). The resources are taken into consideration when entering the activity duration. And the project is controlled by the activity's base calendar. The work begins according to the relationship ties, i.e., its logic, and continues until its duration has elapsed. These activities are coded as "task" activities.

There are, however, a few projects requiring *resource driven* schedules. Each resource has its specific calendar and a specified amount of work to accomplish. There is no specified duration for activities; P6 calculates durations based on the resources driving the work. These activities are coded as an "independent" type.

Level of Detail

How much detail do you need for your schedule? Keep in mind that while more detail can make the schedule larger and more cumbersome to manage, insufficient detail in a schedule makes it less useful for forecasting work and documenting delays. Many contracts require that no activities be longer than a specified number of days in order to force the contractor to break long activities into several smaller ones, thereby putting more detail in the schedule. Be aware of the contract requirements bearing upon the schedule.

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Resources, etc.

As previously mentioned, each project has 4 basic resources: time, money, equipment, and manpower. P6 allows you several ways of assigning these resources to specific activities.

The activity's duration or resources control time, depending on the type of activity. The three remaining resources are assigned by setting up a resource dictionary and inserting resources into each activity.

Resource assignment will be discussed in more detail later on.

Work Breakdown Structure (WBS)

A well-planned project contains organized activity data to divide work into meaningful groups. P6 uses Work Breakdown Structure (WBS) as a method of organizing a project. WBS is extremely useful for grouping, sorting, and filtering activities. The WBS can be used to categorize activities such as area, department, phase, location, or type of work. You can also use the activity codes for each project or define your own coding structure for additional organization (responsibility, etc).

We will discuss how to develop and use the WBS and activity codes in greater detail later on.

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Chapter 3
Introduction to P6

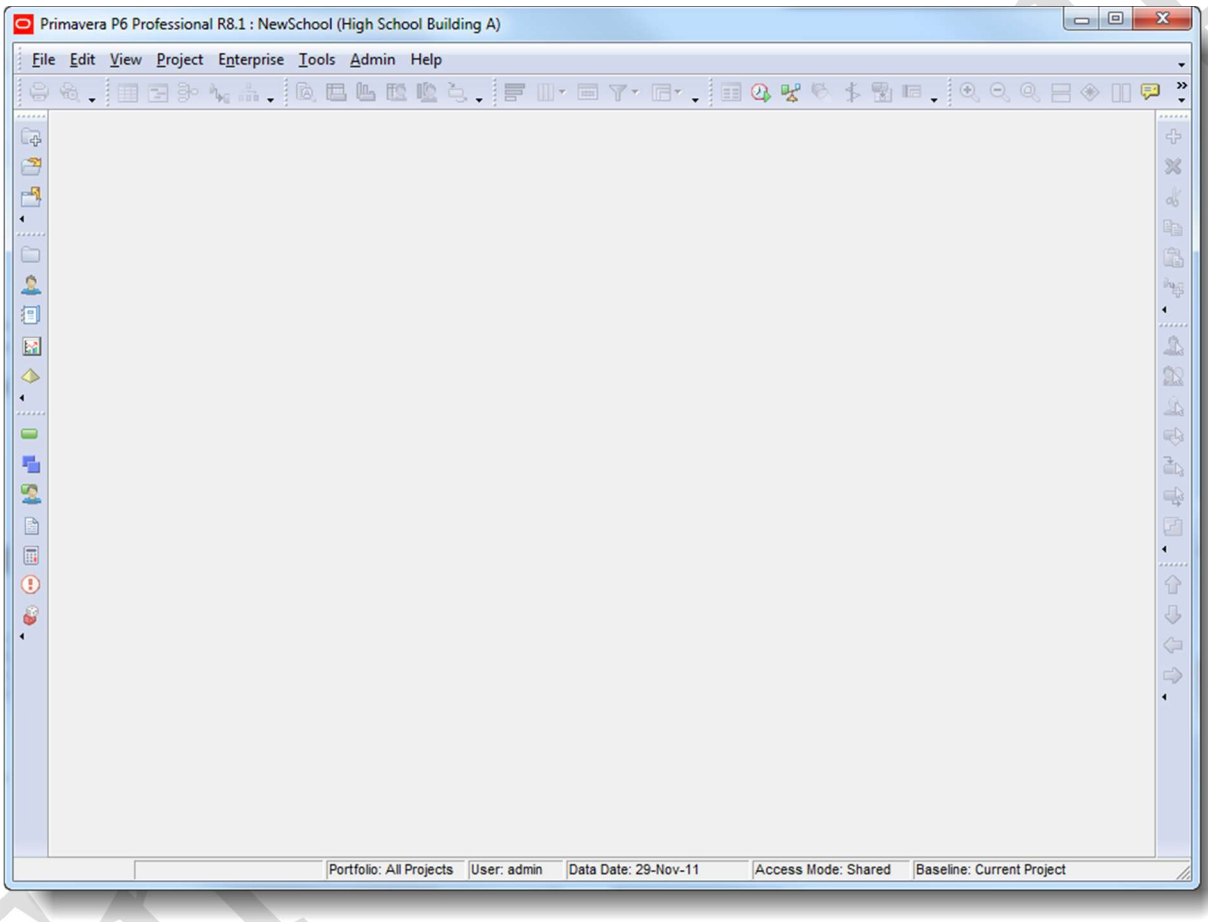
In this chapter you'll be introduced to Primavera P6. We'll then look at P6's menus at the open screen and project screen to aquaint you with the program's capabilities.



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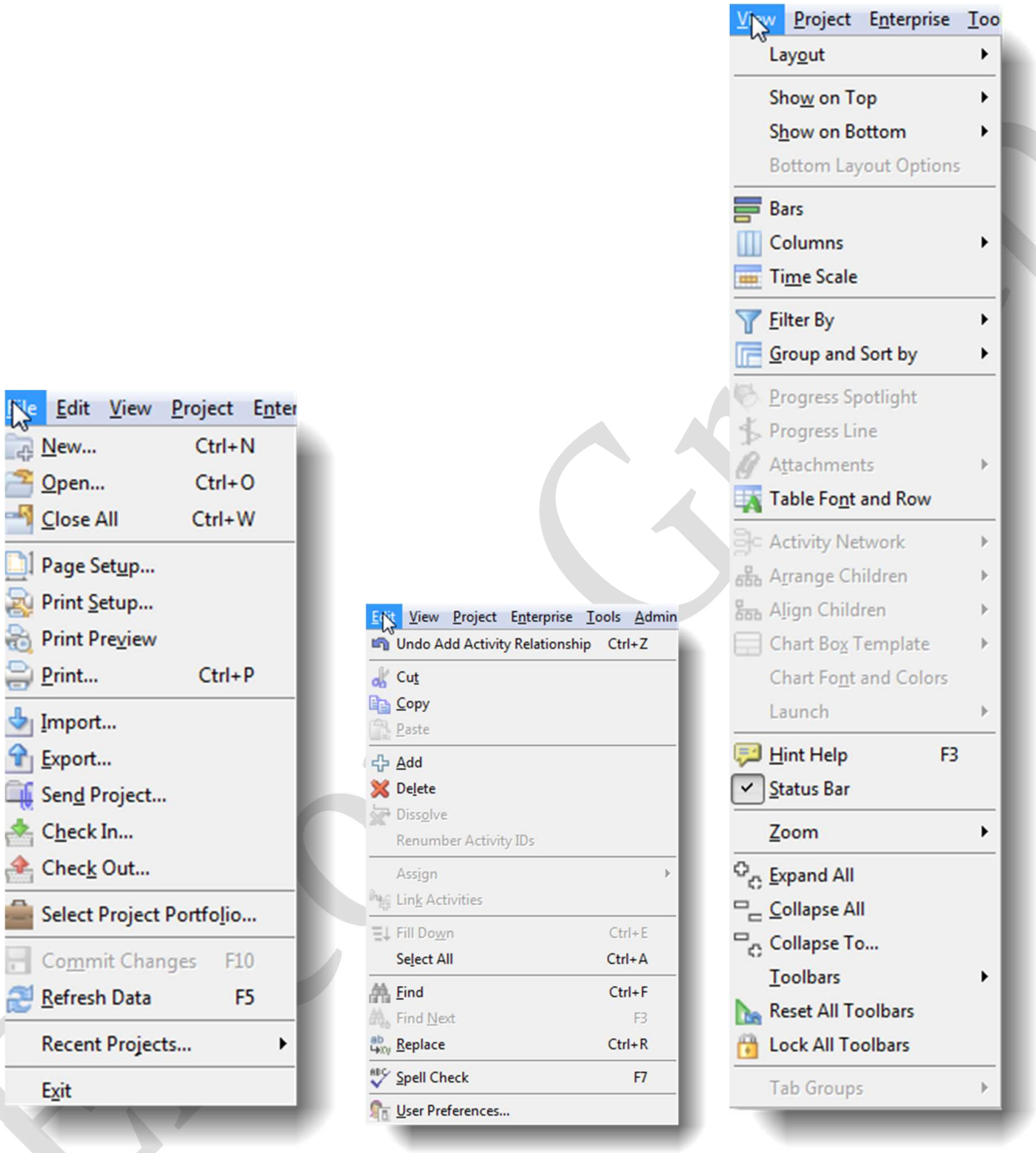
Primavera Overview

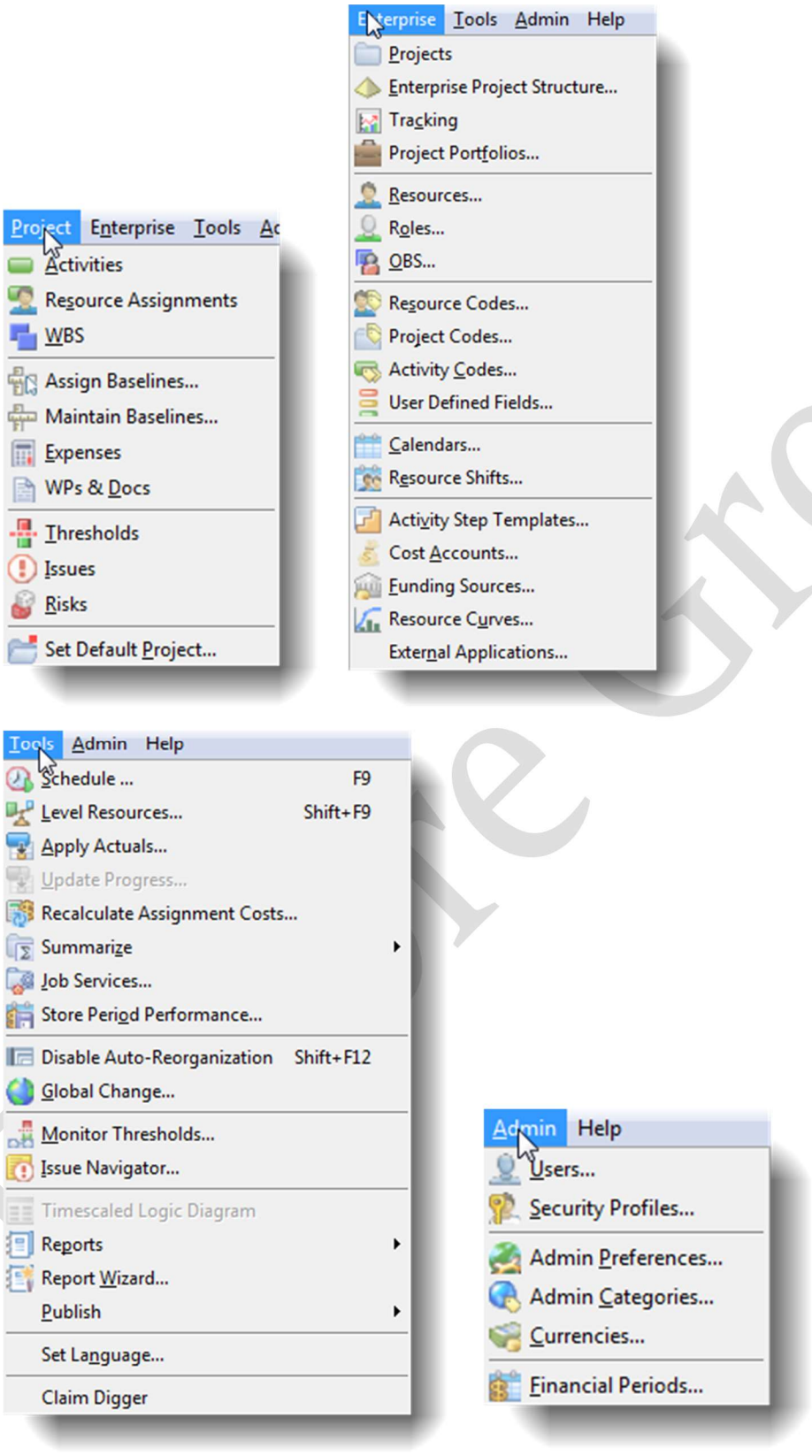
Start Primavera P6 by double-clicking on the desktop program icon. Once you've started P6, you can begin working with a project. The initial startup menu allows the user to open new or existing projects and perform various project utilities such as backing up and restoring files.



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The P6 Menus and Toolbars

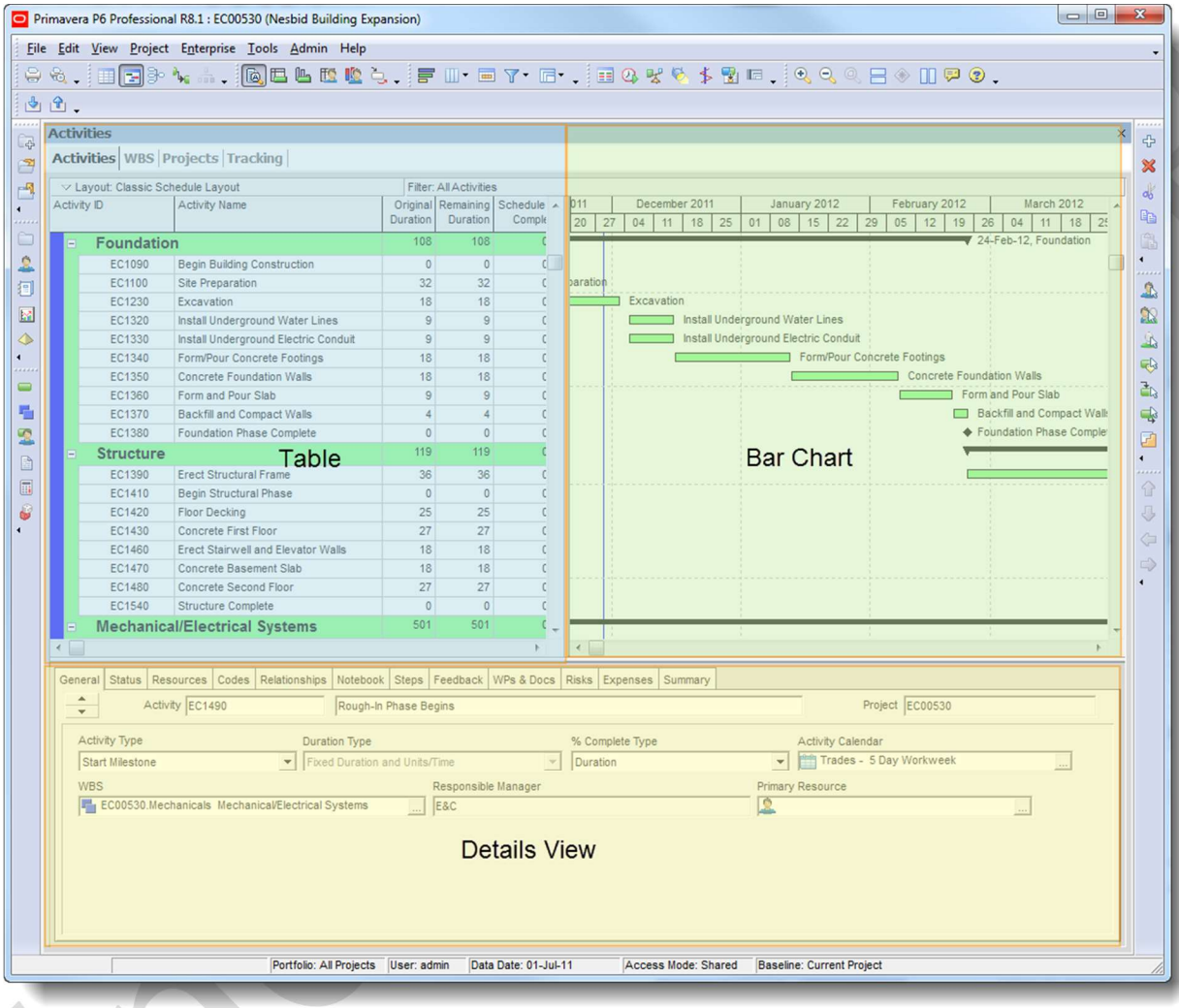



Project Enterprise Tools Ac
Activities
Resource Assignments
WBS
Assign Baselines...
Maintain Baselines...
Expenses
WPs & Docs
Thresholds
Issues
Risks
Set Default Project...
Enterprise Tools Admin Help
Projects
Enterprise Project Structure...
Tracking
Project Portfolios...
Resources...
Roles...
OBS...
Resource Codes...
Project Codes...
Activity Codes...
User Defined Fields...
Calendars...
Resource Shifts...
Activity Step Templates...
Cost Accounts...
Funding Sources...
Resource Curves...
External Applications...
Tools Admin Help
Schedule ... F9
Level Resources... Shift+F9
Apply Actuals...
Update Progress...
Recalculate Assignment Costs...
Summarize
Job Services...
Store Period Performance...
Disable Auto-Reorganization Shift+F12
Global Change...
Monitor Thresholds...
Issue Navigator...
Timescaled Logic Diagram
Reports
Report Wizard...
Publish
Set Language...
Claim Digger
Admin Help
Users...
Security Profiles...
Admin Preferences...
Admin Categories...
Currencies...
Financial Periods...

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Viewing your Schedule in P6

There are three areas of the Primavera P6 window that you should become very familiar with. They are the **Table**, **Bar Chart**, and **Details Form (Bottom Layout)**.



The Enterprise Project Structure (EPS)

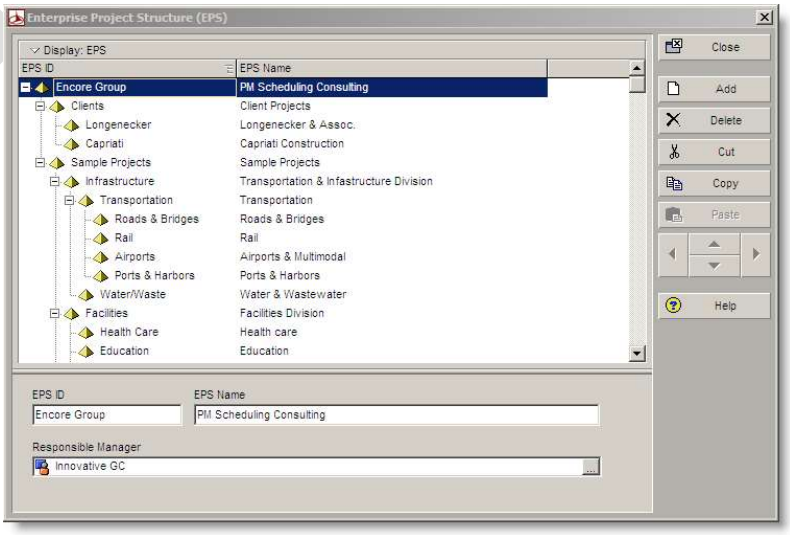
The Enterprise Project Structure is the highest level of the P6 project hierarchy.

- **Enterprise Project Structure (EPS)**
  - **Project**
    - **Work Breakdown Structure (WBS)**
      - **Activities**

Create the hierarchical establishment of the EPS identifying company-wide projects and enabling organization and management of these projects. (Select Projects from the Home workspace).

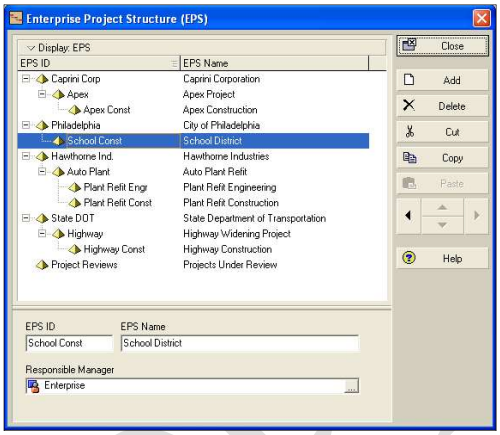
What is the EPS

a. Highest-level nodes or roots can represent divisions of a company or groupings of an organization.
b. Lowest level nodes or roots normally represent individual projects or sub-jobs.
c. A pyramid symbol represents nodes or roots containing other "sub" nodes and projects.
d. A "+" symbol represents summarization or hidden additional levels rolled up beneath. Click the "+" or double-click the node itself to expand the hidden additional levels.
e. A "-" symbol represents levels that are expanded. Click the "-" or double-click the node itself to collapse or roll up the level.
f. A folder symbol represents the project or lowest level in the EPS hierarchy.
g. EPS nodes can be assigned dates as well as budget data.



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Setting Up the EPS



a. Highest-level nodes or roots can represent divisions of a company or groupings of an organization.
b. Lowest level nodes or roots normally represent individual projects or sub-jobs.
c. A pyramid symbol represents nodes or roots containing other "sub" nodes and projects.
d. A "+" symbol represents summarization or hidden additional levels rolled up beneath. Click the "+" or double-click the node itself to expand the hidden additional levels.
e. A "-" symbol represents levels that are expanded. Click the "-" or double-click the node itself to collapse or roll up the level.
f. A folder symbol represents the project or lowest level in the EPS hierarchy.
g. EPS nodes can be assigned dates as well as budget data.

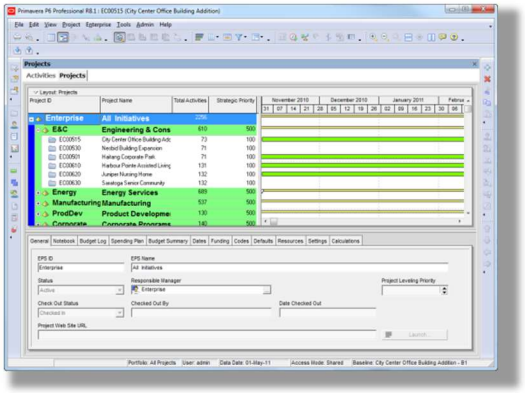
Exercise: Create an EPS Node

Create the following EPS node:

EPS ID: **School Const** EPS Name: **School Construction** Responsible Manager: **(select)**

Defining Additional Information

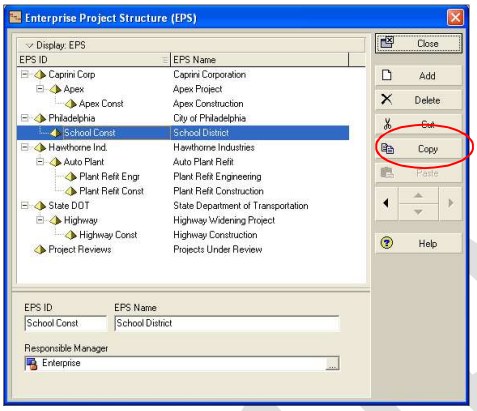
Use Project Details to define additional data (i.e. new projects, anticipated dates, budgets, etc.) to each node.



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Copying Additional EPS Nodes/Projects

Nodes or Projects can be copied to be used again using the Copy key to the left in the Enterprise Project Structure window. Select the node level into which the copied node is to be pasted.



Deleting an EPS Node

Deleting nodes (the Enterprise menu, Enterprise Project Structure, Delete) deletes all projects under that node hierarchy. If they are not to be deleted, they should first be moved or copied to another node.

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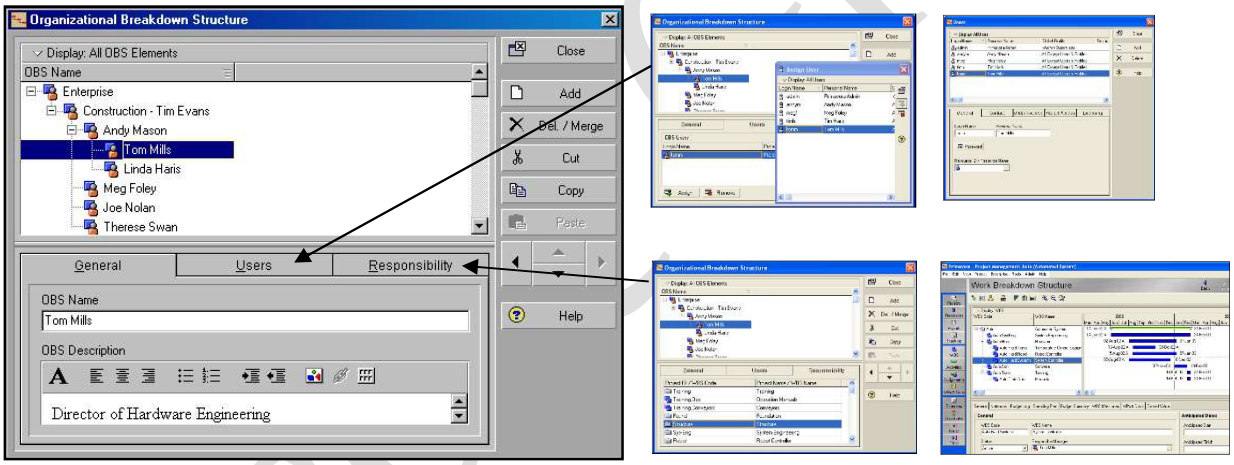
The Organizational Breakdown Structure (OBS)

Create the hierarchical establishment of a company's management structure (either as roles or individuals). The OBS implements user access and data privilege into the Enterprise Project Structure (EPS) hierarchy. Each user is assigned his or her role in the EPS / WBS either by node or by project just as resources are assigned to activities. Users assigned to particular elements of the OBS are given access to the corresponding EPS level or project, based upon their security profile.

An EPS node can have one or many OBS levels assigned dependent upon the company's management structure. However, an OBS can have one root element only.

Projects added to a specific EPS node automatically inherit the OBS assigned to the node.

Setup up Security Profiles under Admin, Security Profiles and then assigned to users under Admin, Users.



Creating the OBS

Creating the OBS by selecting the Enterprise menu, OBS (use either the Chart View or Table View under the Display bar).

1. Highlight the OBS element directly above, normally of the same hierarchy level, and select Add.
2. Assign a meaningful name to OBS level in the General tab with a brief description.
3. Select the Users tab and assign or view the Login Name of the user(s) and the Project Security Profile.
4. Select the Responsibility tab to view assigned responsibilities throughout the enterprise.

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Editing the OBS Element

Editing an OBS element is normally accomplished in the General tab or by changing its position in the hierarchy using the arrow keys to the right.

Deleting the OBS Element

Deleting an OBS element normally deletes all items (WBS, issues, thresholds, risks, etc.) contained in that level OR the items contained can be merged with the next higher-level OBS element.

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A Project in the Enterprise

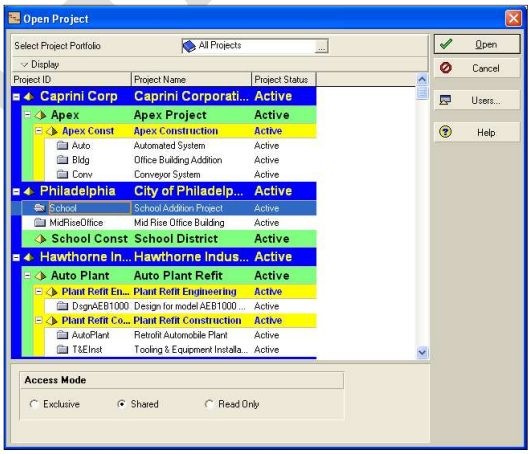
Projects are a group of activities and their associated elements that constitute a plan for providing a product or service.

Defining Project Properties

1) Start date and Finish date
2) WBS hierarchy
3) Activities and their logic structure, expenses, risks, issues, thresholds, work products/documents, and a baseline for comparison
4) Resource assignments
5) Project specific calendars

Opening the Project

1) Highlight the project(s) to be opened.
2) Select the File menu, Open or right-click.
3) Projects can be opened with one of three Access Modes (depending on user access):
4) Exclusive (One user can access the Exclusive mode at a time.)
5) Shared
6) Read Only
7) Users accessing projects can be viewed by selecting the Users button in the Open Project dialog box.



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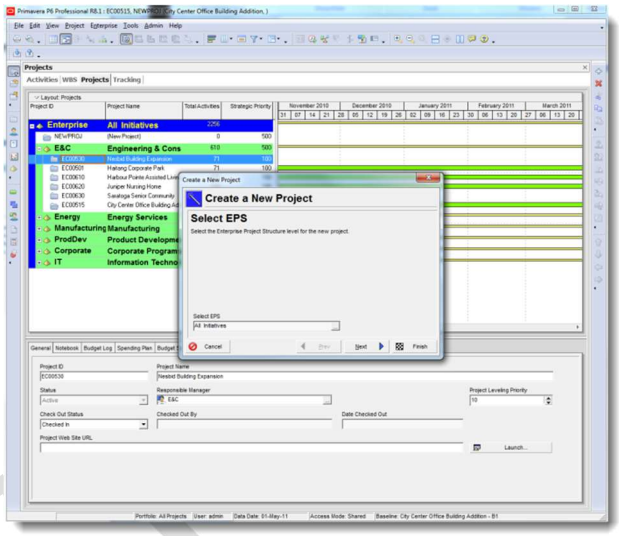
Chapter 4
Creating a Project in P6

In this chapter you will create a project in P6 and begin building a construction schedule.



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Creating a New Project in P6



1. Select the Enterprise menu, Projects OR from the Home workspace, select Projects.
2. Highlight the EPS node (or the root EPS node) under which the project is to be added. *Projects cannot be added below existing projects.*
3. Select Add to initiate the New Project Wizard (select Finish to add the project without using the wizard and entering applicable information in the General tab in the Project Details) OR select the File menu, New Project (*Refer to* Copying the Project.)
4. Verify or change the EPS node under which the project is added.
5. Enter a Project ID and Project Name.
6. Enter the planned start and must finish by dates.
7. Enter the responsible manager.

Adjusting the Project hierarchy

Using the mouse, Projects or EPS nodes can be moved from one level to another.

1. Highlight the project or EPS node to be moved.
2. Hold the mouse to the far left of the Project hierarchy to view the move option.
3. Click and drag the project or EPS node to the desired hierarchical level.

Exercise: Create a Project in P6

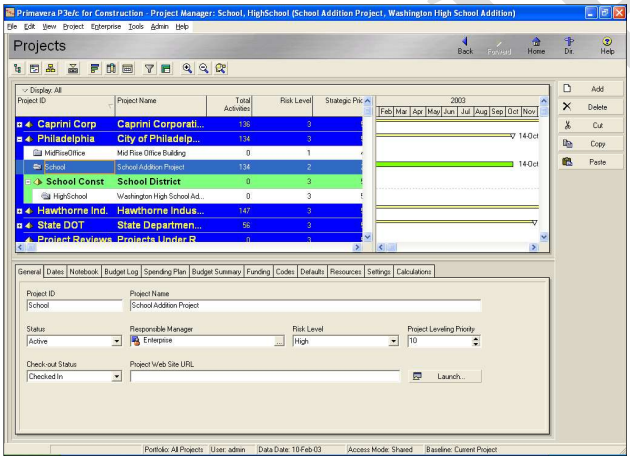
Create the following project under the selected EPS node: Project ID: **NewSchool** Project Name: **High School Addition** Planned Start Date: ***today's date – (leave the Must Finish By date empty)*** Responsible Manager: **Select a Responsible Manager**

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Viewing Project Details (or EPS detail)

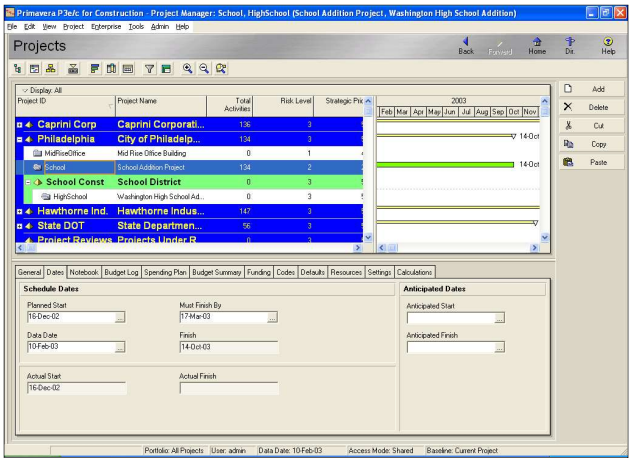
Project Details consist of specific information pertaining to the highlighted project of EPS node. Select the Display bar, Show on Bottom, Project Details to set details and defaults used throughout a project. Default settings for EPS nodes in the hierarchy are established using Project Details. Customize the Project Detail tabs by right-clicking in the Details window.

General tab – edit information about the selected project or node including:



1. ID and Name
2. Status – Active or current projects, Inactive or closed projects, What-If for analysis or what-if scenarios, Planned during project planning phase
3. Responsible Manager – assigned OBS element. *Required* for each level of the EPS.
4. Risk Level – overall risk level to complete the node or project.
5. Leveling Priority – used to deduct from resource availability across multiple projects during leveling. Enter a level from 1-100, with 1 being the highest leveling priority.
   Closed projects can be included when leveling resource assignments. Select Tools, Level Resources. Mark the option, Consider assignments in other projects with priority equal/higher than. All projects with this level priority number in the General tab of Project Details will be considered when leveling. Include Project Leveling Priority under Leveling priorities for a tiebreaker.

Dates tab – edit schedule information including:



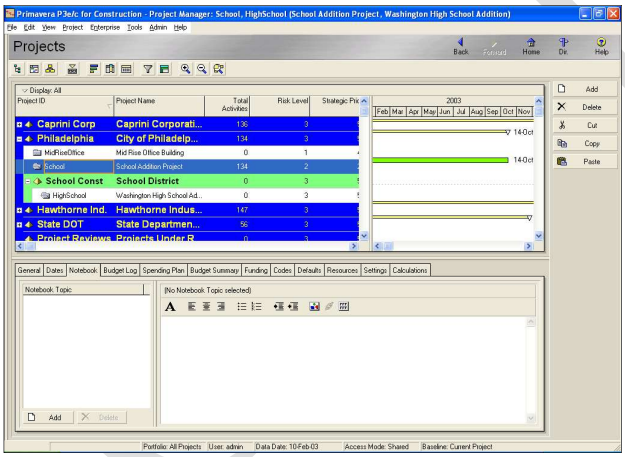
1. Planned Start – start date of the project
2. Must Finish By – constraint placed on the finish date of the project
3. Data Date – calculation date placed on the project
4. Finish – last scheduled latest early finish date calculated
5. Actual Dates – Actual Start/Finish dates of the project if started/finished (With a finish date

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included, all activities in the project should have actual start/finish dates as well.)

6. Anticipated Dates – expected start/finish dates defined by the user normally during the project planning stage. If there are no actual start dates or the project is in the planning stage, set the Planned Start/Must Finish by dates to the Anticipated Start/Finish dates. Once scheduled, the Anticipated Start/Finish is overwritten. (EPS nodes have Anticipated Dates only.)

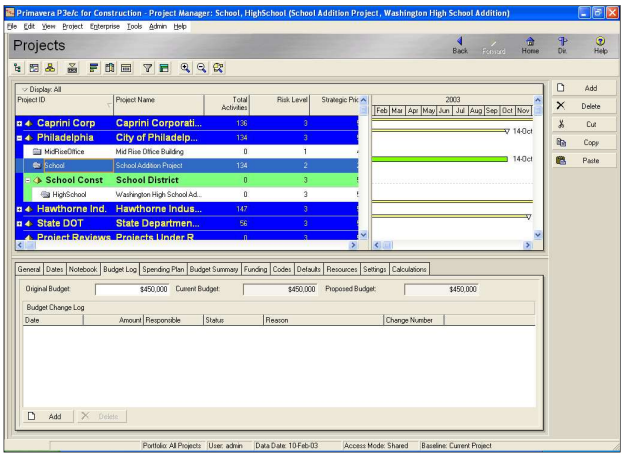
Notebook tab – assign Notebook Topics and Current Status details to a selected node or project. (Available Notebook Topics are defined under Admin, Admin Categories, Notebook Topics.)



1. Add the applicable topic in the left window and enter the topic description in the right window.
2. Format features including text format, justification, bullets and numbering, indent and outdent, picture inserting, hyperlinks, and tables.

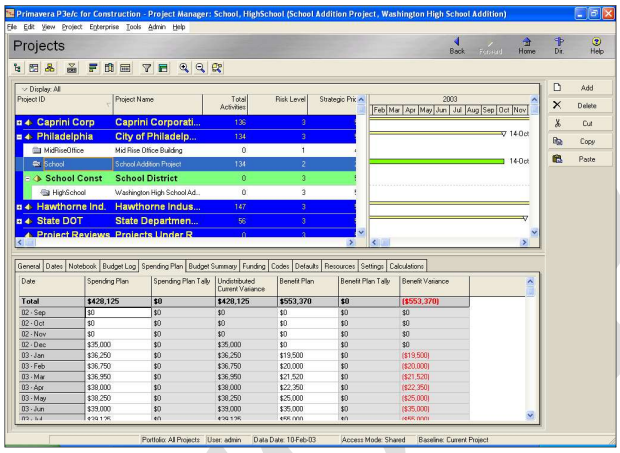
Budget Log tab – edit budget information for the project/node.

1. Original Budget – estimated budgeted amount required including funding contributions
2. Current Budget – Original Budget + Approved budgets
3. Proposed Budget - Original Budget + Approved/Pending budgets
4. Budget Change Log – logged changes to the Original Budget including dates, amounts, responsibility, status codes, reasons, and change numbers



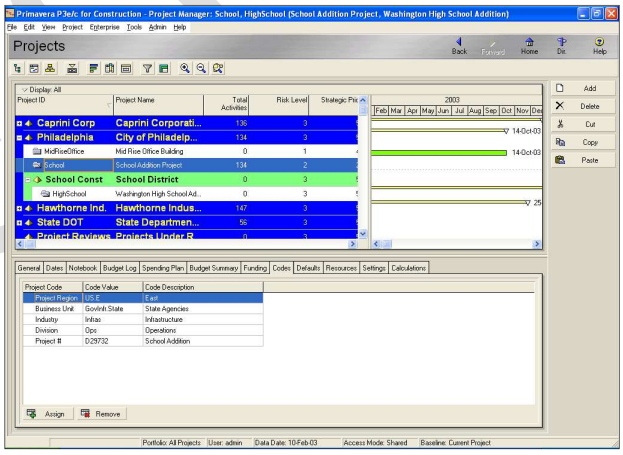
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Spending Plan tab – edit/record spending information



1. Date – monthly range beginning 3 months prior to project start and 4 years after project start
2. Spending Plan – amount spent on a per month basis
3. Spending Plan Tally – project summary when selecting an EPS node
4. Undistributed Current Variance – Spending Plan (monthly value) – Spending Plan Tally (monthly value)
5. Benefit Plan – *profit portion* of the monthly amount normally entered at the end of the project
6. Benefit Plan Tally – summarized benefit plan information when selecting an EPS node

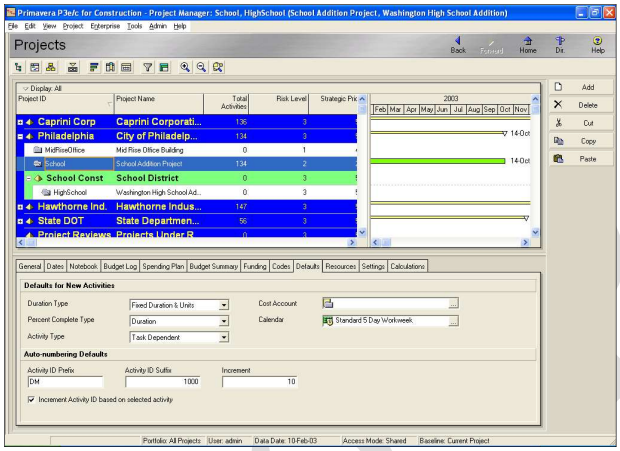
Codes tab – assign or remove applicable project codes. Project codes must first be created (the Enterprise menu, Project Codes). Project Codes *are not* assigned to EPS nodes.



Information in columns can be sorted by selecting the column heading in either ascending or descending order. Sorting can also be changed back to the outline format by selecting the column heading again.

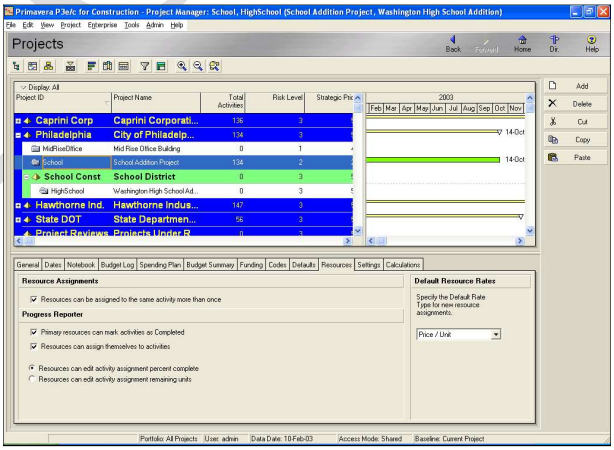
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Defaults tab – specify default settings for the project including:



1. Defaults for New Activities – does not change existing activities
   a. Duration Type and Percent Complete Type
   b. Activity Type
   c. Cost Account applies to resource assignments and project expenses
   d. Calendar default applies to new activities only
2. Auto-numbering Defaults – new activities are generated using auto-numbering settings. Establish Activity ID prefix, suffix, and increment settings. Select to increment activities based upon the highlighted activity.
3. Default settings are not available for other EPS nodes.

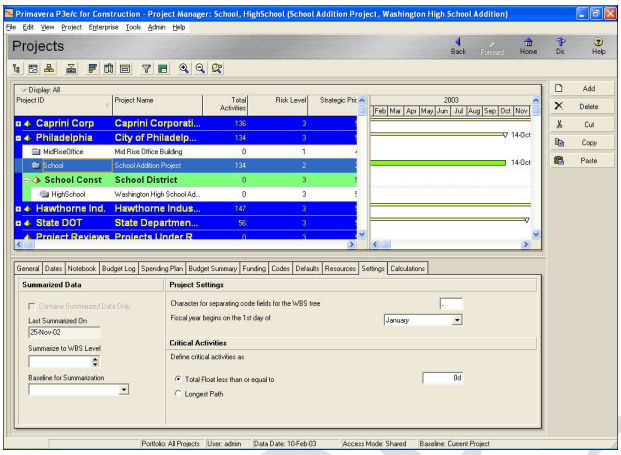
Resources tab – establish resource permissions/settings for resource assignment and the progress reporter.



1. Primary resources can mark activities as Completed.
2. Resources can be allowed to assign themselves to activities.
3. Resources can update/edit activity assignment using either remaining units or percent complete.
4. Select the Default Resource Rates for the default rate type used to determine the price/unit set on the resource assignment. This change does not impact existing resource assignments. (Default Resource Rates are set under Admin, Admin Preferences, Rate Types.)
5. Resource settings are *not* available for other EPS nodes.

Settings tab – define summarization information and project-level settings.

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1. Contains summarized data only – select if only summarized data is to be maintained no individual, specific activities but used for summary-level tracking and reporting. Allows for projects managed outside of Project Manager to be maintained at a summary-level.
2. Last Summarized On – the last time data was summarized, *cannot be edited*
3. Summarize to WBS Level – maximum summary WBS levels users are able to view in Tracking and Portfolio Analyst
4. Baseline for Summarization – current or all projects defined as targets/baselines/benchmark projects to be used for summarization
5. Select the character used to separate codes for WBS elements
6. Select the starting month for the project's fiscal year
7. Select the project's definition of criticality using either Total Float less than or equal to a specific number and time unit OR based on the longest path in the project's network. **Longest Path** is calculated by identifying all activities where early finish = latest calculated early finish. Driving relationships for these activities are identified and traced back to the start date of the project.

8. Settings are not available for other EPS nodes.

Copying the Project

1. Projects or Nodes can be copied for later use by selecting the Copy key to the left.
2. Select the node into which the project is to be pasted.
3. Select other features to be copied such as links to the WBS, OBS, Reports, Documents, etc.

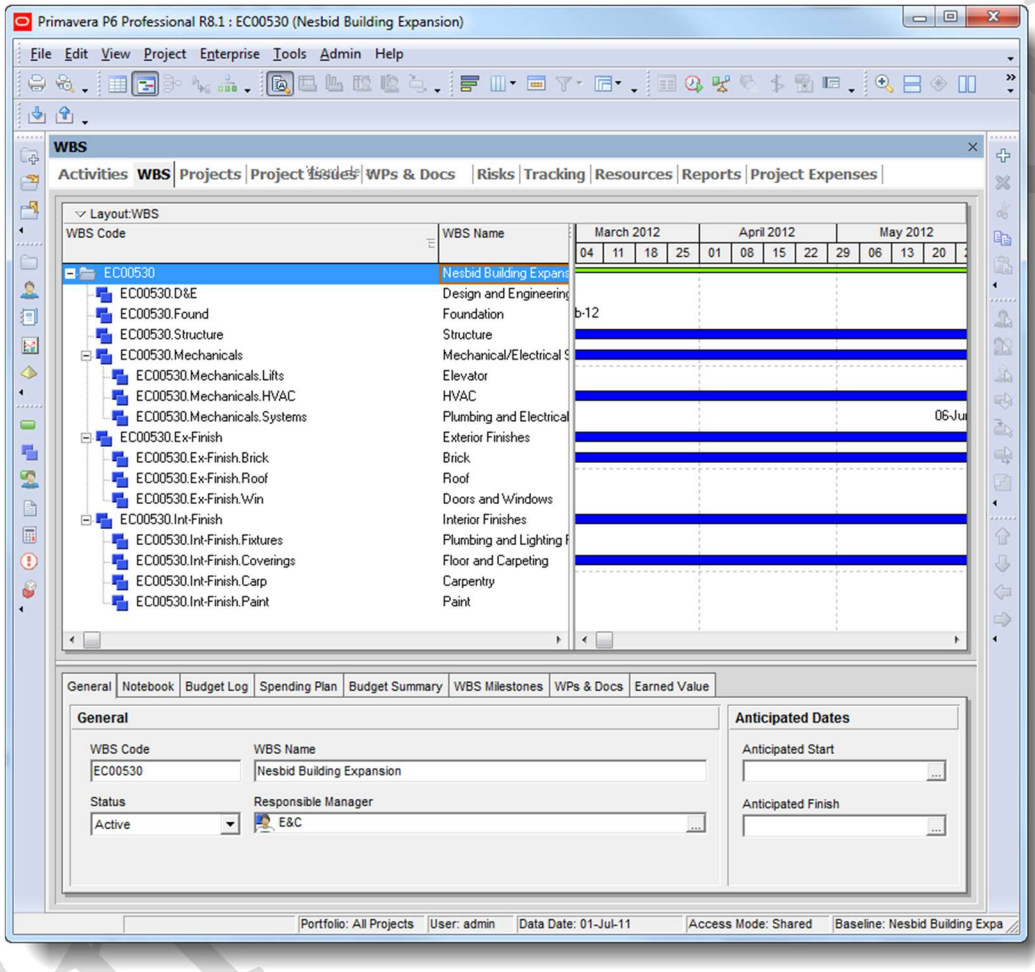
Deleting the Project

1. Delete projects by selecting the Enterprise menu, Enterprise Project Structure, or in the Projects or WBS windows.
2. Highlight the project, and select Delete. (Only projects opened in *Exclusive access mode* can be deleted.)

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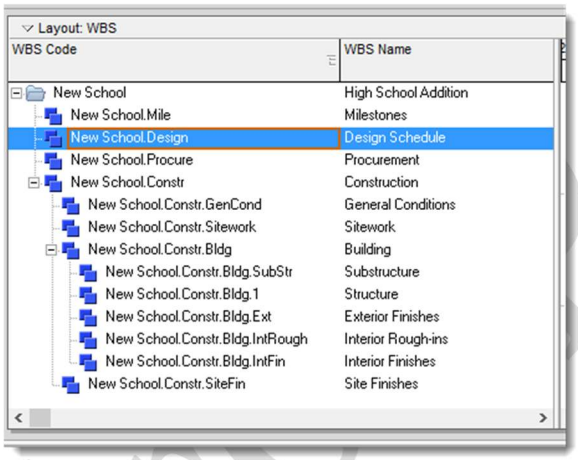
Scheduling with the WBS

The WBS is the hierarchical establishment of products and services produced during a project. It is normally used as the lowest or most detailed level in the EPS hierarchy, taking the EPS down to the activity level. Individual activities are the lowest level in the WBS while the project is the highest level.



Exercise: Build a WBS for a Project

1. Create automatically the highest WBS level when adding a project. This WBS element is given the same name as the project.
2. Create WBS elements by highlighting an existing element at the upper hierarchy level and selecting the **Add** button.
3. Set anticipated or planned dates, budgets, and spending plans at the higher level WBS elements that can then be shared with and used by the project and activity counterparts using the WBS Details.

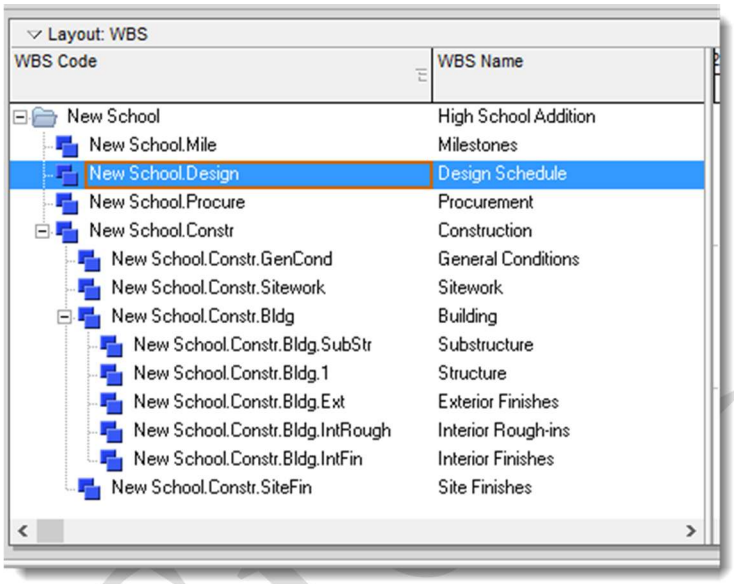


Create the following WBS levels within the High School Addition project:

WBS Code: ***NewSchool*.Mile** WBS Name: **Milestones**
WBS Code: ***NewSchool*.Design** WBS Name: **Design Phase**
WBS Code: ***NewSchool*.Procure** WBS Name: **Procurement**
WBS Code: ***NewSchool*.Constr** WBS Name: **Construction Phase**
WBS Code: ***NewSchool.Constr*.GenCon** WBS Name: **General Conditions**
WBS Code: ***NewSchool.Constr*.SiteWork** WBS Name: **Site Work**
WBS Code: ***NewSchool.Constr*.Bldg** WBS Name: **Building A**
WBS Code: ***NewSchool.Constr.Bldg*.SubStruc** WBS Name: **Substructure**
WBS Code: ***NewSchool.Constr.Bldg*.Structure** WBS Name: **Structure**
WBS Code: ***NewSchool.Constr.Bldg*.Rough** WBS Name: **Interior Rough-ins**
WBS Code: ***NewSchool.Constr.Bldg*.Exterior** WBS Name: **Exteriors**
WBS Code: ***NewSchool.Constr.Bldg*.Finishes** WBS Name: **Interior Finishes**
WBS Code: ***NewSchool.Constr*.Site** WBS Name: **Site Finishes**

Remember that levels can be moved by highlighting the node and with the pointer to the far left, click and drag the node to the desired location.

Viewing the WBS Details

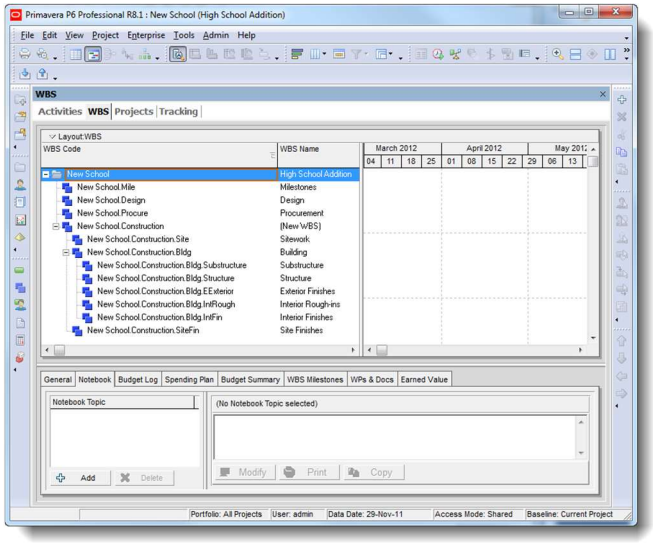


Customize the Details tabs by right clicking in the Details window, Customize WBS Details.

General tab – view/edit WBS level Code, Name, Responsible Manager or root OBS element, and Status. Status considerations include:

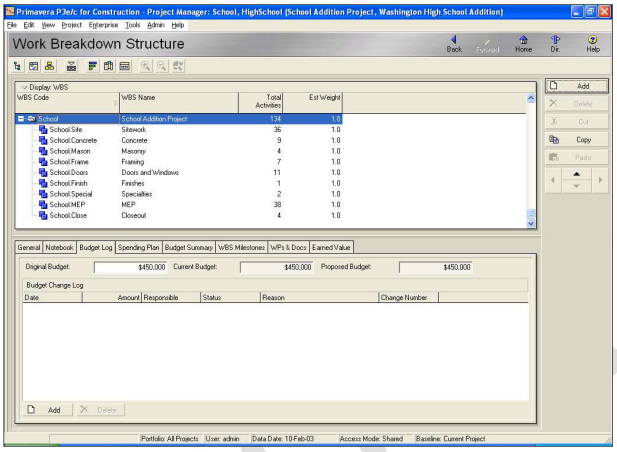
1. Higher-level WBS elements determine the status of their lower level WBS elements.
2. Status other than Active does not allow users access to assigned activities within the timesheet.
3. Status other than What-If are included with a summarization run of activities.
4. Planned Status prevents users in Progress Reporter from viewing any assigned activities and/or assigning themselves to perform work on those activities.
5. Active Status allows users in Progress Reporter to view activities within that element and assign themselves to perform work on those same activities.
6. Inactive Status prevents users in Progress Reporter from viewing any assigned activities and/or assigning themselves to perform work on those activities
7. Activities assigned to an Inactive Status WBS element are calculated with a 0 duration when scheduling though the actual duration does not change. The activity's start date and finish dates are scheduled as the same dates.
8. What-If Status prevents users in Progress Reporter from viewing any assigned activities and/or assigning themselves to perform work on those activities

Notebook tab – add a notebook topic (created under the Admin menu, Admin Categories, Notebook Topics) and enter a project-specific description of the topic.



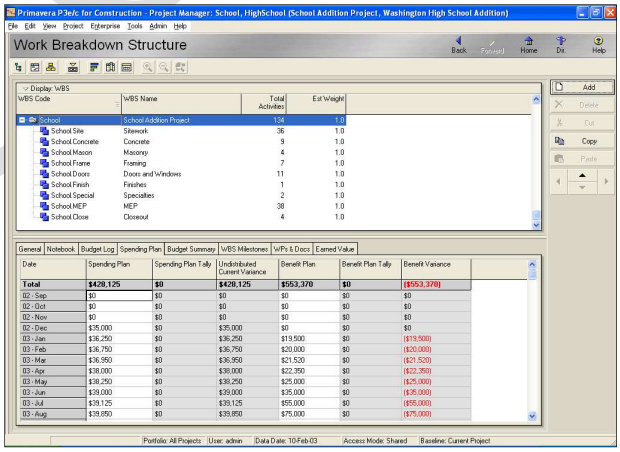
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Budget Log tab



1. Original Budget – estimate of total amount required to complete the project including funding contributions
2. Current Budget – Original Budget + Approved Budgets
3. Proposed Budget – Original Budget + Approved and Pending Budgets
4. Budget Change Log – track changes to the Original Budget
5. Enter the Date change requests were made as well as Amount, Responsible for change, Status of change (which affects totals shown in Current and Proposed Budgets), Reason for change request, and Change Number

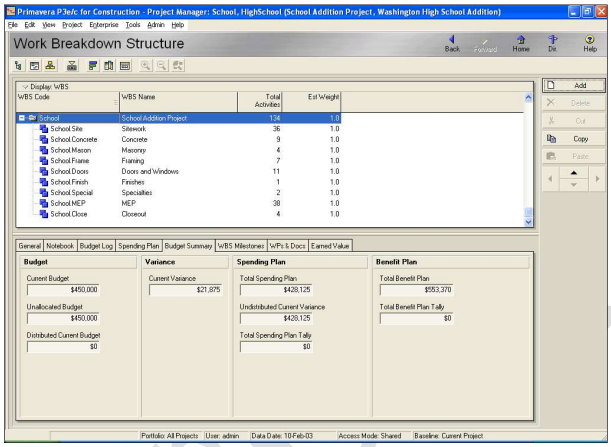
Spending Plan



1. Date – monthly range beginning 3 months prior to project start and 4 years after project start
2. Spending Plan – amount spent on a per month basis
3. Spending Plan Tally – project summary when selecting an EPS node
4. Undistributed Current Variance – Spending Plan (monthly value) – Spending Plan Tally (monthly value)
5. Benefit Plan – profit portion of the monthly amount normally entered at the end of the project
6. Benefit Plan Tally – summarized benefit plan information when selecting an EPS node
7. Benefit Variance – Benefit Plan Tally minus the Benefit Plan Monthly

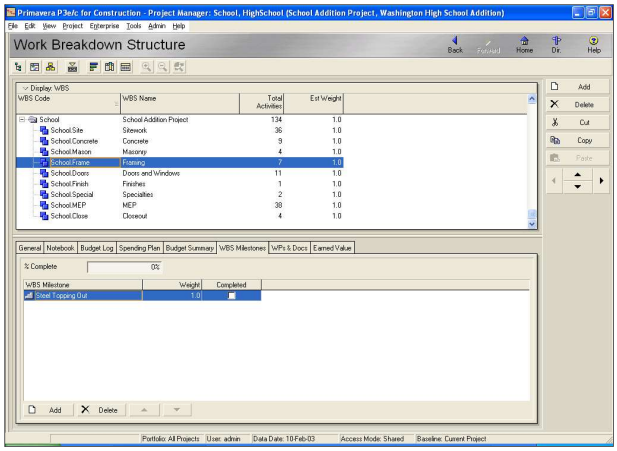
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Budget Summary tab – calculated using amounts entered in the Budget Log and Spending Plan tab



1. Budget
   1. Current Budget – Original Budget + Approved Budgets
   2. Unallocated Budget – Current budget minus distributed current budget
   3. Distributed Current Budget – total of current budget values
2. Variance
   a. Current Variance – Current budget minus total spending plan
3. Spending Plan
   a. Total Spending Plan – total of monthly spending plan
   b. Undistributed Current Variance – Total Spending Plan minus Total Spending Plan Tally
   c. Total Spending Plan Tally – total monthly spending tally
4. Benefit Plan
   a. Total Benefit Plan – total monthly benefit plan
   b. Total Benefit Plan Tally – total of the monthly benefit plan tally

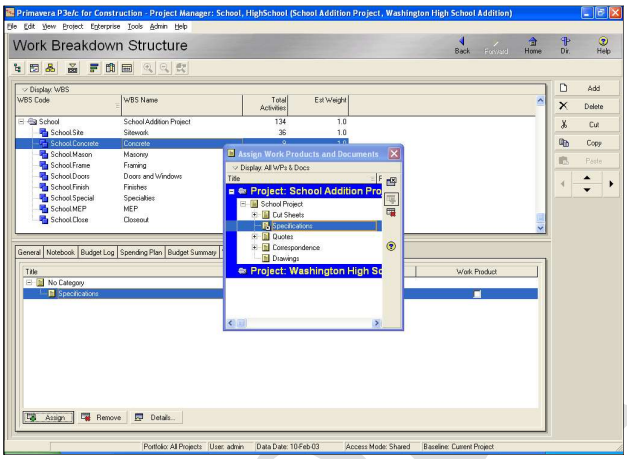
WBS Milestones tab – establish a weight used to indicate importance within the project and to calculate earned value or activity performance percent complete.



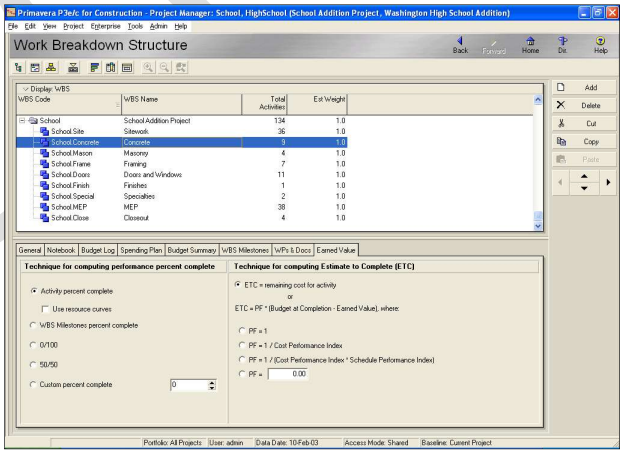
1. Performance % Complete will remain o with Milestones having no assigned activities
   HINT: Use a dummy activity to the WBS element/milestone to calculate performance % complete.
2. Add unlimited WBS milestones and assign a weight to be used for calculating performance % complete for all activities assigned within the WBS element *when the milestone is marked as Completed.*
3. *Actual Weight of Completed Milestones / Total Possible Weight of All Milestones = Performance % Complete.*

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WPs & Docs tab – assign work products and documents.



Earned Value – used to measure project performance to project costs and the project schedule by comparing budgeted cost to actual cost. Comparisons can be performed at the WBS element level or activity / activity group level. (Define default Earned Value settings for all WBS elements by selecting the Admin menu, Admin Preferences, Earned Value.)



1. Earned Value Cost = Budget At Completion (BAC) x Performance % Complete
2. Select the Technique for computing performance percent complete
   a. Activity percent complete – calculated based upon the current activity's completion percentages and percent complete type (Activity Details, General tab)
   b. Use resource curves – overrides Activity Percent Complete type for activities with a resource curve assigned to one or more resource assignments – uses Units Percent Complete x Budget at Completion = Earned Value
   c. WBS Milestones percent complete – calculated based on WBS element's weighted milestones
   d. 0/100 – percent complete used to calculate Earned Value at 100 % after the activity is completed and 0% before completion
   e. 50/50 – percent complete used to calculate Earned Value as 50 % after the start of the activity and 100% after the activity is completed
   f. Custom percent complete – enter a percentage to calculate Earned Value to be applied only after the activity begins and up until its completion when its earned value is 100 %
3. Select the Technique for computing Estimate to Complete (ETC)

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a. ETC = remaining cost for activity – ETC values are calculated as remaining cost to complete the activity (ETC = remaining duration x applicable resource rates)
OR

b. ETC – PF * (Budget at Completion – Earned Value), where:
   1. PF = 1: ETC = BAC – Earned Value Cost for optimistic result
   2. PF = 1/CPI: ETC values calculated according to a Performance Factor (PF) of 1 / Cost Performance Index (CPI) for most actual result
   3. PF = 1/(CPI*SPI): ETC Values calculated according to a PF of 1 divided by the product of the CPI and the Schedule Performance Index (SPI) for a more pessimistic result
   4. PF =: ETC values calculated according to a specified PF value

Deleting the WBS Element

Deleting WBS elements allows for activity assignment deletion OR reassigning / merging activity assignments to the element's higher-level WBS. *Deleting higher-level WBS elements will delete all lower levels contained within that element.*

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Chapter 5
Planning and Scheduling in P6

In this chapter you will begin to schedule your project in a typical order of scheduling.



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Calendars

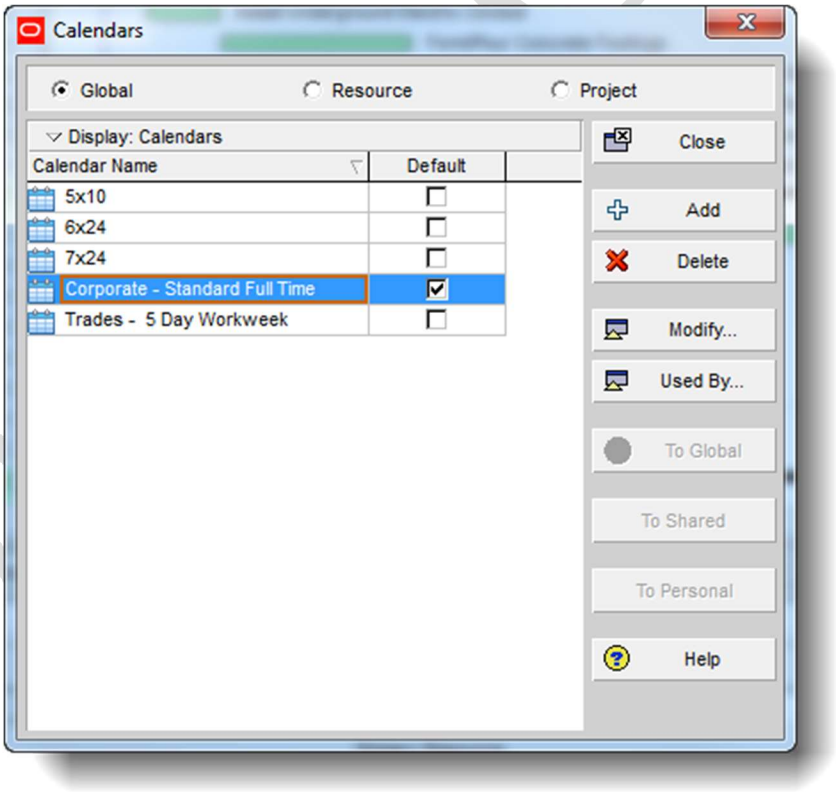
Calendars assigned to each resource and each activity define available workhours per calendar day as well as non-workdays including holidays and resource vacations.

Creating and Assigning Calendars

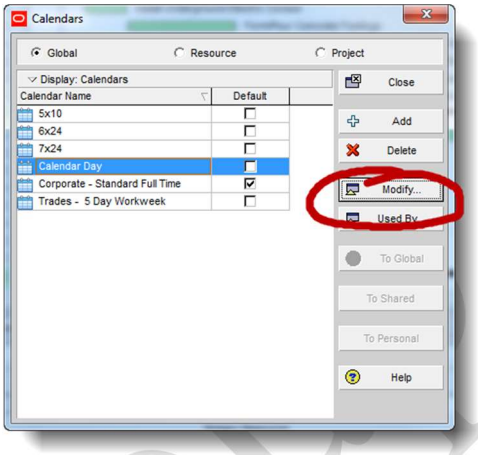
Creating calendars defines an unlimited number of calendars for varying work patterns.

1. Create project calendars for each project and resource calendars for each resource in the EPS.

Select the Enterprise menu, Calendars. For *project* calendars, first open the applicable project.
Choose either resource or Project, then select the **Add** button.
Highlight the calendar to be copied, then select the **Select** button.
Enter the calendar's name.



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Select the **Modify** button to edit the highlighted calendar.
Select the **Used By** option to view the calendar's assignments.
Click the **To Global** button to convert a resource or project calendar to a global calendar.

2. Link Resource and project calendars to control changes to entire groups of calendars. These global calendars apply to all projects in the database.

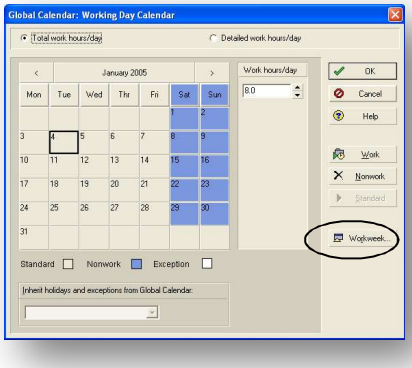
    a. Select the Enterprise menu, Calendars, Global, Add.
    b. Highlight the calendar to be copied and select the **Select** button.
    c. Enter the calendar's name and mark the Default option to make the calendar the default global calendar for activities and resources.
    d. Select **Modify** to edit the highlighted calendar.
    e. Select the **Used By** option to view the calendar's assignments.

Scheduling with Calendars

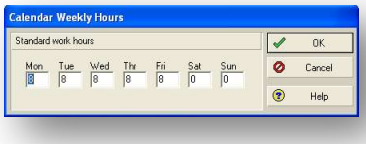
Schedule and track activities and level resources using calendar assignments.

1. Calendars consist of a standard workweek and a list of exceptions. Work/non-work time is based on the calendars regular workweek.

    a. Select the Enterprise menu, Calendars, and choose the calendar type (Global, Resource, or Project).



Highlight the calendar to edit and select **Modify**.
Use the arrows (right and left) to scroll through each month.
Click the month/year to select another month or use the arrows (right and left) to scroll to another year.
Select **Workweek** to modify the number of workhours for each day. Changes made to the regular workweek are reflected in the global, resource, or project calendar dialog box.
Work, nonwork, and exceptions are displayed using varying colors.



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Apply the same change to a specific weekday in the displayed month by selecting the weekday's column label and applying the change (Work, Nonwork, Standard).

2. Use more than one calendar when the project contains activities occurring on different schedules (i.e. Working days, Calendar days).
3. Assign each activity to the specific calendar to indicate the worktime available for performing that activity.
4. Create multiple calendars to control the periods when work is performed by resources throughout the organization and associate holidays/exceptions from the global calendar for each work cycle to indicate resource availability.

Exercise: Build Project Calendars

Create the following two Project Calendars:

Working Day Project Calendar based on the Global Calendar's Standard 5 Day Workweek. Enter all holidays for the next year.

Calendar Day Project Calendar based on the Global Calendar's 24/7 Workweek Schedule.

Calendars and Activity Types

Activity type determines whether the activity uses the calendar of an assigned resource or its activity calendar.

Deleting Calendars

3. Delete existing calendars by opening the Enterprise menu, Calendars.
4. Choose Global, Resource, or Project, select the calendar to delete, click the Delete button. If activities or resources are assigned to the calendar the Calendars in Use dialog box appears to link the calendars assignments to another calendar.

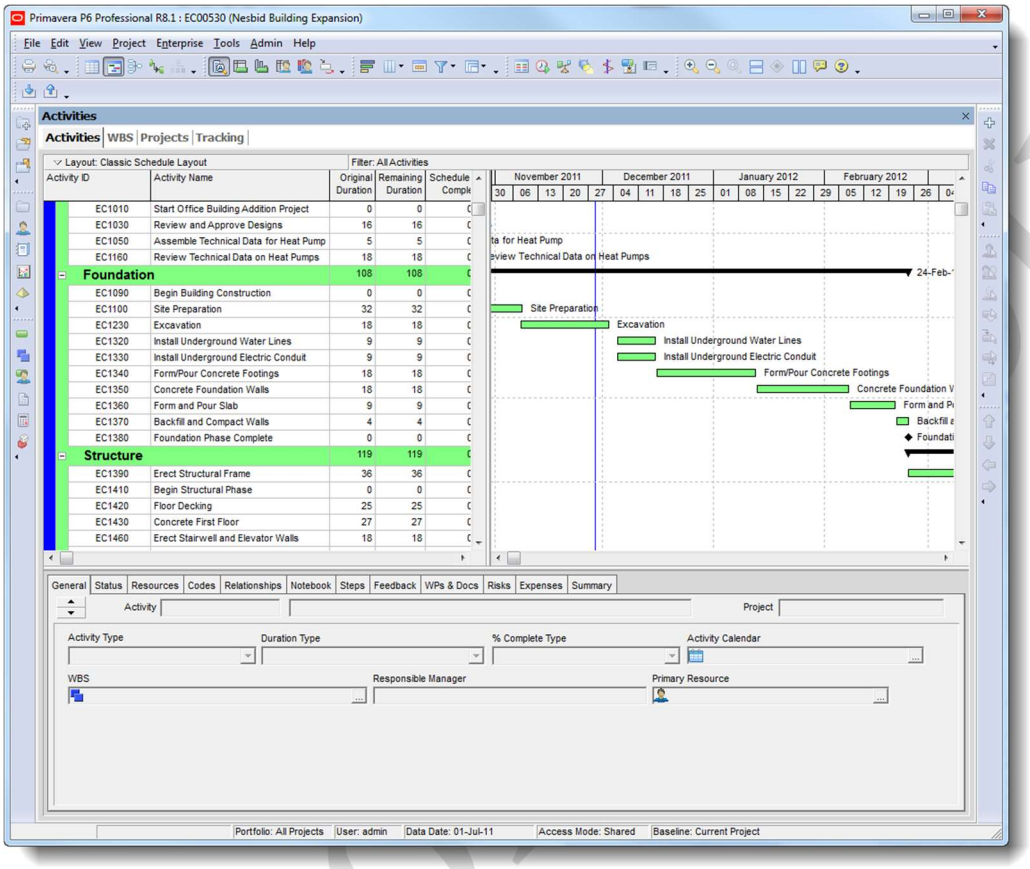
    a. Choose the Select Replacement Calendar option and select a replacement calendar.
    or

Choose Link to Default Global Calendar to delete the calendar and link its assignments to the default global calendar.

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Activities

Activities are the fundamental work elements of the project having a determined amount of time as well as being the lowest level of the WBS.



Adding Activities

Use the Activity Table or Activity Network layouts to add activities within the project.

1. Using the upper and lower layouts – show the Activity Table, Gantt Chart, Activity Usage Spreadsheet, or Activity Network in the top layout, and/or show Activity Details, the Activity Table, Gantt Chart, Activity or Resource Usage Spreadsheet or Profile, or the Trace Logic option in the bottom layout. Choose the Layout bar, Show on Top/Show on Bottom. Layouts can be customized for ease of use.
2. Select an existing activity above where the new activity is to be created and select Add. Assign an activity to an existing group in the Activity Table or Activity Network while creating the activity by selecting the group row and selecting Add.

3. Enter the Name as well as other pertinent information directly in the column cells for the new activity OR use the Activity Details Layout to enter activity information.

Exercise: Enter Project Activities

Add the following activities to the New High School Project:

WBS	Task Description	Duration
Milestones	Notice to Proceed	0
Milestones	Substantial Completion	0
Design	Prepare Final Design	5
Procurement	Submittals	10
General Cond	Mobilization	2
General Cond	Trailer Setup and Fencing	3
General Cond	Final Clean	5
Site Work	Survey and Staking	2
Site Work	Excavate and Grade	7
Substructure	Layout and Grade Pad	5
Substructure	Underground Utilities	5
Substructure	Form/Pour Footings	3
Substructure	Form/Rebar/ Pour Slab	4
Substructure	Cure Slab	3
Structure	Structural Steel	5
Structure	Roof Joists & Decking	5
Exterior	Exterior Framing	8
Exterior	Door/Window Frame	5
Exterior	Lath & Plaster Ext Walls	10
Exterior	Roof Membrane	5
Int Rough-in	Interior Frame	5
Int Rough-in	Rough Plumbing	5
Int Rough-in	Rough Mechanical	5
Int Rough-in	Rough Electrical	5
Int Rough-in	Insulation	3
Int Finishes	Drywall	5
Int Finishes	Tape/Texture	3
Int Finishes	Ceiling Grid	3
Int Finishes	Acoustical Ceiling Tile	2
Int Finishes	Paint	5
Int Finishes	Millwork	5
Int Finishes	Interior Doors	3
Int Finishes	Finish Plumbing	3
Int Finishes	Finish Mechanical	5
Int Finishes	Finish Electrical	7
Int Finishes	Flooring	5
Int Finishes	RR Partitions & Accessories	3
Int Finishes	Test & Balance	5
Site Finishes	Site Finish Grade	10
Site Finishes	Base & Pave Parking Lot	5

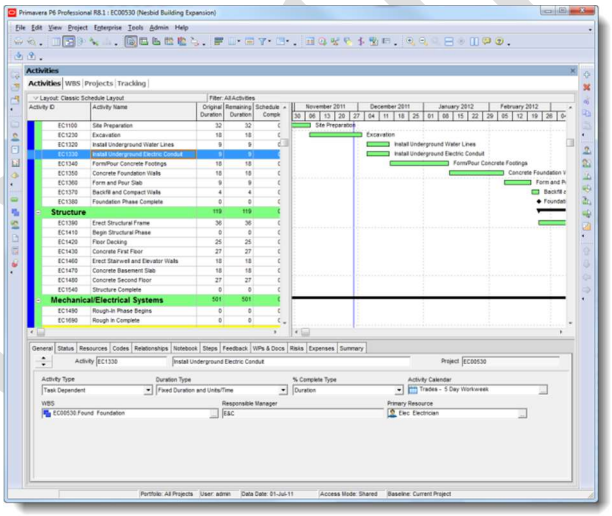
Assign the appropriate calendars to each activity.

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Viewing Activity Details

Open the Activity Details by clicking the Layout bar, Show on Bottom, Activity Details. Customize the Bottom Layout (the tabs shown and their order) by selecting the Layout bar, Bottom Layout Options OR right-click in the bottom layout to Customize Activity Details.

1. **General tab – define duration type, WBS assignment, primary resource, activity type, and activity calendar.**



   1. Duration Type – determines whether resources, the schedule, or costs are most important when activities are updated. *Only applies when resources are assigned to the activity.* The duration type controls which variables of the following formula are calculated when values are changed: Remaining Resource Units = Units/Time x Remaining Activity Duration (e.g. Erect Structural Steel is assigned 8 hours/day for 5 days, the remaining units or work effort is 40 hours).
      1. *Fixed Duration & Units/Time* – indicates the schedule is the limiting factor in the project. The duration does not change when modifying or updating the activity regardless of the assigned resources. For task dependent activities.
         1. Calculate either the remaining units or the units per time period when updating remaining duration.
         2. Recalculate the remaining units while the units/time for the resource stays constant.
         3. Remaining Units = Units/Time x Remaining Duration.
      2. *Fixed Duration & Units* – indicates the schedule is the limiting factor in the project. The duration does not change when modifying or updating the activity regardless of the assigned resources. For task dependent activities.
         1. Calculate either the remaining units or the units per time period when updating remaining duration.
         2. Recalculate units/time while the remaining units stays constant.
         3. Units/Time = Remaining Units/Remaining Duration.
      3. *Fixed Units/Time* – indicates resource availability is the most critical factor of your project.

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1. Units/time or resource rate remains constant while the activity's duration or work effort may change.
2. For resource dependent activities.

4. *Fixed Units* – indicates that the budget (units or cost) is the limiting factor or the total amount of work is fixed.
    1. Work effort required to complete the activity remains constant while the activity's duration or the resource rate may change.
    2. For use in conjunction with resource dependent activities so that increasing resources can decrease the activity's duration.

2. Activity Type – determines the activity's function within the project as well as the calendar that is used during scheduling.
    1. *Task Dependent* – the activity's resources are scheduled according to the activity calendar.
    2. *Resource Dependent* – each resource assigned to the activity is scheduled based on the resource's own calendar.
    Level of Effort (LOE) – the activity's duration is dependent on its predecessor and/or successor activities. These activities have no constraints and are considered ongoing (e.g. project management tasks, reviews, and meetings). LOE activities are not included when leveling resources.

FS and SS predecessors / SF and SS successors drive the start date of the LOE activity. FF and SF predecessors / FS and FF successors drive the finish date of the LOE activity.

    3. *Start Milestone* – indicates the beginning of a major stage in the project. There are no time-based costs or resource assignments other than a primary resource. Milestones have zero duration.
    4. *Finish Milestone* – indicates the ending of a major stage in the project. There are no time-based costs or resource assignments other than a primary resource. Milestones have zero duration.

3. % Complete Type – calculate an activity's percent complete according to the duration, activity units, or a physical percent complete entered per activity.
    1. *Duration* – the activity's percent complete is calculated from the actual and remaining duration.
    2. *Physical* – the activity's percent complete is entered by the user.
    3. *Units* – the activity's percent complete is calculated from the actual and remaining units.

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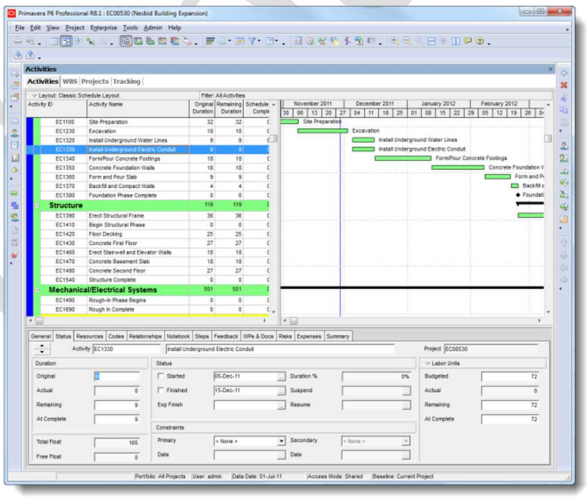
4. Activity Calendar – select the activity's calendar

5. WBS – assign a WBS element

6. Responsible Manager – the name of the responsible manager (OBS) for the activity.

7. Primary Resource – the name of the person responsible for the overall work on the activity as well as updating the activity status.

2. **Status tab** – view and edit detailed scheduled information (i.e. actual dates, float types, constraints, and duration; also labor and nonlabor unit values).



1. Time values and periods are recalculated according to the project's calendar and the standard time period. Choose the Admin menu, Admin Preferences, Time Periods to view time period abbreviations.

2. Duration
   1. *Original* – expected number of work periods to complete the activity.
   2. *Actual* – actual number of work periods spent to complete the activity.
   3. *Remaining* – remaining number of work periods remaining to complete the activity.
   4. *At Complete* – estimate of the duration at the completion of the activity. (At Complete = Actual Duration + Remaining Duration)

c. Status
   1. *Started* – indicates the start of the activity as well as the planned start date, or if started, the actual start date.
   2. *Finished* – indicates the finish of the activity as well as the planned finish date, or if completed, the actual finish date.
   3. *Exp Finish* – the expected end date entered by the primary resource. When adding the Expected Finish date, the calendar defaults to the date and time of the activity's Early Finish date. When editing the Expected Finish date, the calendar defaults to the date and time of the current Exp Finish date.

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If planned dates differ from current scheduled dates a constraint must be applied to hold the activity to the dates. Otherwise the dates are overwritten during scheduling.

4. *Duration %* – activity must be started to edit
    1. Calculated from the actual and remaining duration if the % complete type is Duration.
    2. Displays unit % complete calculated from the actual and remaining units if the % complete type is Units.
    3. Displays physical % complete manually entered if the % complete type is Physical.
5. *Total Float* – amount of time the activity can be delayed without delaying the finish date of the project.
6. *Free Float* – amount of time the activity can be delayed without delaying the immediate successor activities.

d. Constraints – for activities that must start or finish on a specific date.
    1. *Start on constraint* – imposed start date can delay an early start or accelerate a late start satisfying the imposed date maintaining network logic.
    2. *Start on or after constraint* – imposed restriction limiting the earliest time the activity can start. Used in the forward pass only if the calculated early start date is earlier than the imposed date. The early start date cannot be earlier than the imposed date. This constraint affects only early dates.
    3. *Start on or before constraint* - imposed restriction limiting the latest time the activity can start. Used in the backward pass only if the calculated late start date is later than the imposed date. This constraint can decrease total float and only affects late dates.
    4. *Finish on constraint* - imposed finish date can delay an early finish or accelerate a late finish satisfying the imposed date maintaining network logic.
    5. *Finish on or after constraint* – imposed restriction limiting the earliest time an activity can be completed. Reduces float to coordinate parallel activities to ensure the finish is not scheduled before the specified date. Usually applied to activities with few predecessors that must finish before the next phase of the project.
    6. *Finish on or before constraint* – imposed restriction limiting the latest time the activity can be finished. Affects only late dates. Use to ensure that the late finish date of an activity is no later than the date imposed.
    7. *As late as possible constraint* – imposed restriction imposed on an activity or work unit with positive float allowing it to start as late as possible without delaying its successors. The early dates are set as late as possible without affecting successor activities.

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8. *Mandatory start constraint* – imposed restriction setting the early and late start dates equal to the specified dates regardless of its effect on network logic. Can affect late dates for all activities leading up to the constrained activity and those activities leading from the constrained activity.
   NOTE: When Mandatory constraints are placed on calendar non-worktime periods, early and late dates are not equal to each other. The early date is moved forward to the next available worktime while the late date is moved backward to the first valid worktime. Possible negative float.

9. *Mandatory finish constraint* – imposed restriction setting the early and late finish dates equal to the specified dates regardless of its effect on network logic. Can affect late dates for all activities leading up to the constrained activity and early dates for those activities leading from the constrained activity.

When Mandatory constraints are placed on calendar non-worktime periods, early and late dates are not equal to each other. The early date is moved forward to the next available worktime while the late date is moved backward to the first valid worktime. Possible negative float.

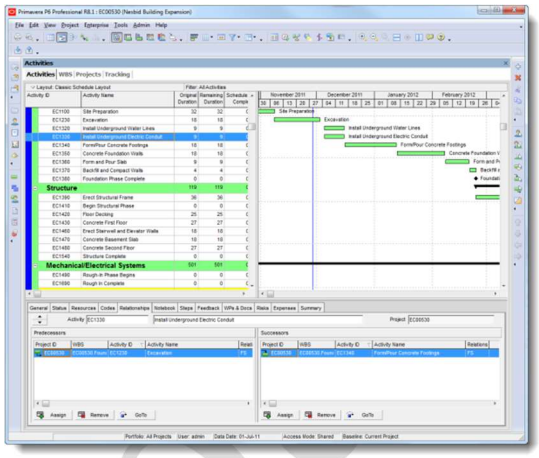
10. *Date* – the activity's constrained date.
11. *Secondary* – possible secondary constraint type.
12. *Date* – the activity's constrained date.

e. <u>(Non)Labor Units/ (Non)Labor Cost</u> – click the Labor Units bar to select Labor or Nonlabor units or costs. Labor Unit amounts total amounts for all resources assigned in the Resources tab.
   1. *Budgeted* – planned or expected number of labor / nonlabor units used.
   2. *Actual* – actual number of labor / nonlabor units or cost used. If the activity has begun enter a new actual value.
   3. *Remaining* – remaining number of labor / nonlabor units or cost used. If the activity is underway enter a new remaining value.
   4. *At Complete* – estimate of the labor / nonlabor units or cost at the completion of the activity. (At Complete Units[Costs] = Actual Units[Costs] + Remaining Units[Costs]). If the activity is underway enter a new At Complete estimate.

3. **Relationships (Predecessors / Successors tabs)** – determine whether an activity can begin only after other activities start or finish to calculate early and late dates for each activity. Relationships can be created between activities in the same project or established between activities in multiple projects in the EPS.

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a. Four Relationship Types exist between from predecessor to successor activities:
   1. ***Finish to start*** – the successor activity can only begin when the predecessor activity is completed
   2. ***Finish to finish*** – the finish of the successor activity depends on the finish of the predecessor activity
   3. ***Start to start*** – the start of the successor activity depends on the start of the predecessor activity
   4. ***Start to finish*** – the successor activity cannot finish until the predecessor activity starts



b. Lag time – number of time units from the start or finish of an activity to the start or finish of its successor OR the minimum variance between the two activities start or finish dates. Lag is calculated based on the successor's calendar.

c. Dissolving Activities – deletes an activity and creates a FS relationship to its predecessor and successor activities. Select the activity to dissolve, Edit, Dissolve.

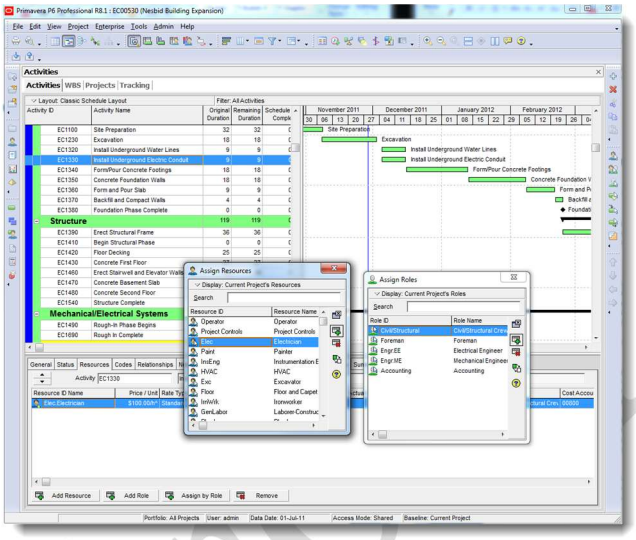
d. Assign inter-project relationships by selecting the Display bar in the Assign Predecessors / Successors dialog boxes, Project, then select applicable activities and relationship ties.

e. Trace Logic layout (Layout bar, Show on Bottom, Trace Logic, with the Gantt Chart or Activity Network on the Top) examines a path or chain or activities moving along the path by selecting the predecessor or successor activities. Choose the number of predecessor/successor levels displayed by selecting the Layout bar, Bottom Layout Options.

4. **Resources tab** – view/edit the personnel, equipment and material used to perform work on activities.

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a. Select Add Resource to assign a resource (Hold down the Ctrl button to select multiple resources to be assigned.)



1. Replace an existing assigned resource by highlighting the resource to replace and select Add Resource, Replace.

2. View additional columns by right-clicking in the columns area, Customize Resource Columns.

3. Add the Cost Units Linked column to calculate costs from the units indicated or to recalculate units based on actual cost. If marked costs are recalculated dependent upon the activity's progress. If no progress has occurred the Budgeted Cost/Units, Remaining Cost/Units, and At Completion Cost/Units are equal. If progress has occurred (i.e. actual start is assigned) and actual quantity values or an actual cost value for the resource assignment has been recorded then the Budgeted Cost/Units value is retained and the Remaining and At Complete Cost/Units are recalculated.

b. Click Add Role to assign a role. (Hold down the Ctrl button to select multiple roles to be assigned.)

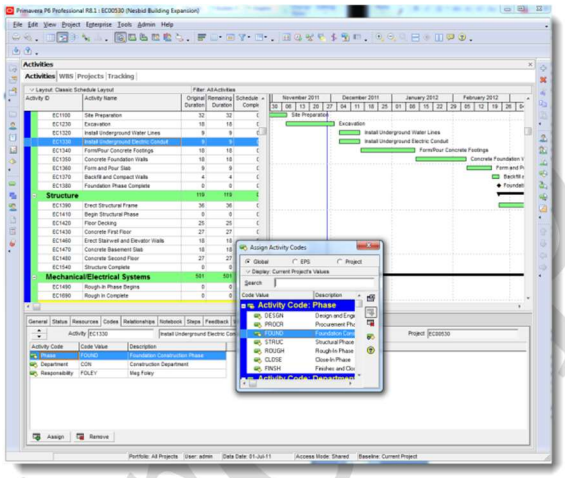
1. Assigned resources to roles with specific skills for schedule and cost planning until resources are established. Replace the role with the correct resource based on the proficiency requirements established for the role.
2. Roles can only be removed from an activity once a resource has been assigned.

3. Assign resources by role by first adding at least one role to the activity.
   a. Click Assign by Role to open the Assign Resources by Role dialog box.
   b. Choose the resource(s) to be assigned and click the Select button.
   c. View the Role Usage Profile in the bottom portion of the layout to analyze role use per time-period.
   d. View the weekly breakdown or resource use each month through each activity's duration under the Activity Usage Spreadsheet organized by resource.

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5. **Codes tab** – view codes used to categorize activities according to the organizational and project needs.

   a. Assign codes and their applicable values.
   b. Hold the Ctrl key to select multiple codes and values.



6. **Notebook tab** – view additional information describing the activity by specific categories of information.

   a. Assign applicable notebook topics (predefined in the Admin menu, Admin Categories, Notebook Topics).
   b. Hold the Ctrl key to assign multiple topics.
   c. Enter a brief description for the highlighted topic.

7. **Steps tab** – divide the activity into smaller units and apply weights to each step linked to the activity's physical percent complete to complete each step.

   a. Select Add and enter the name of the new step as well as a brief description of the step.

   b. Use the up and down arrows to move the step to an earlier or later stage of the activity.

   c. Enter the weight (0.0 to 999999.0) of the step relative to other listed steps. The Percent corresponds to the step's weight. *Use the Percent Complete Type of Physical, General tab, to use weighted steps in calculating the activity's percent complete AND mark the checkbox in the Project Details Defaults tab Activity Percent Complete Based on Activity Steps.*

   d. Use weighted steps to status activities when the project consists of large activities encompassing distinct tasks.

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e. Example: An activity has 3 weighted steps: The first with a weight of 2, the second and third with a weight of 1. When the first activity is complete, the percent complete is 50, and when the second activity is complete the percent complete is 75.

8. **Feedback** – exchange notes with the primary resource through the Progress Reporter.

   a. Review notes created by the primary resource through Progress Reporter.

   b. Mark new if the item created is new since last reviewed and clear the checkbox once the item has been reviewed.

   c. Enter any comments to the resources working on the activity in the Notes to Resources area that are then posted to Progress Reporter.

9. **WPs & Docs** – catalog and track project-related work products and documents (e.g. guidelines, procedures, standards, plans, design templates, worksheets, deliverables, etc.).

   a. Assign multiple documents holding the Ctrl key.

   b. Mark the checkbox to indicate whether the document is a work product.

   c. View detailed information or open a specific document by selecting the Details button.

   d. Establish categories for WPs & Docs under the Admin menu, Admin Categories, Document Categories.

10. **Expenses** – one-time expenditures for nonreusable items categorized by predefined cost accounts with expenses. Right-click in the columns to Customize Expense Columns.

    a. Add the activity's expense and enter the expense's name.

    b. Assign the category to each expense by double-clicking in the item's expense category listing.

    c. Assign the accrual type to each expense by double-clicking in the item's Accrual Type cell.

    d. Enter the budgeted/planned units amount.

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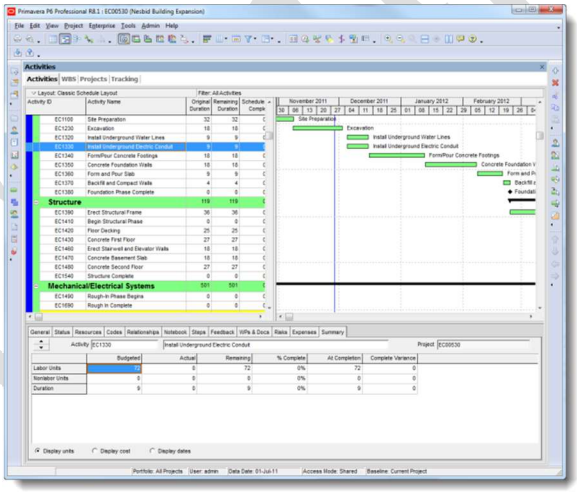
e. Enter the cost per unit to calculate the expense's budgeted/planned cost in the Budgeted/Planned Cost field (Budgeted[planned] units x price per unit).

f. Mark the Auto Compute Actuals checkbox to calculate an expense's actual cost based on the activity's planned completion percentage.

g. Enter actual expense costs already incurred in the Actual Cost field.
h. Enter the Vendor name to which the expense is payable.

11. **Summary** – view detailed cost and unit information.

a. Select Display Units to view summary information about the activity's units.
b. Select Display Cost to view summary information about the activity's cost.
c. Select Display Dates to view all start and finish dates.



3 Methods of Assigning Relationships

1. ***Use the Linking Tool***–drag the mouse between any two activities to define relationships. Double-click the relationship line to edit or delete the tie.
2. ***Use the Relationships tab*** OR Predecessor / Successor tabs to assign relationship types. Double-click to modify the Relationship Type, and Lag.
3. ***Use the Link Activity*** Select activities, the choose Edit, Link Acticities. FS relationships are added based on the order of selection.

Exercise: Add logic to connect the activities

Using the above methods for assigning relationships, assign relationships in 3 stages:

1. ***Build the relationships within each WBS element;***
2. ***Add relationships to the starting activity of each WBS element;***
3. ***Build Relationships to the ending activities of each WBS element.***

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Exercise: Copy and Paste WBS Elements

Expand the Schedule to include a second building – Copy WBS for Building A to New WBS for Building B (add relationships to start and finish of new building activities)

The Critical Path

The project duration is controlled by those activities with the least amount of float or the longest critical path. Highlighting the critical activity bars using distinctive colors or endpoints can identify the critical path. P6 highlights critical activities red automatically.

In order to calculate negative float, there must be an imposed project finish date or other constraint on the schedule completion, like a Finish Milestone with a late finish constraint.

Exercise: Run a Filter for the Critical Path

To view critical path activities select Filters and select Critical Path.

Final Adjustments

After scheduling you may verify and refine activity durations by stretching or shrinking activity bars in the Bar chart view. Duration may be adjusted directly on the activity by positioning the mouse (which will appear as ⇔) on the end of the bar and dragging the bar to the right or left. To move the activity itself without adjusting the duration, position the mouse (which should appear as +) on the activity bar and move the bar either to the right or left. However, calculating the schedule will readjust activities according to their ties and durations.

Congratulations! You've created an As-Planned or Baseline Schedule.
Now Export the file so it can be emailed.

Exercise: Exporting a File

Select File, Export
Select Primavera XER on the Export Format Window; select Project for Export Type; select the Project you wish to export for Projects to Export; select where to save the file and the Filename for File Name.

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Chapter 6
Customizing P6

In this chapter you will learn additional tips to organize and manage the project schedule



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Importing Schedules

Import a schedule by selecting FILE, Import. The import wizard will walk you through the steps including selecting the schedule file type.

From the Import Type window select to import a project, resources only, or Roles only. Select Project. The File Name window will provide browse button to select the file to import.

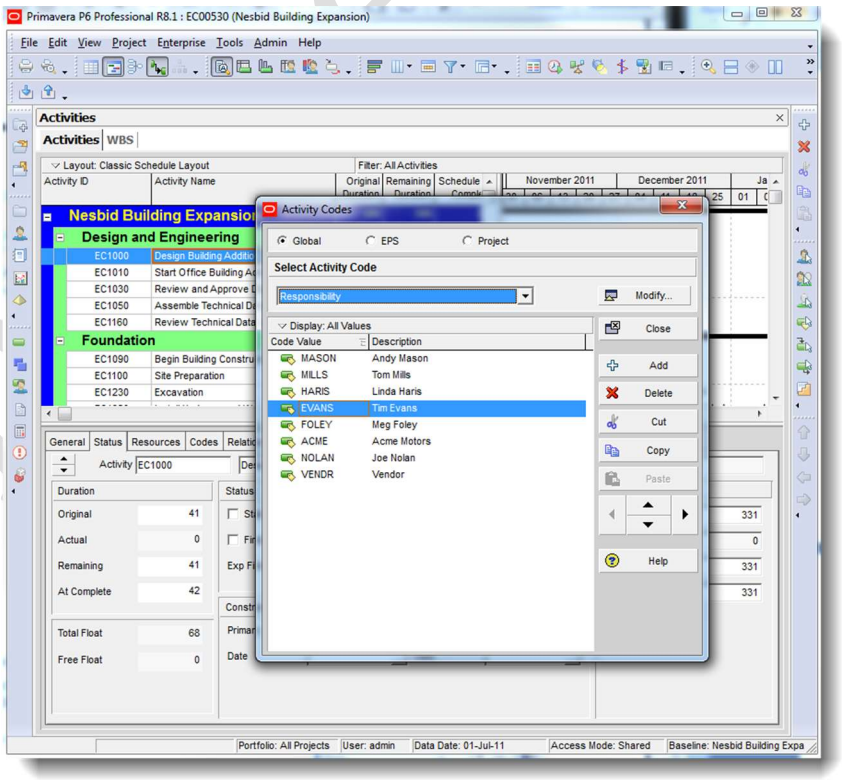
On the import project options select the project ID and the Action. From the Action selection you can choose to create a new project, overwrite an existing project, or replace an existing project. Finally select the EPS Node to import the project to.

Select Default Configuration on the Update Project Options Window for the layout name.

Select Finish to complete the import.

Activity Codes

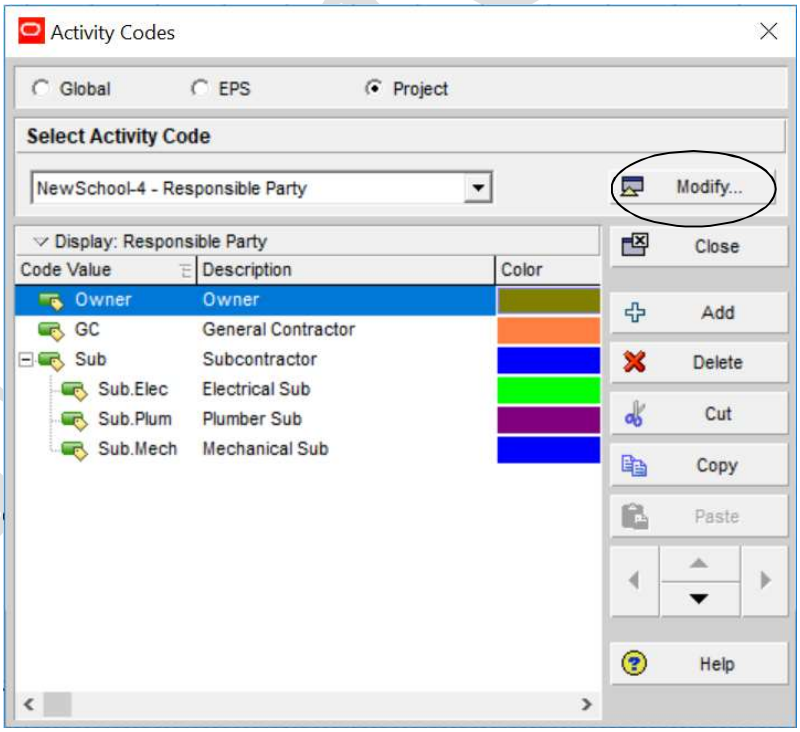
Activity Codes categorize activities in the project using specific values to further describe coding



Creating Global Activity Codes

Create Global activity codes to organize (sort, filter, and group) project activities throughout the EPS.

1. Select the Enterprise menu, Activity Codes
2. Choose Global and select the **Modify** button.
3. Click the **Add** button and enter the name of the global activity code.
4. Enter the maximum number of characters for the activity code's values.
5. Project activity codes are listed beneath the name of the opened project.
6. Establish global activity code values by highlighting the activity code for which values are to be created and select **Add**.
7. Enter the value's name and description paying attention to the maximum character length set for the activity code.
8. Using the arrow keys move the value to its correct location or placement within the Activity Codes hierarchy. Indent for a more detailed position or outdent for a more general position.



Exercise: Create Activity Codes in P6

Select Enterprise, Activity Codes
Select "Project" radio button for Project Activity Codes
Select Modify to add a Project Code "Responsible Party"

Create the Project Activity Code and Values as shown on the previous chart. Indent the Electrical, Plumbing, and Mechanical under the subheading Subcontractors:

Converting Project Activity Codes to Global Codes

Convert project activity codes and their values to global activity codes and values.

1. Highlight the project activity code to convert and click **Modify**.
2. Select the Make Global option and click **Yes** to make the conversion.
3. Changes made to activity codes and their values are applied to all activity assignments as well as converting project activity codes to global activity codes.

Exercise: Assign Activity Codes

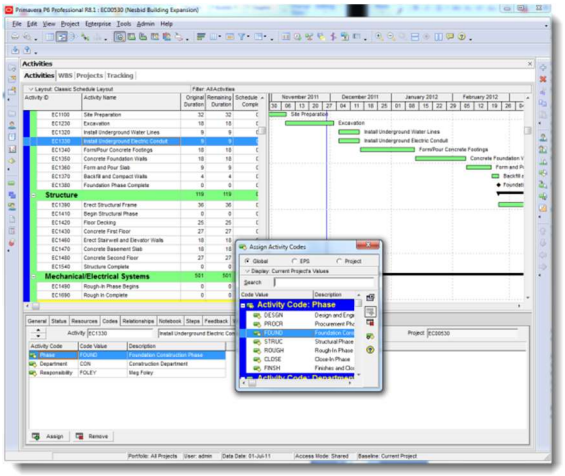
Activities can be assigned to multiple Activity Codes and their values.

Open the Codes tab on the Activity Details Layout.

Select the Assign button to open the Assign Activity Codes window.

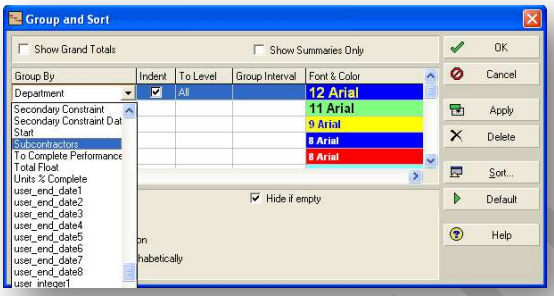
Choose either Global or Project to determine which codes are to be assigned to the Activity.

Highlight the activity code value and select the Assign button



Grouping and Sorting Activities with Headings

Group activities and projects using the Activity Table, Gantt Chart, and Activity Network layouts.



1. Select the Layout bar, Group and Sort.

2. Under the column Group By click the first available line and select the applicable project or global activity code.

3. Select the lowest level to include in the layout to group by a hierarchical element.

4. Each group's data is summarized by default displaying summary bars in the Gantt Chart.

   a. Select the Layout bar, Show on Top, Gantt Chart.

   b. Select the Layout bar, Bars, and mark the Display option next to summary to show a summary bar for each group band in the layout.

   c. Double-click the band to exclude the project details and show only a summarized band. A plus (+) sign to the left of the group name indicates a group is summarized.

   d. Click the Layout bar, Collapse All to summarize all groups in the layout.

   e. Summarizing displays the earliest start date of all activities with the code value as well as displaying the latest early finish date of all activities in the code value.

5. View values assigned to all open projects by selecting the Display bar, Filter By, Current Projects' Values.

Exercise: Group and Sort Activities by Responsible Party

Select View, Group and Sort by;
From the Pulldown under Group by, select Responsible Party;
To organize by Responsible Party and WBS, select WBS on the second Group by Line.

Grouping, Sorting, and Filtering

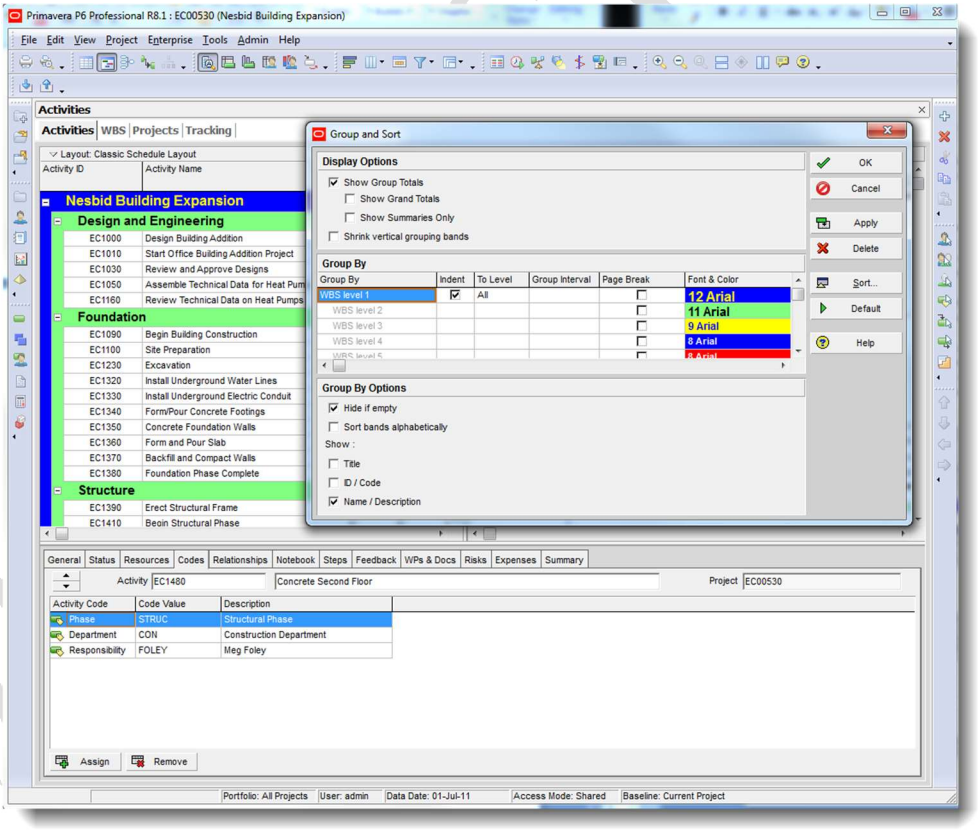
Activities or project that share common attributes can be grouped into categories and filtered to narrow the selection to a specific data group.

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Grouping Data

Use predefined grouping and sorting options OR customize group criteria in single or multiple-level data items. Group criteria can be arranged in up to 20 levels of hierarchy. Decide whether to indent at each level of the hierarchy and specify up to which level to show. (Limiting the number of levels, allows for grouping by additional data items.)

1. Customize grouping criteria in the Activities window, Layout bar, Group and Sort. Click the Display bar from the Projects or Resources window, Group and Sort By, Customize.
2. Select the data item to group by under the Group By column.
3. Mark the indent checkbox in the Group and Sort window to indent each level (such as EPS/WBS) and specify how many levels to show.
4. Choose an interval in the Group Interval column when grouping by numbers, dates, or durations.



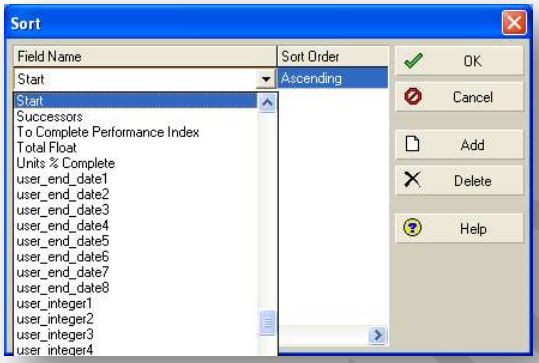
Sorting Data

Select the sequence (either alphabetically, numerically, or chronologically) in which projects, activities, or resources are listed.

1. Specify a particular sort order in the Activities window, Layout bar, Group and Sort, Sort, or in the Projects or Resources window, Display bar, Group and Sort By, Customize, Sort.

2. Choose **Add** and double-click the cell under the Field Name column to select the data item.

3. Double-click the cell under the Sort Order column to sort Ascending or Descending.



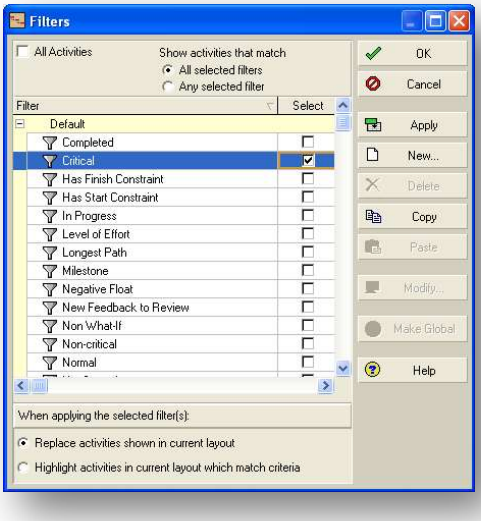
Filtering Data

Specify a set of parameters to determine which data is displayed in the current window. Create either user-defined filters available only to the user who created them or global filters available to all users.

Running a Predefined Filter

1. Filter selected activities from the currently open project using the Activities window, Layout bar, Filters or in the Projects window, Display bar, Filter By.

2. Mark the checkbox under the Select column of the filters to apply. (The predefined list of filters displayed is dependent upon the layout window currently open – Activities window OR Projects window.)



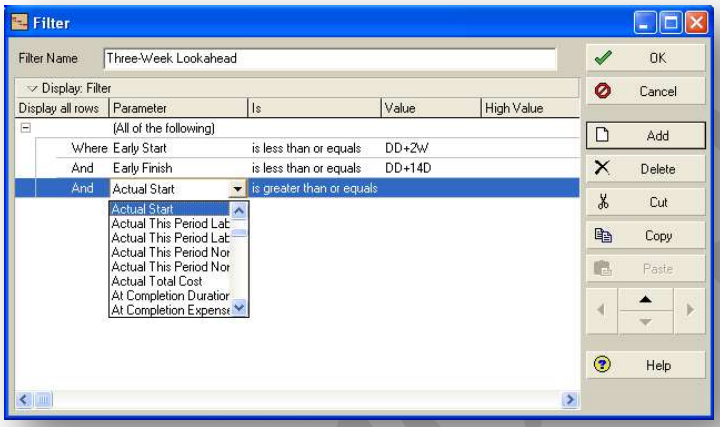
3. Check the All Activities checkbox to remove any filter and view all activities. Choose All(AND) / Any(OR) selected filters to narrow the selected data.

4. Choose whether to Replace activities shown in current layout or only Highlight (a subset of) activities in the layout.
5. View the filter before running it by selecting the Apply button.

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6. Convert a user-accessible filter to a globally accessible filter by selecting Make Global.

7. To view existing criteria for a pre-defined filter, Copy and Paste the filter and select Modify.



Creating a User-Defined Filter

1. Filter selected activities from the currently open project using the Activities window, Layout bar, Filters or in the Projects window, Display bar, Filter By, Customize.

2. Select **New** and enter the Filter Name.

3. Double-click the cell under the Parameter column to select a data item.

4. Double-click the Is cell to select filtering criteria and specify a value in the Value field.

5. Click **Add** to define multiple selection criteria and specify whether all criteria must be met or at least one criteria using And / Or statements.

6. "Nest" criteria for multiple levels of selections using the arrow keys at the right.

7. Enter the topmost parameter of All of the Following or Any of the Following to determine how each level is selected.

Exercise: Create a Filter for all Subcontractor Activities

Select View, Filter;
Select New;
In the New Filter window enter Subcontractors as the Filter Name;
Complete adding the following Parameters – Where Responsible Party Is Under Subcontractor (Subconractor can be selected when the Responsible Party list is opened.

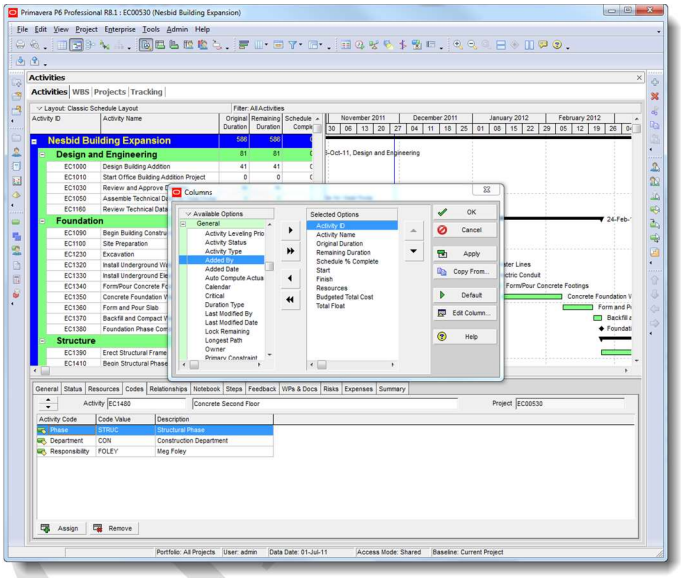
***Only* user-defined filters can be deleted by selecting the Layout bar, Filter, Delete.**

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Formatting Layouts

Customizing Columns

Modify the look and content of columns included in tables and spreadsheets in the Activities window. Select the columns to be included, modify the widths and the order in which the columns are displayed and row height. Edit fonts and colors and column titles.



1. A Column can be moved by dragging the column heading to its new location.

2. Change the sort order from ascending to descending by clicking on the column heading.

3. Modify which columns are displayed in the Activities window, Layout bar, Columns.

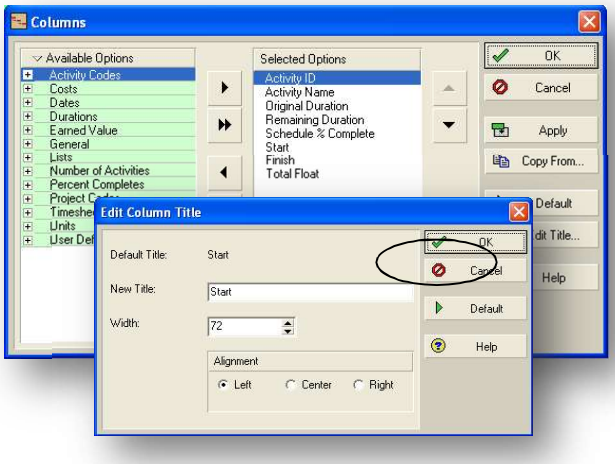
4. Use the arrows in the center to move columns from the Available Options to the Selected Options columns.

5. Specify the sequence of columns to be displayed using the arrows to the right.

Copying Column Formatting

Column formats can be copied from one layout to another by selecting the Copy From button in the Columns window and choosing the layout.

Editing Column Titles



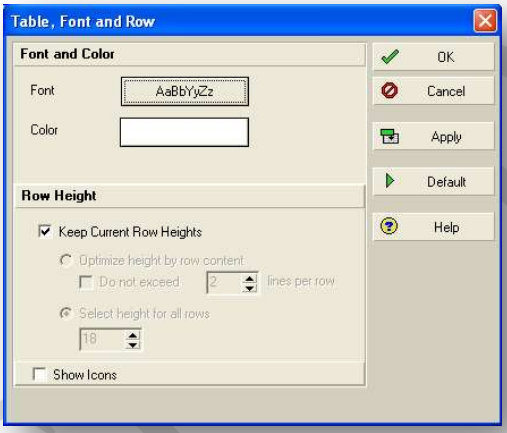
1. Select the column heading to be changed in either the Available Options or Selected Options columns.

2. Click the Edit Title button and enter the new name and specify the maximum number of characters for the column width.

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3. Select the column heading alignment.

Modifying Column Fonts, Colors, and Row Height

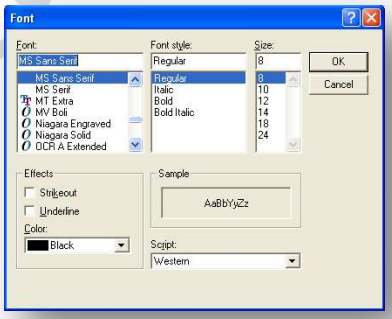


1. Specify the format changes to be applied in the Activities window, Layout bar, Table Font and Row option.

2. Click the Font button and select the new font or select the Color button to choose a new color.

3. Restore the standard table fonts and colors by selecting the Default button.

4. Specify the height for rows by either selecting:
   a. *Keep Current Row Heights option* – retain all custom row heights set in the current layout.



   b. *Clear the Keep Current Row Heights option* – automatically sizes each row based on cell content, font size, and column width OR based on a specified row height.

   c. *Choose Optimize Height* by Row Content to set a value limiting the amount of lines per row automatically adjusted during text wrapping.

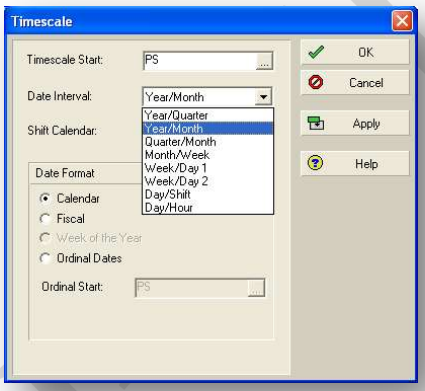


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Modifying The Timescale

The timescale is displayed in Gantt Charts, Activity and Resource Usage Spreadsheets, and Activity and Resource Usage Profiles. Display the timescale in different time units (i.e. days, weeks, months, quarters, years). Expand and condense the timescale to control the bar size and column view in the layout.

Edit the Timescale in a Profile, Spreadsheet, or Gantt Chart



1. Manually expand/condense the timescale by clicking a dragging the timescale.

2. Right-click in the Bar area, Timescale (in the Activities and Projects window) to change the timescale settings OR click the Layout bar, Timescale in the Activities window.

3. Select the start date for the timescale in the Timescale Start field and enter the date intervals at which data is displayed.

4. Choose the format to display date intervals: Calendar, Fiscal, Week of the Year, or Ordinal Dates (with a selected start date).

5. These settings apply to both the top and bottom layouts.

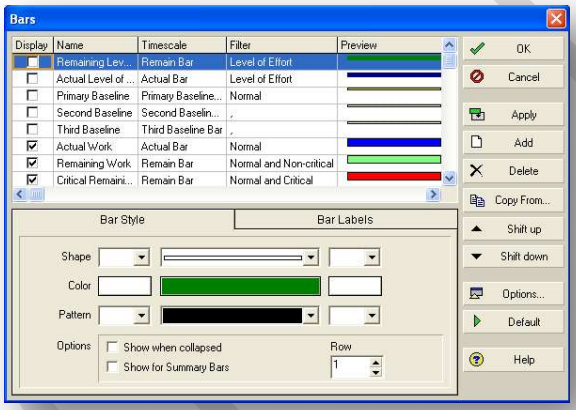
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Formatting the Gantt Chart

Changing the Bars

Format the bars within a Gantt Chart to visually distinguish different data. The Gantt Chart can be viewed in the Activities and Projects window.

Adding Bars



1. Add new bars by selecting the Layout bar in the Activities window or the Display bar in the Projects window, Bars.

2. Select the Add button and enter a name for the new bar.

3. Determine the time span the bar represents in the Timescale drop-down list:
   a. Current Bar = Start Date to Finish Date

   b. % Complete Bar = Percent Complete

   c. Plan Bar = Planned Start Date to Planned Finish Date

   d. Actual Bar = Actual Start Date to Actual Finish Date

   e. Remain Bar = Remaining Start Date to Remaining Finish Date

   f. Baseline Bar = Baseline Planned Start Date to Baseline Planned Finish Date

   g. Early Bar = Early Start Date to Early Finish Date

   h. Late Bar = Late Start Date to Late Finish Date

   i. Float Bar = Remaining Finish Date to Late Finish Date

4. Double-click the field under the Filter column to select which filter is to be applied.

5. Mark the field under the Display column to show or hide the bar.

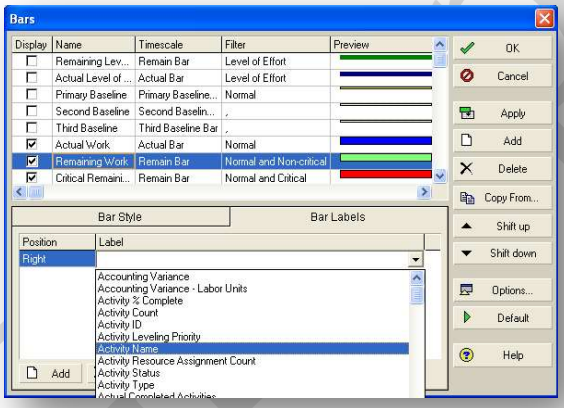
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6. Modify the bar Shape, Color, and Pattern in the Bar Style tab and determine how the bar is displayed when collapsed.

7. Specify the position of the bar in the Row field.

Adding or Changing a Bar's Label

Select the Gantt Chart bar to be changed in the Bars dialog box. Click the Bar Labels tab and select Add to create a new label. Double-click the Label field and select the label value. Modify the label's position on the bar by double-clicking in the Position column and selecting the new position. Click Delete to remove an existing highlighted label.



Copying Bar Settings

Copy the Gantt Chart bar settings from another layout by selecting the Copy From button in the Bars dialog box and specifying the layout settings to be applied.

Deleting a Bar

Delete an existing bar by selecting the bar and clicking the Delete button.

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Bar Chart Options

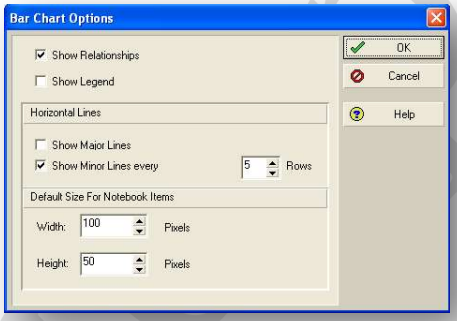
1. Show or hide relationship lines in the Gantt Chart by selecting the Options button in the Bars dialog box.

2. Check or uncheck the Show Relationships box to display or hide activity relationship ties.

3. Click the Relationship Lines button in the toolbar to hide/show relationship lines.

4. Mark the Show Legend in the Bars dialog box, Options to view the Gantt Chart legend.

5. Mark the Show Major Lines to display background lines before every Summary bar. Mark the Show Minor Sight Lines to show background horizontal lines before every specified number of rows.



Saving Layouts

Formatting the Activity Network Layouts

Modify the Activity Box Template in the Activity Network layout by selecting the Layout bar, Activity Network Options, Activity Box Template.

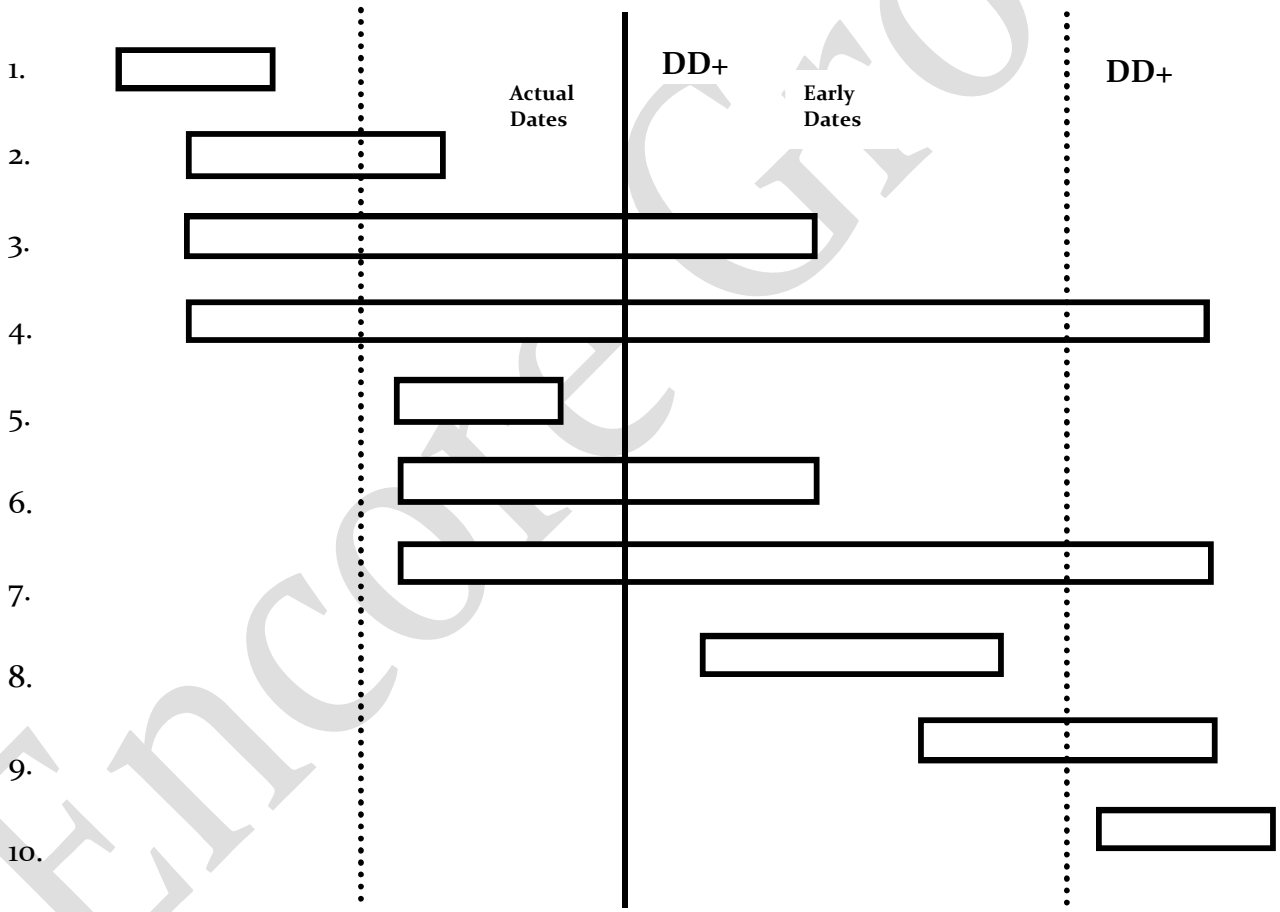
Exercise: Create the following Layouts with Filters

1. **Save the Subcontractor Layout as a Project Layout– with Subcontractor Group and Sort and Filter;**
2. **Create a Detail Bar Chart layout as a User Layout – Group and Sort by WBS, Filter "All Activities";**
3. **Create a Critical Path layout as a User Layout – Group and Sort by WBS, Filter by Critical Activities;**
4. **Create a 4 Week Lookahead as a User Layout**
   - **Group and Sort by WBS;**
   - **Create a Filter with 3 Criteria;**
     1. **Activity starts within range of DD-1W and DD+3w**
     2. **Activity Finishes within range of DD-1W and DD+3w**
     3. **Activity Status equals In Progress.**

Using Bar Chart Options, setup sight lines for week/day and stretch timeline to fit window.

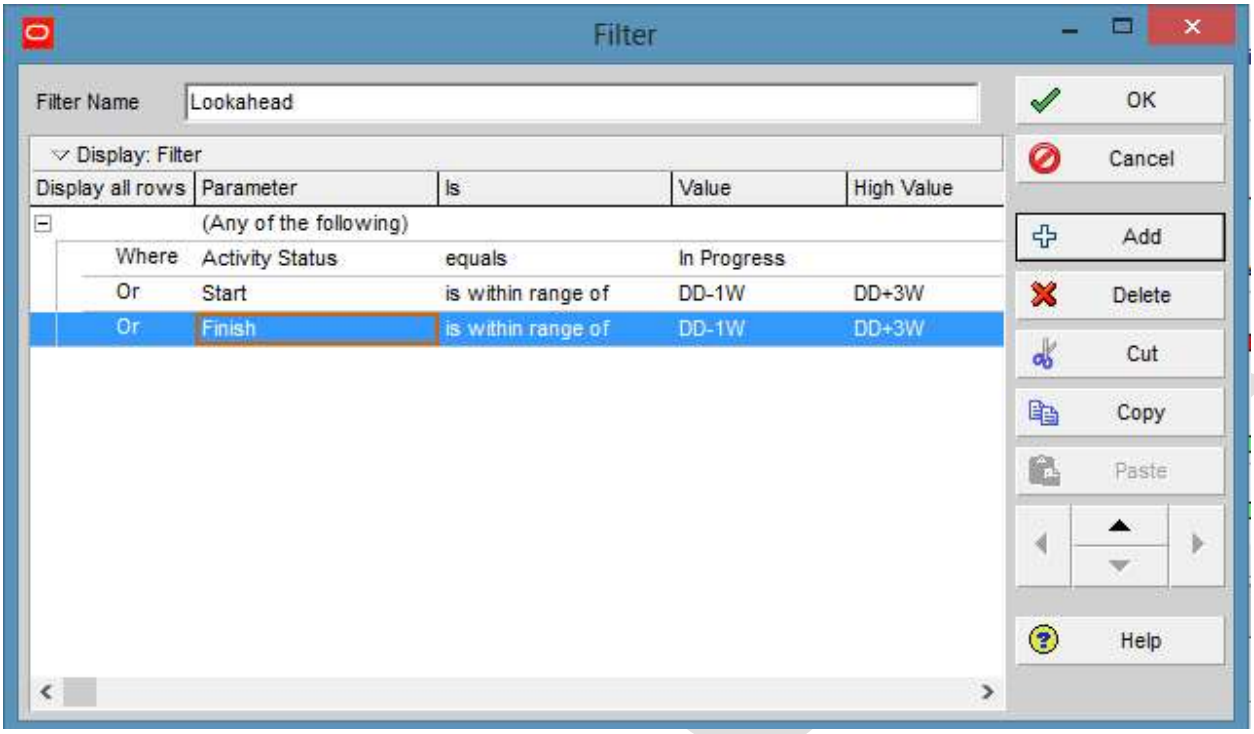
Lookaheads

Use the **Filter**, **Timescale**, and **Page Setup** windows. Using the layout on the previous page, setup a three-week lookahead for your project.



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Set up a Lookahead Filter



Hint: Select Where Early Start is within range of DD-7D and DD+21. Select Where Early Finish is within range of DD-7D and DD+21. Don't forget Percent Complete.

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Chapter 7
Updating Your Schedule

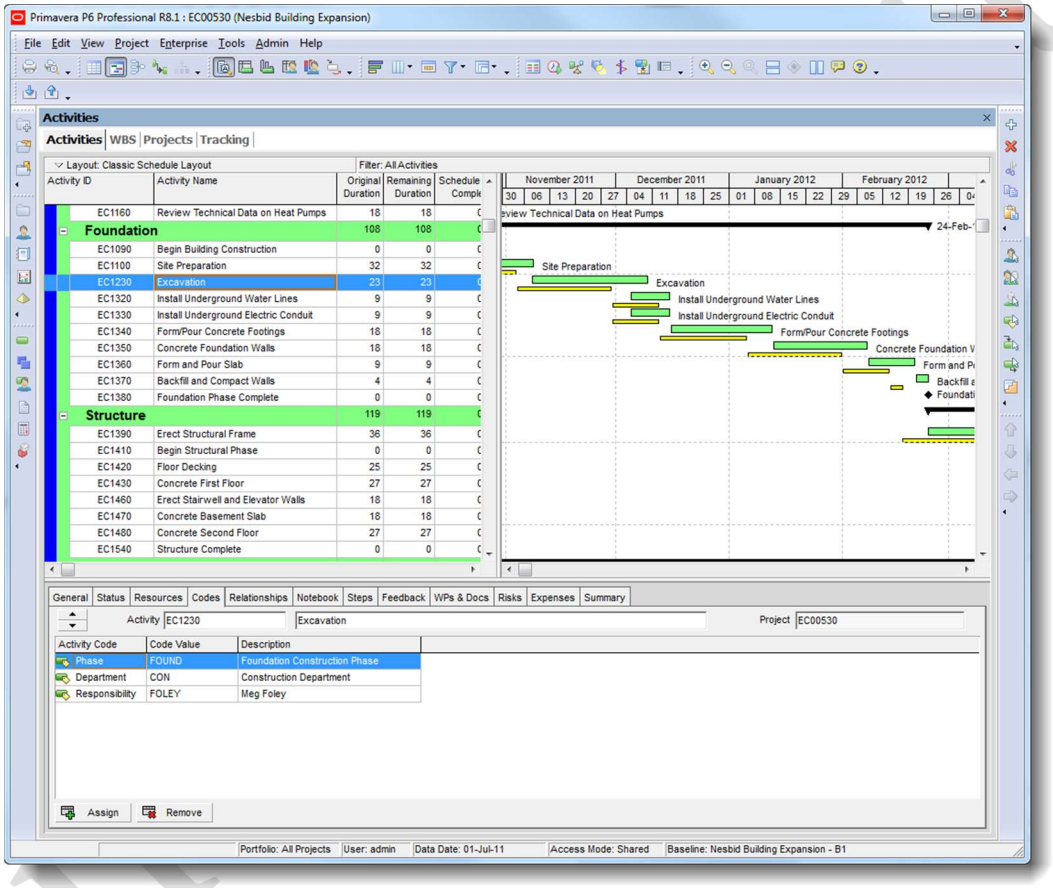
In this chapter you will learn to update and progress activities and resources in the construction schedule.



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Setting a Baseline or Target Schedule

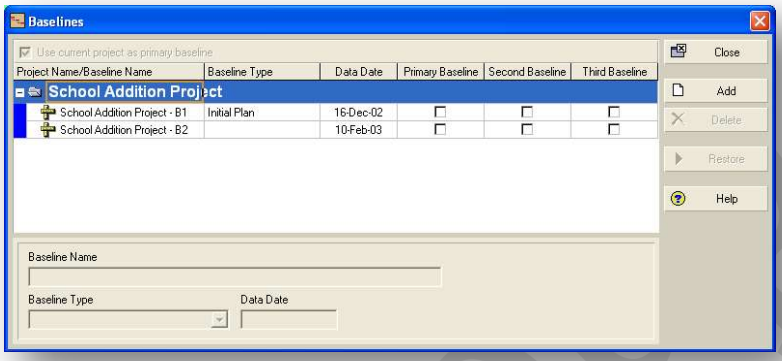
The Baseline or Target Schedule provides a point of reference on which to measure and control the underway project. Create a Baseline copy of the schedule before the first update. This will be the target against which the project's cost, schedule, and performance are tracked. Assign each baseline a category to define its purpose (i.e. initial planning baseline, what-if project plan baseline, mid project status baseline, monthly update comparison baseline). Categories are created under the Admin menu, Admin Categories, Baseline Types.



Open at least one project to access the project's baselines. Baselines cannot be accessed as a separate project. They must be unlinked from their current projects in order to be copied or modified.

Creating the project baseline

1. Open the project for which the baseline is to be created.



2. Select the Project menu, Maintain Baselines to view the open projects and any assigned baseline projects below.

3. Click **Add** to create the baseline project by saving a copy of the current project as a baseline or converting another project to a new baseline.

4. Use the Ctrl button to create baselines from multiple open projects. It's recommended to copy the project before converting it to a baseline; baselines are no longer available in the project hierarchy. Projects that are to be converted cannot be open nor can they have assigned baselines.

5. A copied project as a baseline is created with the name and data date as the original project with a "B1" appended to the name. This appendage is incremented per new baseline. The baseline can be renamed to make it unique.

6. Assign a Baseline Type to categorize baselines across multiple projects.

7. Assign the baseline to use for the current project.

8. Mark the checkbox Use under the Primary Baseline for the current project. The Primary Baseline is required before the second and third can be used. If the Primary baseline is selected, the current project is used. The Second and Third Baselines are not required.

9. Baseline assignments are user-specific, each user can choose a different version for their active baseline to the current project.

Exercise: Create a Baseline schedule for the High School Addition Project with an Initial Plan Category.

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Deleting the project baseline

1. Open the project for which the baseline is to be deleted.

2. Select the Project menu, Baselines to view the open projects and any assigned baseline project below.

3. Highlight the baseline to be deleted and click Delete.

Modifying the project baseline – update the baseline project as the project progresses and changes occur.

1. Restore a baseline project as a separate project in the project hierarchy.

2. Select the Project menu, Baselines to view the open projects and any assigned baseline project below.

3. Highlight the baseline to be restored then click Restore. The restored project is placed in the same node as the project to which it was linked.

4. Make the necessary changes (e.g. changes to original scope) to the restored baseline project then return it as a baseline to retain the changes for comparison against the current project.

Comparing the baseline to the current project – evaluate progress and performance after the current project is updated by opening a layout showing both the baseline and current bars. This type of comparison identifies tasks that start or finish later than planned and, using the Gantt Chart, indicates how the current schedule is progressing according to the original (baseline) plan. Activity bars can be customized to display target and variance data. Columns shown can be customized to include planned value, actual costs to date, and earned value to identify tasks that are behind schedule or over budget. Create activity matrix reports as well as resource and cost graphics for detailed reporting. The Earned Value report analyzes cost and schedule variance. (BL appears before any data item available from the baseline project.)

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Updating and Scheduling

Update regularly to record progress and identify potential problems. Apply actual data to activities directly or via timesheets using Progress Reporter.

Schedule the project after updating the data to calculate the early/late start and finish dates. Level resources to ensure that the resource demand does not exceed resource availability. Leveling could potentially delay each activity until sufficient resources are available.

Updating the project schedule

Update the schedule regularly and compare it to the baseline to ensure that resources are being used effectively, project costs are being monitored against the original budget, that there is a knowledge of actual durations and costs in order to initiate contingency plans if necessary.

Ask the following questions when updating:

- What information needs to be collected to update the project and what method(s) will be used to collect information?
- Who will gather the necessary information used for the project update?
- How often should the project be updated?
- Are resources local or offsite?
- Who will need to see the results of the update and when are these results needed?
- What types of information will be generated after each update in order to communicate progress before the next update period?

Identify the type of information to collect – depends on whether activities or individual resource assignments are being updated.

Update activities by assigning actual dates and adjusting remaining duration.

Update resource assignments by entering actual hours to date and the hours remaining.

What method(s) will be used to collect information – will data be entered through timesheet entries via Progress Reporter OR will data be imported from other systems (i.e. accounting systems) OR will updates be handwritten on printouts of the schedule and entered in the project management application.

How often should the project be updated – depending on the changes that occur in the project, updates might occur monthly, weekly, or even daily. Recommendation: If

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the project never seems to be accurate, it is not being updated often enough or the scope of activities is too broad – divide the activities into smaller ones.

Analyze and communicate data – examine updated project schedules using display and print options: onscreen layouts for immediate analyses, reports for more detail, baseline/target comparison to pinpoint potential problems, resource usage profile to track resource use. Focus on critical activities, resource and cost overloads, and slippages, identifying actual and required future progress.

Methods for updating

1. Update progress for all activities and resources as a whole or
2. Update activities and resources individually
3. Update progress from timesheets
4. Use a combination of the above methods

Projects rarely progress as planned –

1. Activities start out-of-sequence, take more or less time to complete than originally planned, or resource use exceeds planned use. In these cases it is usually recommended to update activities and resources individually in order to forecast the effects of unforeseen progress or the lack of progress and to take the appropriate corrective action necessary.
2. Estimation of progress is required to "auto compute actuals" by specifying the data date and applying actual data. The data date (the date up to which progress is reported) is set to the project start date before the first update. This data date is used to determine which activities have progress and how much. It is also used to calculate the remaining duration of activities that have started. Activities that have been completed have a remaining duration of zero and % complete = 100.
3. Some activities occur as planned and some do not. Allow for automatic calculation of the project schedule as if the project were progressing exactly as planned while individually updating those activities and resources that have deviated from the plan.

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The update process should include:

1. an established standard update procedure to include a method used to record progress while setting calculation variables for percent complete type and duration type.

2. a baseline plan.

3. recorded progress of activities entered automatically or manually to include:
    a. actual start/finish dates
    b. actual resource use/cost to date

    c. estimated remaining work to complete

    d. if applicable, approve timesheet from Progress Reporter

4. applied project actuals.

5. calculating the schedule and leveling resources.

6. comparing the current and baseline schedules to identify variances.

7. analyses of data through layouts and reports.

8. making necessary adjustments and communicate the schedule updates.

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Updating Activities Manually

Record actual dates, actual resource use, and costs incurred up to the data date based upon activity types, percent complete types, and duration types.

Set percent complete type – to calculate the activity's percent complete according to the activity's duration, activity units, or a physical percent complete for each activity.

1. Select the Project menu, Activities while displaying Activity Details in the lower layout.

2. Select the percent complete type in the Activities General tab or by adding a column for Percent Complete Type in the Activity Table.
    - Select Duration when activity progress can be easily reported in terms of actual calendar days of work remaining.

Duration % Complete = ([Original(or Planned) Duration – Rem Duration] / Original(or Planned) Duration) x 100

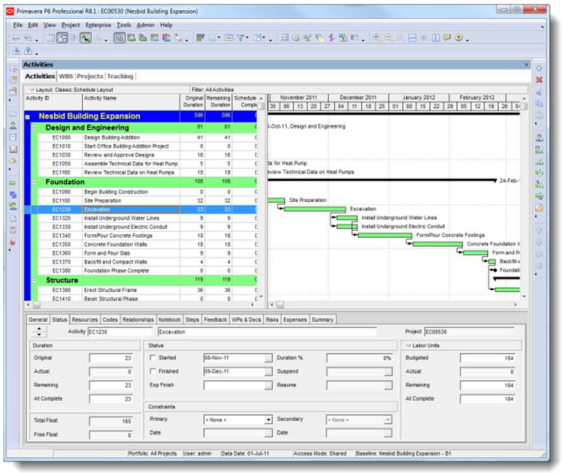
- Select Physical Percent Complete when activity progress can most easily be reported based on personal judgment. Enter the activity percent complete.
- Select Units when progress is best reported based on the work effort that has been accomplished and how much effort remains. Enter the actual and remaining units.
  Units % Complete = (Actual Units / At Completion Units) x 100

Update actual dates by setting the activity's start and finish date as well as other status information.

1. Select the Project menu, Activities while displaying Activity Details in the lower layout.

2. Select Status tab, mark the Started checkbox and specify the actual start date.

3. Mark the Finished checkbox (if applicable) and specify the actual finish date.



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Four Steps to Update Manually

1. Set Automatic Schedule OFF

2. Status: Enter Actual Dates
   Remaining Duration
   Logic Changes

3. Schedule – **F9**
4. Move the **Data Date** to the **Current Data Date**.

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Updating Activities Using Update Progress

Turn Auto Schedule ON

Step 1 – Change Durations and Sequence (Logical Relationships) so that the schedule matches the current progress.

Step 2 – Run Global Change to make Planned Dates = Early Dates.

Step 3 – Select Tools, Update Progress to assign a new data date.

What happens?

2. All data before the data date become actual dates.
3. All projections are maintained.

What are the advantages?

1. Logic is as-built
2. What-if-scenarios can be run prior to changing data date.

Making Global Changes to Activities

Edit existing values, assign new values, and/or delete groups of activities using the Global Change feature under the Tools menu, Global Change. This feature must be used by opening the project with Exclusive access. (Select Exclusive in the Access Mode in the Open Project dialog box before opening the project.)

Understanding If, Then, Else Statements

1. *If* statements control which project data changes.
2. *Then* and *Else* statements control what changes are to be made. (At least one *Then* statement is required.) (An *Else* statement requires and *If* statement.)
3. *Specify* whether Any or All conditions must be met.
4. View the last specifications set when opening a Global Change (Tools, Global Change).

When to Add Activities

When should you add an activity? How long should an activity duration continue to increase?

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Check the Float

Set up a layout to sort by **Total Float** / **Early Start** to view the effect the update had on your schedule's critical path.

Key Questions for Updating

- **Is the project progressing according to plan?**

Response - target comparison

- **Which activities have slipped?**

Response - A variance report or manual analysis

- **What is the status of the critical activities?**

Response - total float report

- **Which activities are critical or near critical?**

Response - Lookahead schedule

- **What is scheduled to begin in the next month?**

Response - Lookahead schedule

- **When will the project be finished?**

Response - Summary Bar chart layout. The following example shows the current projects in a financial institution; each phase within each project is summarized. Choose **Format**, **Summarization** and summarize data within each group. From either an onscreen or printed display, you can report when work started and when it will finish, how much progress has been made to date, and how much everything will cost.

- **When will each milestone be achieved?**

Utilizing the same process as the above example you can summarize by milestones.

- **What procurement activities are the most critical?**

With procurement codes the schedule can be organized to show the most critical submittals.

Applying Actuals

Applying actuals by moving the data date schedules activities with progress. Activities within the specified timescale (between the previous data date and the new data date) are scheduled, and progress is calculated for those activities that are set to automatically calculate actuals.

Apply actuals under the Tools menu, Apply Actuals by setting the same data date for all projects or using a different data date for each project.

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Scheduling the Project

The Critical Path Method scheduling technique is used to calculate the project schedule. Durations and relationships between activities are used in calculating project dates. Two passes are run through the activities in the project. One project or all projects within an EPS node can be scheduled.

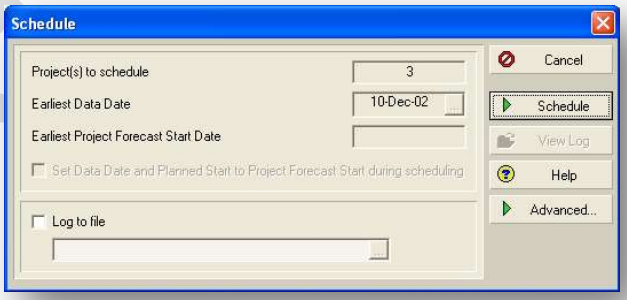
First pass or "forward pass" calculates early dates for each activity based on the start/finish dates of predecessor activities and the duration of the activity itself.

Second pass or "backward pass" calculates late dates for each activity based on the start/finish dates of successor activities and the duration of the activity itself.

Free float and total float for each activity are then recalculated.

Schedule the project by selecting the Tools menu, Schedule to view the number of projects to be scheduled as well as the current data date of all open projects. These data dates can be changed for specific projects by applying actuals (Tools, Apply Actuals). Modify/view the default scheduling settings under the Tools menu, Schedule, Advanced.

A forecast start date is a manually changed start date of the project by dragging the project bar to a new timeframe in Portfolio Analyst or Tracking layouts. Choose this new "start" date and current data date to schedule the project by marking the Set Data Date and Planned Start to Project Forecast Start during Scheduling option. If more than one project is open the Earliest Data Date is shown. Use the Log to File option to record scheduling results.



Automatic scheduling and leveling calculates the schedule each time a significant change is made to an activity, relationship, or resource. Mark the Schedule Automatically When a Change Affects Dates checkbox in the Advanced Schedule Options dialog box. Mark the Level Resources During Scheduling checkbox to level resource during automatic scheduling.

Must finish by date under the Project Details, Dates tab calculates the schedule based upon the must finish by date rather than the schedule end date during the backward pass.

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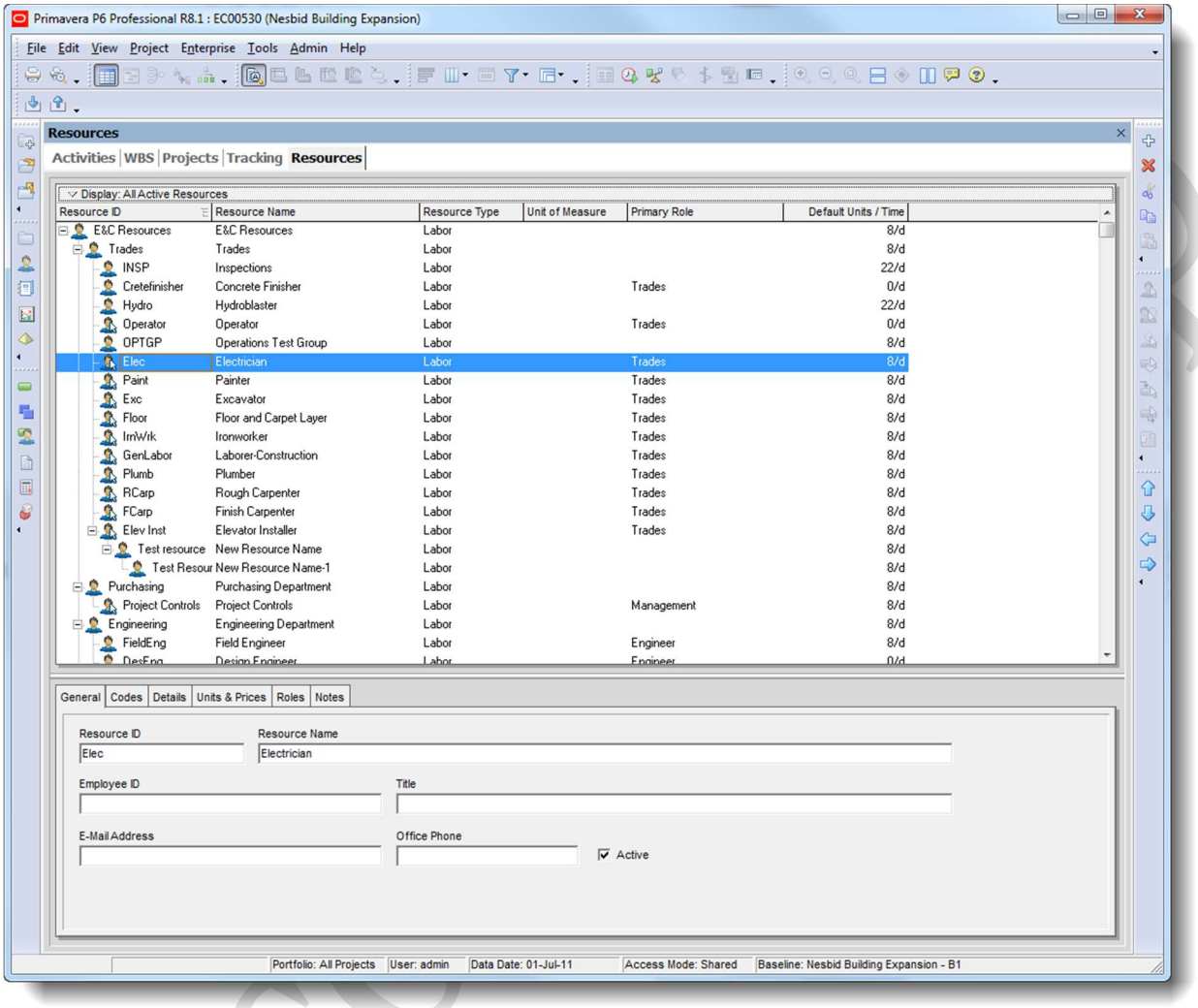
Chapter 8
Resources and Roles

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Resources



Resources are the personnel performing work across all projects and/or equipment used in work performance. The resource hierarchy reflects the organization's resource structure and the ability to assign multiple resources to activities. Establish the basic resource structure defining the resources necessary to complete projects within the enterprise. Set each activity's *primary resource* to coordinate and update the activity's status, and manage the activity's nonlabor resources (i.e. hours/units). Each resource will have availability limits, unit prices, and an assigned calendar determining work periods and downtime or periods of inactivity.

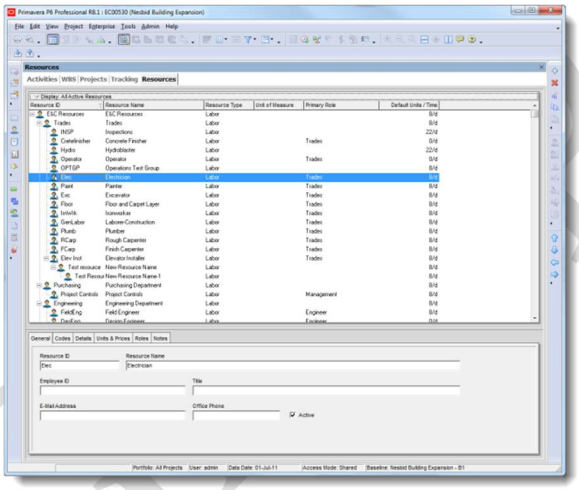
Reviewing Existing Resources

5. Choose the Enterprise menu, Resources to display the Resource Breakdown Structure (RBS).
   COOL TOOL: View resources in their hierarchy levels or click on a column heading to sort ascending or descending order or return to the hierarchy structure.

6. Select the Display bar, Details to view detailed information about the highlighted resource.

7. Resources can be viewed as a chart by choosing Chart View under the Display bar.

8. Customize the columns to be shown using the Columns, Customize feature under the Display bar.



9. Filter resources to be displayed by selecting the Display bar, Filter By, All Active Resources, All Resources, Current Project's Resources.
10. Group and sort resources using either the predefined groupings or a customized grouping under the Display bar, Group and Sort By.

11. Using the arrow keys move the resource to its correct location or placement within the RBS hierarchy. Indent (►) for a more detailed position or outdent (◄) for a more general position.

Creating Resources

1. Select the Enterprise menu, Resources to structure resource hierarchy (RBS). **COOL TOOL**: The resource window can be opened at the global level without projects open. Select Enterprise, Resources OR choose Resources on the Home workspace.

2. Add root elements to serve as the lead individuals.

3. Highlight an existing resource at the same hierarchy level of the resource to be created and select the Add button to activate the New Resource wizard.

4. If the wizard is not used select the Display bar to view resource Details and enter the necessary information for the newly created resource.

Viewing Resource Details

Add resource information, view or edit the selected resource in the Resource Details tab.

General tab – enter the Resource ID and Name.

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1. Employee ID and Title – the identifier corresponding to the resource/employee and their title within the enterprise (e.g. social security number, job description).

2. E-mail Address – should correspond to the e-mail address on the Contact tab of the Users dialog box in Primavera's Progress Reporter application.

3. Active – check if the resource/employee is available for assignment or uncheck if unavailable or inactive.

Codes tab – establishes resource code configuration by setting codes and values (e.g. Classification Level – Management, Engineering) to organize resources for grouping, filtering, and sorting.

1. Resource Code – view and/or assign resource codes

2. Code Value – view values for assigned codes

3. Code Description – view the description for each code

Details tab – select the resource's labor classification, currency and overtime options, and resource profile settings.

1. Labor Classification – Labor for personnel and Nonlabor for equipment/materials

2. Currency and Overtime
   a. Currency – set the currency associated with the resource. (The default currency is displayed as defined under the Edit menu, User Preferences, Currency.)

   b. Overtime Allowed – check the option to allow the resource to log overtime hours in the Primavera Progress Reporter application.

   c. Overtime Factor – a multiplication factor, 1 to 10, to calculate the resource's overtime price (overtime price = standard price x overtime factor).

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3. Profile
   a. Calendar – select a global or resource calendar for the resource for resource leveling and tracking.

   b. Default Units / Time – units / time amount the resource is available to work on an activity. Enter a percentage (100% = available to work full-time) in the Default Units / Time field or a numeric value/time duration (e.g. 8 hours per day = 8.00h/d). Five individuals as a crew or department might be entered as 500% or 40.00h/d. This calculation is used along with the assigned calendar to calculate the resource allocation/distribution when scheduling and leveling.

   c. Auto Compute Actuals – check the option to calculate the resource's actual quantity according to the project plan; uncheck if using Primavera's Progress Reporter to report hours and update actuals.

   d. Calculate costs from units – mark to indicate that any new assignments for this resource will have its costs recalculated whenever quantity changes occur. Resources with this option appear with a checkmark when adding the customized column "Calculate Costs From Units". Recalculation is set under the Resources tab of Project Details and Activity Details.

Units & Prices tab – specify available resource quantity limits to identify over allocation.

Limits delay the start of the resource in the project schedule until it becomes available using the effective dates and max units/time.

1. Shift Calendar – display the shift calendar assigned to the resource.

2. Shift – enter the shift number to apply limits when the shift calendar has more than one shift.

3. Effective Date – double-click to enter the effective start date to set varying limits and prices over time.

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4. Max Units / Time – double-click to set the maximum units per time available for the resource during each work period (hour, day, week, or month). Enter a percentage or a numeric value / time duration (e.g. 8 hours per day = 8.00h/d OR 100% = available to work full-time).

In order to level resources, availability limits must be set using Max Units/ Time.

5. Price / Unit – double-click to set the resource's price / unit associated with the price.
6. Roles – view role information for the resource.
7. Role ID – assign / view identification code for the resource's role.
8. Role Name – view the name of the assigned role.
9. Proficiency – view the skill level or proficiency level for each assigned role.
10. Primary Role – check to indicate whether this is the resource's main role or uncheck if only a secondary role.

Notes tab – enter comments concerning the resource, including pictures and pasting information from other documents.

Exercise: Add Resources to Activities

- In the Resources Tab, add a resource for Crane
  - With ID and Name (ID may be an equipment number)
  - Select Non-Labor
  - Default Units/Time and Max Units/Time – "1/D"
- In the Activities tab, select an activity, select the Resources tab in the bottom layout, and open the assign window.
- Assign the Crane resource to the structural steel, roofing, and exterior framing activities.

HINT: Once the resource is assigned, select View, Show on Bottom, Resource Profile for a view of the Crane usage.

Adding Resource Codes

1. Select the Enterprise menu, Resource Codes.

2. Select Modify to create resource code definitions or broad categories of resources and enter the maximum number of characters for each resource value under the code.

3. Add specific codes and their values to create a coding hierarchy.

4. Using the arrow keys move the code or value to its correct location or placement within the coding hierarchy. Indent (►) for a more detailed position or outdent (◄) for a more general position.

5. Assign the code values under the Codes tab in the Details window under Resources.

6. Use the Resource Usage Spreadsheet to view resources grouped or sorted. Right-click in the Resource Usage Spreadsheet layout, Group and Sort By, Customize to modify grouping and sorting options.

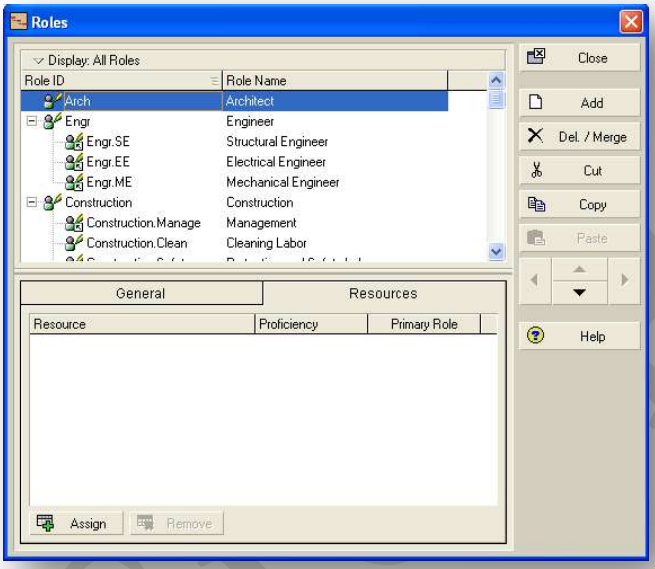
Customize the columns in the Resource window to add Resource Codes / Values.

Grouping Resources

Group resources into categories for greater control and organization using resource codes and values (e.g. Classification Level – Management, Engineering). Organize resources for grouping, filtering, and sorting.

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Resource and Roles



Assign resources in the form of *roles* before the project begins to represent a type of resource with skills or a level of proficiency (i.e. job titles, work categories). An activity's skill requirements are determined by the role assigned to it.

Viewing Roles

1. Open the Enterprise menu, Roles.
2. Select the Display bar, Filter By, to determine which roles are shown.
3. Look at roles for specific resources by selecting the Enterprise menu, Resources, and with the Details showing select the Roles tab.
4. Assign roles to resources using either the Details options, Roles tab in the Resources window and mark the applicable settings OR in the Roles window, Resource tab.

Creating Roles

1. Open the Enterprise menu, Roles.
2. Highlight an existing role at the same hierarchy of the role to be created and select Add.
3. Enter the new role's ID and Name.
4. Using the arrow keys move the role to its correct location or placement within the role hierarchy. Indent (►) for a more detailed position or outdent (◄) for a more general position.
5. Enter any necessary information about the role including a description, its responsibilities and include any applicable pictures and copied documents.

Leveling Resources

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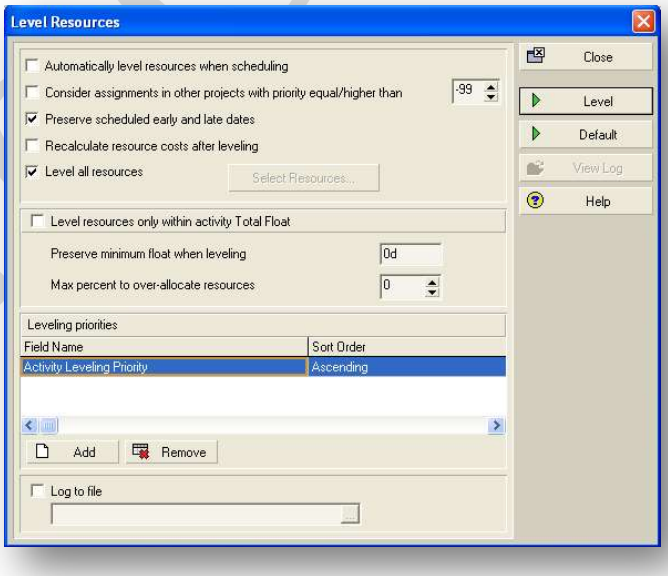
Ensure that resource demands do not exceed availability by leveling resources. Resources required for all scheduled activities are compared to the maximum resource quantities available at the time of leveling. An activity's start date is delayed if too few resources are available at any time during the activity's duration.

Define the resource's Max Units/Time in the Units & Prices tab in the Resource Details window to determine the maximum amount of work that the resource is able to perform for a given time period.

Expenses are not including when resources are leveled.

Select those resources to be leveled and add leveling priorities to specify which project or activity is leveled first should a conflict occur.

Process of leveling resources begins by opening the project(s) containing the resources to be leveled. Choose the Tools menu, Level Resources and set the resource leveling options to meet specified requirements. Select the Level button to complete the process.



Mark the Automatically level resource when scheduling to automatically level resources each time the project is scheduled.

Mark the Level all resources option to level all resources or clear the option to use the Select Resources option, and choose which resources are to be leveled.

Mark the Level resources only within activity Total Float to delay activities with resource conflicts only up to their late finish dates. Enter the minimum amount of total float in the Preserve minimum float when leveling. Specify the maximum percentage by which the resource availability can be increased in the Max percent to over-allocate resources field.

Use the leveling priority number assigned to each project when added (Project Details, General tab) to determine how resource assignments are considered in comparison to

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other projects. Enter the priority number to be considered when leveling in the Consider assignments in other projects with priority equal/higher than field.

Prioritizing activities during leveling specifies which project or activity is leveled first to resolve resource conflicts during scheduling. Add a leveling priority in the Leveling Priorities section and select the type of priority. Specify the order in which the item in the priority will be leveled. Enter priorities in the order in which the scheduling process should consider them. Double-click the column information to edit the data or highlight a priority and select Remove to take out the priority level.

Select the Preserve scheduled early and late dates to retain the project's current early dates before leveling. The schedule is *forward leveled only*, the early dates of activities for the start to the finish of the project are scheduled. Clear this box to also perform a backward leveling, moving scheduled activities up to their late dates (so as not to delay the project). This feature starts from the project's late finish and works backward towards the beginning of the project. If resources are insufficient for an activity on its late dates, that activity is then advanced to an earlier date. This feature will then update the project's early/late start/finish dates by leveling forward and backward.

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Chapter 9
Reporting

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Creating Reports with Layouts

Most project reporting is accomplished using the Layouts and customizing the look using the saved Layout features and page layout features (also saved in Layouts) from the Print Preview Screen.

Exercise: Setup the printing features for each of the Layouts previously created.

- From each Layout select the Print Preview Screen and open the Page Layout window:
- From the Page tab, select Portrait or Landscape; Select Fit Timescale to 1 page wide;
- From the Footer Tab select an image file for the footer section;
- From the Options tab, select the appropriate Timescale Start and Finish, select which items to show on print;

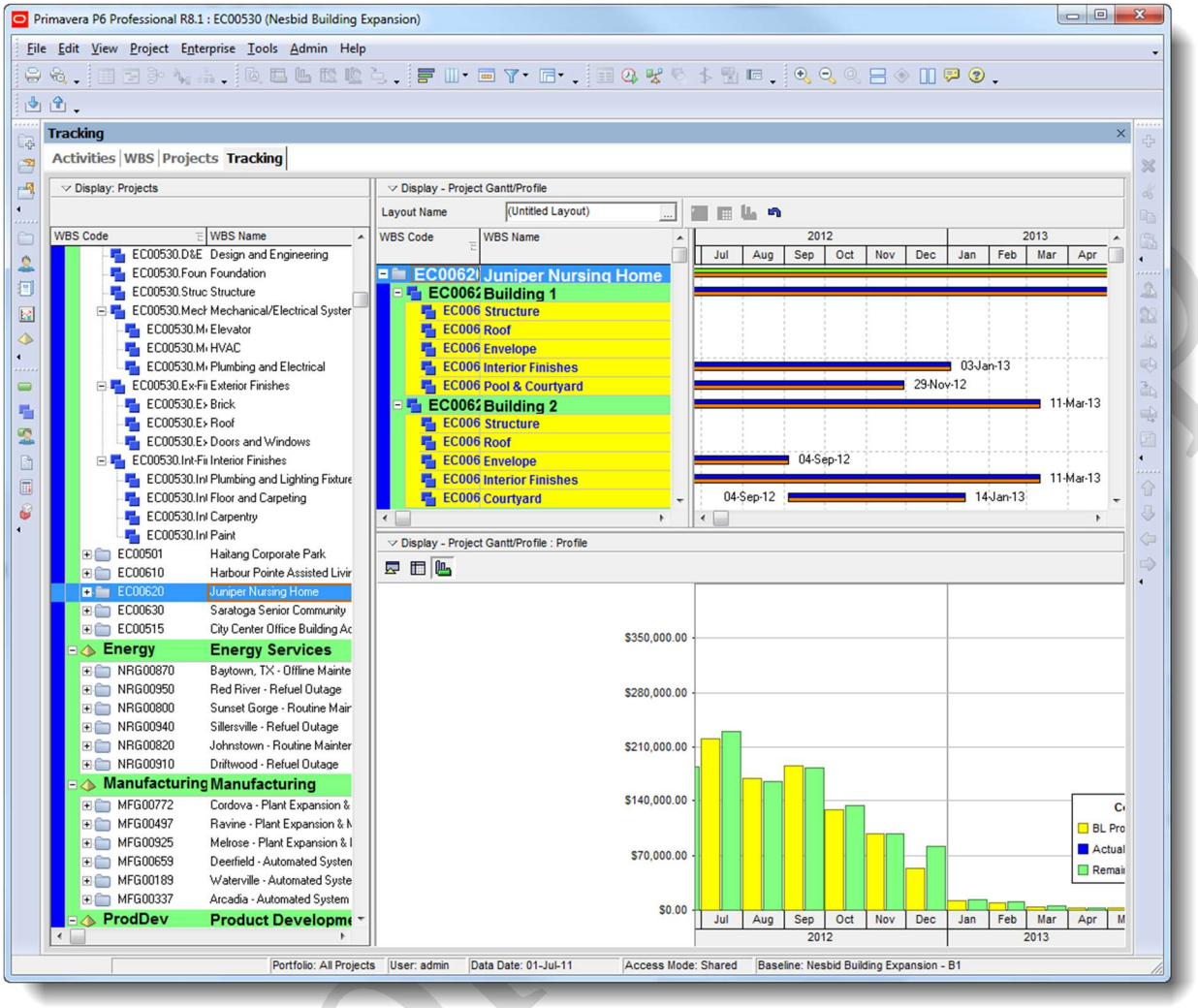
HINT: A Resource Profile Print Layout can be saved by select only the Profile option with the resource profile open in the bottom layout.

Tracking Layouts

Access, display, and manipulate summarized or live project data through various formats to perform schedule, cost, and resource analyses.

Tracking layouts view the project at a more comprehensive level to the more detailed levels of the EPS, project, WBS, OBS, phase, etc. Modify the format and level of information shown by grouping and filtering data.

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Assign tracking layouts to thresholds and issues to monitor a project's problems.

Two or more panes make up the tracking layout window. The upper left pane, the Project Explorer window displays the EPS with information about the available projects. The upper right pane, Top Layout window, displays the current layout with options. These panes can be split horizontally displaying additional panes including the Resource Explorer window, lower left, and the Bottom Layout window, lower right.

Summarized data is displayed when selecting closed projects in the Project Explorer window. Live data is displayed when selecting open projects in the Project Explorer window. Choose one of the project options in the Edit menu, User Preferences, Resource analysis tab to display information for closed projects.

Review additional information about a project by right-clicking a project in the Project Explorer and opening Project Details.

Four types of tracking layouts can be created:

Project Tables – display data in a table format (Top Layout window only)

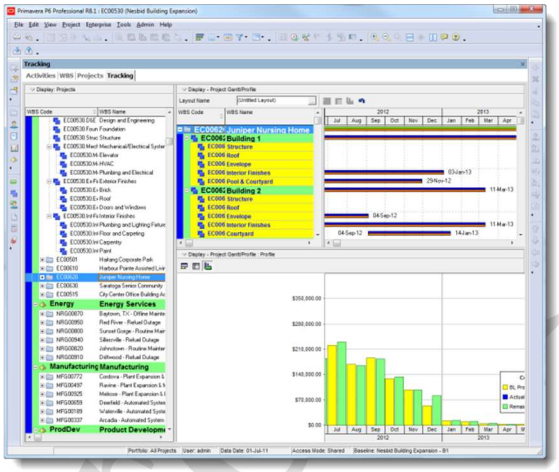
Project Bar Charts – display data in a horizontal Bar Chart format (Top Layout window only)

Project Gantt/Profiles – display data in columns and Gantt Chart form (Top Layout window only) and time-distributed data in either spreadsheet or profile format (in the Bottom Layout window)

Resource Analysis – display resource/project usage data in columns and Gantt Chart format (Top Layout window) and time-distributed total resource allocation information in either spreadsheet or profile format (in the Bottom Layout window)

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Viewing Tracking Layouts



1. Open a tracking layout using the Enterprise menu, Tracking and select the project or EPS node for which the layout will be used.
2. Click the Browse button in the Layout Name field in the Top Layout window to select the layout to be opened OR use the Display bar in the Top Layout window, Layout, Open.

Creating tracking layouts

1. Select the Enterprise menu, Tracking to create tracking layouts.
2. Highlight the project or EPS node for which the layout is to be created and click the Display bar in the Top Layout window, Layout, New.
3. Enter the Layout Name, and who has access to the layout (either the Current User or All Users).
4. Select one of the four Display Types and customize it.

Copying / Deleting Tracking Layouts

1. Create a copy of the presentation or formatting options of a layout by selecting the Display bar in the Top Layout window, Layout, Save As.
2. Enter the new name of the layout and whether access to the layout has changed to All Users or Another User.
3. Layouts can be exported to a file and then e-mailed to other users OR published to a Web site.

Delete the layout in using the Delete button in the Open Layout window.

Importing / Exporting Layouts

Layouts can be exported by selecting Export in the Open Layout window and specifying the drive and folder to which the layout is to be exported. The layout is saved in a .PLF format. Or import the layout using the Import button in the Open Layout window, browsing to the correct *.PLF layout.

Customizing Tracking Layouts

Customize the **Project table** by modifying the columns viewed (Top Layout window, Display bar, Top Layout Options, Columns).

Customize the **Project Bar Chart** by modifying the bars displayed (Top Layout window, Display bar, Top Layout Options, Edit Bars). Up to three bars can be used to display various project data.

Customize the **Project Gantt/Profile** by modifying the columns viewed in the Project Gantt Chart (Top Layout window) as well as the data/timescale appearing the in the profile (Bottom Layout window). In the Top Layout window, select the Display bar, Top Layout options, Columns for Top Layout options. In the Bottom Layout, select the Display bar, Bottom Layout Options, Profile Settings or Timescale Settings. Modify the timescale's time period using the mouse pointer as a hand to drag the timescale to show the desired period.

Additional Customizations

Customize a tracking layout by customizing columns and bars in the Top Layout window, and the data/timescale in the profile (Bottom Layout window).

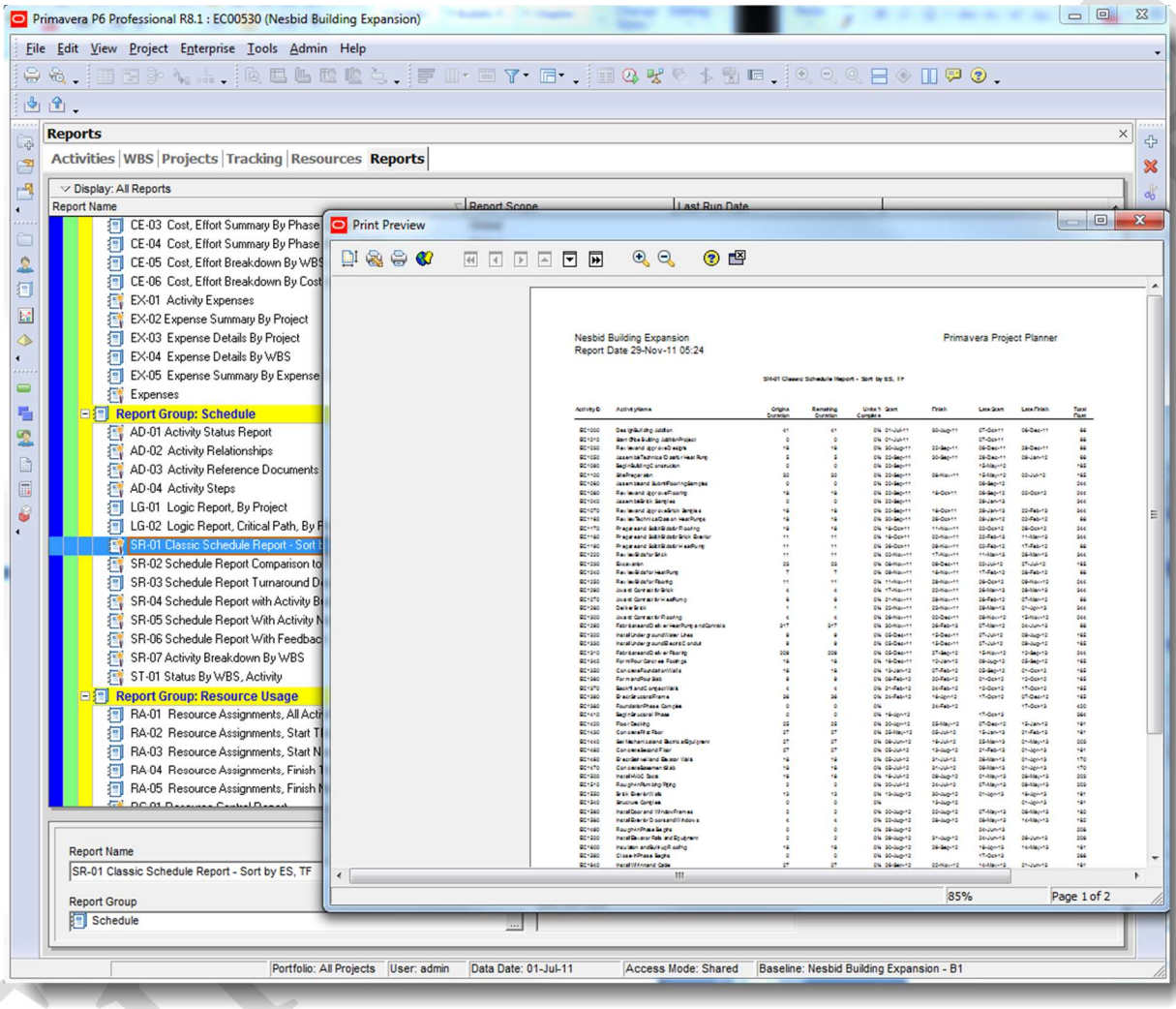
Group data into bands through using common attributes (i.e. hierarchy levels, code values, or resources) by clicking the Display bar in the Top Layout window, Group and Sort By. Select predefined grouping criteria or choose Customize the grouping. Click Sort to arrange the sort order for the grouping. Click Apply to preview the settings and OK to save them.

Display only desired information by creating a formula to restrict the data shown under the Display bar in the Project Explorer window, Filters. Add a new filter and select the Parameter field and applicable values. Double-click the Is cell to select filter criteria and enter the value. Add multiple rows as necessary. Use the "Any of the following" to equate to OR and "All of the following" to equate to AND. Apply the filter to preview changes and OK to save the changes.

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Customizing Reports

Monitor and communicate progress through reporting. Use the existing library of standard reports or create new reports/edit existing ones using the Report Wizard or Report Editor.



Access the Reports window to create, edit, run, and delete global and project reports under the Tools menu, Reports, Reports. Export and import reports to and from other installations of the project management application. The symbol to the left of the reports indicates how the report was created (either through the Report Wizard or Report Editor). Each report can belong to a specific Report Group (Tools menu, Reports, Report Groups). Print a single report or select Run Batch to print multiple reports.

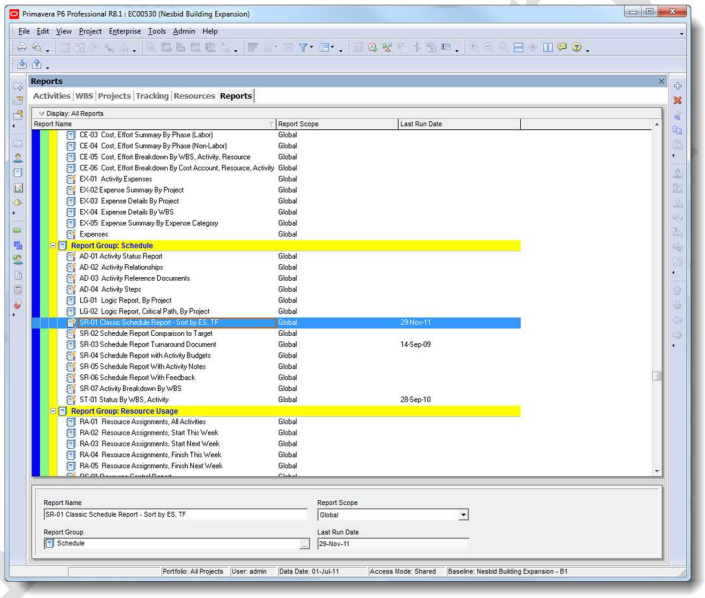
Click the Modify button to access the Report Editor to build reports.

Creating Reports

Click the Wizard button to create ad hoc reports. Use the Wizard to modify reports created by the wizard. To add a report within a particular report group, select the report group first then activate the wizard.

Creating Report Groups

Organize global and project reports using hierarchical report groups.



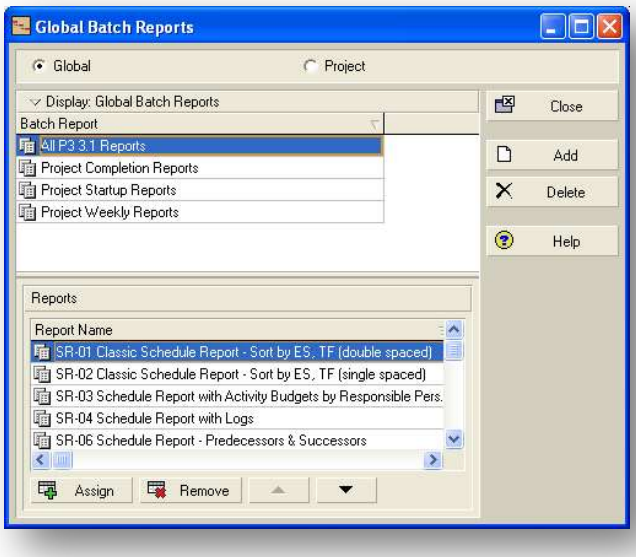
1. Create new groups under the Tools menu, Reports, Report Groups.

2. Enter the name of the new report group.

3. Report groups can have multiple levels by using the arrows to the right to move the group to its respective hierarchy level.

Batch Reports

Create batch groups to run a series of reports at one time under the Tools menu, Reports, Batch Reports. Select Global or Project and enter the name of the new group. Assign reports to this group by clicking the Assign button. One report can be assigned to one batch group at a time. Run a batch report group from the Reports window by clicking Run Batch and selecting the batch to be printed.

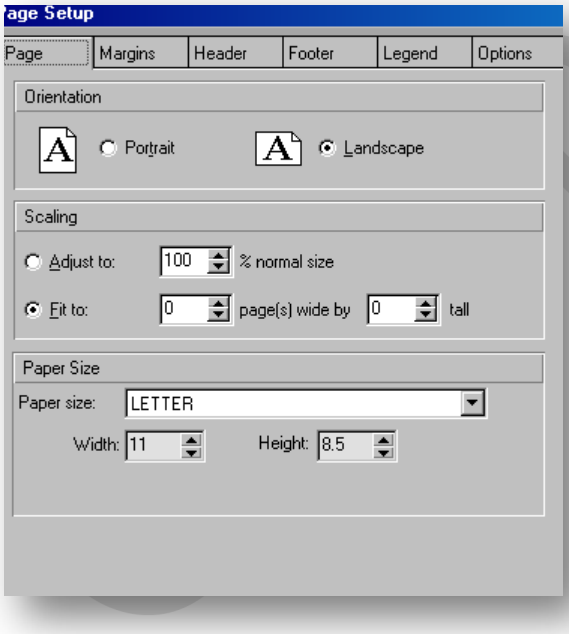


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Page Settings

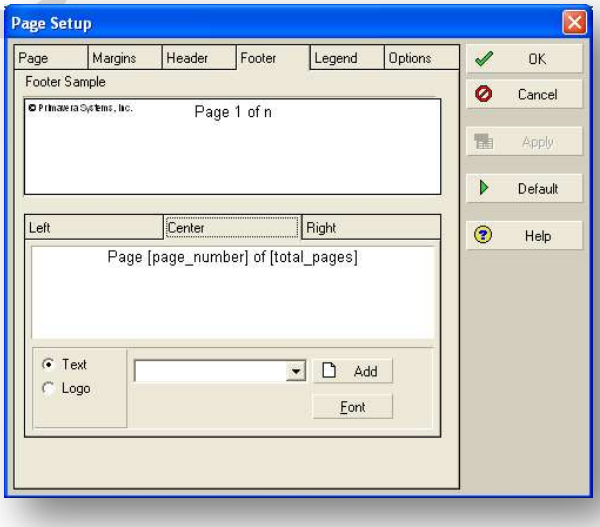
Define page settings (page orientation, margins, and header/footer settings) by selecting File, (Print Preview), Page Setup, Page. Select the desired Orientation, scaling, and paper size options.

Set the values for each margin under the Margins tab.



Headers and Footers

Add headers and footers using a standard header/footer (to be displayed on every page) under the Header/Footer tabs. Select Left, Center, and Right to defined values to be inserted in their respective positions within the header/footer. Insert graphics such as logos by clicking the Logo option and browsing to the file. Add additional text by typing directly in the space. Insert global variables such as project name, project start, data date, etc. Modify the font used.



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Define a legend to be included in the printed layout and specify how often it should be printed under the Legend tab. Add a logo to the legend by selecting Logo and browsing to the file. The Legend tab is not available when printing reports.

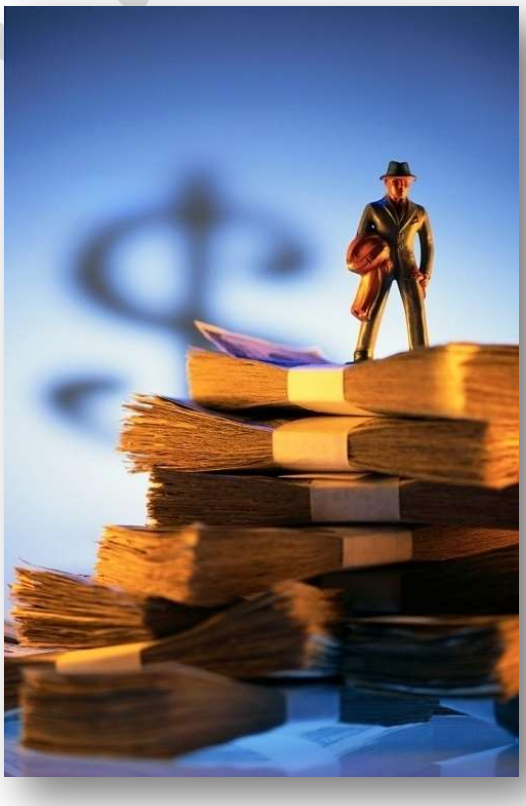
Specify layout options and select layout areas and timeframe values to include in the printed layout under the Options tab. Mark the Break Page Every Group to insert a page break between each group. The Options tab is not available when printing reports.

Changes made using the Print Preview are applied to the current printing. Save changes to the report page setup to make permanent changes (Report Editor, Properties palette).

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Chapter 10
Cost Accounts

Track activity costs and earned value using cost accounts.



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Cost Accounts and Project Expenses

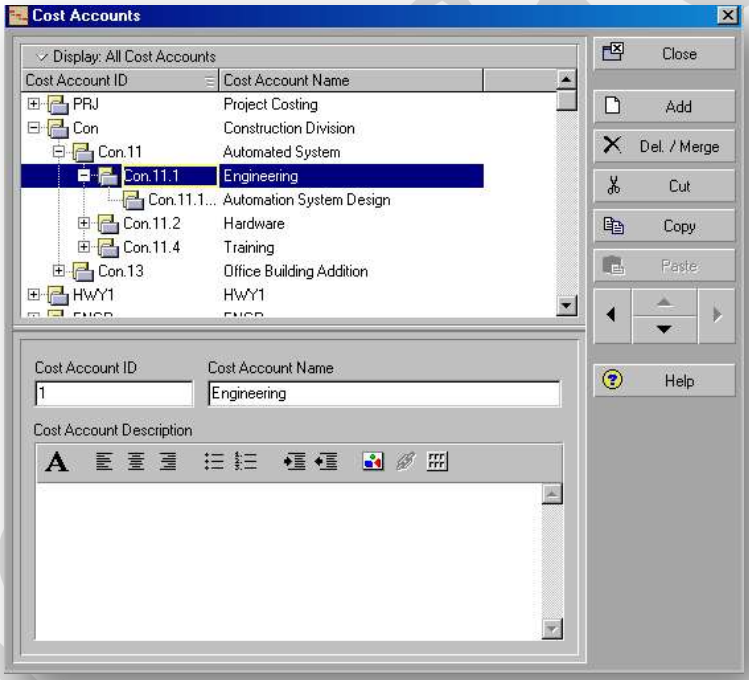
Track activity costs and earned value using cost accounts.

Setting the Project's Default Cost Account

Set the default cost account at the project level to be automatically assigned to the project's activities. This default parent cost account is used for new resources assigned to activities and new project expenses, but does not affect existing resource assignments / project expenses.

1. Select the Enterprise menu, Projects, and click the Display bar, Project Details.

2. Under the Defaults tab, enter an item in the Cost Account field.



Creating Cost Accounts

1. Select the Enterprise menu, Cost Accounts.

2. Highlight an existing cost account at the same hierarchy level of the cost account to be created and select Add.

3. Enter the cost account's ID and name. (The cost account's ID and name normally represents some component within the project hierarchy.) Cost accounts that are indented below a higher level within the hierarchy inherit a prefix from the higher level's ID.

4. Using the arrow keys move the cost account to its correct location or placement within the Cost Accounts hierarchy. Indent (►) for a more detailed position or outdent (◄) for a more general position.

5. Type a brief summary of the cost account in the Cost Account Description.

Copying Cost Accounts

1. Copy cost account from one hierarchy level in the hierarchy display to another by highlighting the cost account and selecting the Copy option.

2. Select the level above where the account is to be pasted and click Paste. The copied cost account includes and lower-level cost account as well but does not include its project assignments.

3. Use the Cut, Paste options to move cost accounts within the hierarchy. The project assignments are maintained.

Editing a Cost Account

1. Select the Enterprise menu, Cost Accounts, and highlight the cost account to be edited.

2. Enter a new cost account ID and/or name. Changing the cost account's ID and/or name applies to all activity assignments.

3. Modify its position within the hierarchy using the arrows to the right.

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Deleting a Cost Account

1. Select the Enterprise menu, Cost Accounts, and highlight the cost account to be deleted.

2. Click the Del/Merge button.

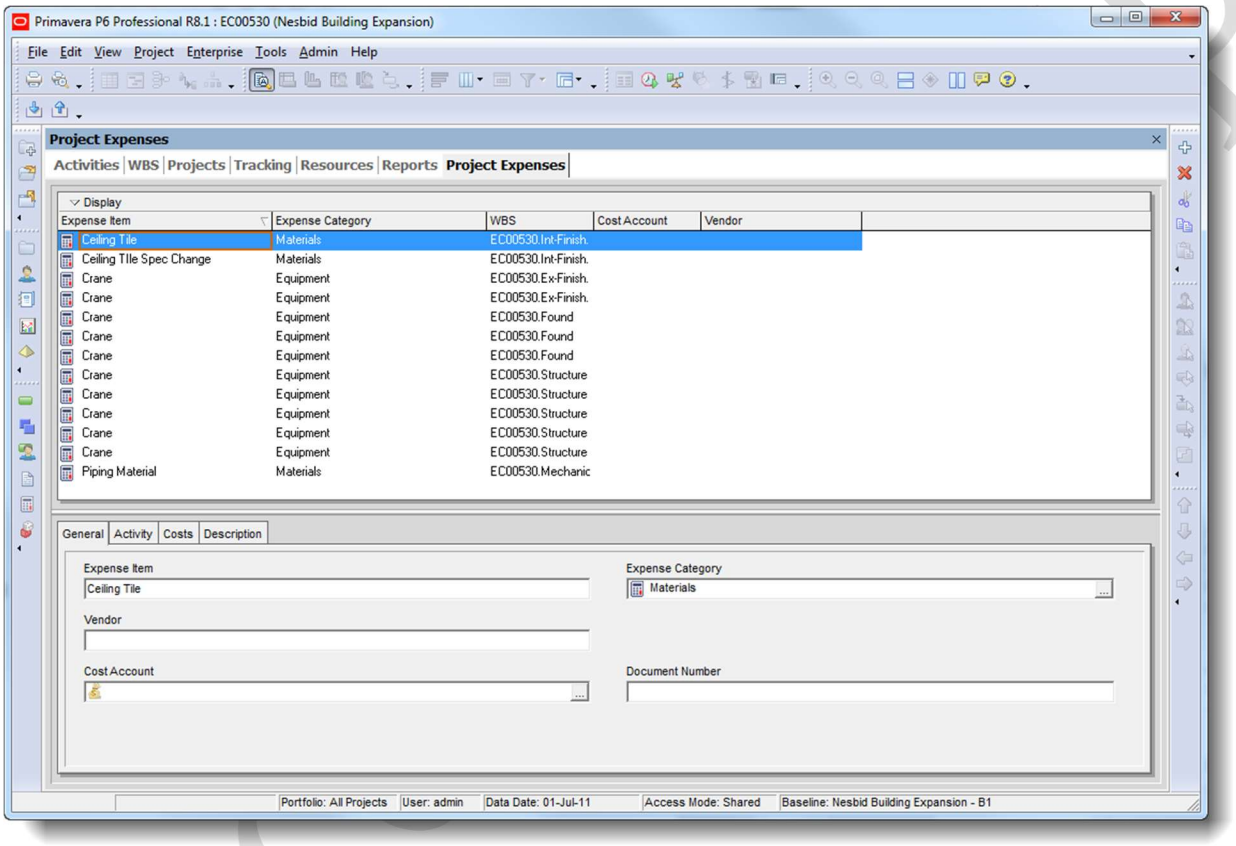
3. If the cost account has activity assignments the Cost Account in Use dialog is displayed. Choose Select Replacement Account to specify a replacement for activity assignments.

4. Delete the cost account without replacing the item by selecting Delete Account(s).

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Expenses

Usually one-time expenditures for nonreusable items are non-resource costs associated with the project and assigned to the project's activities. Examples include materials, facilities, travel, consulting, and training. Determine whether the expense accrues at the activity's start or end or uniformly over its duration. Each expense has a budgeted/planned actual, remaining, and at completion value for both cost and units.



Expenses vs. Resources

Expenses differ from resources (personnel and equipment) in that resources are time-based and usually extend across several activities and/or projects. Expenses are project-specific and not time-based. Expenses are not included during leveling.

Adding Expenses

1. Choose the Project menu, Expenses, Add.

2. Select the activity that incurs the expense. Activities can be grouped for greater organization.

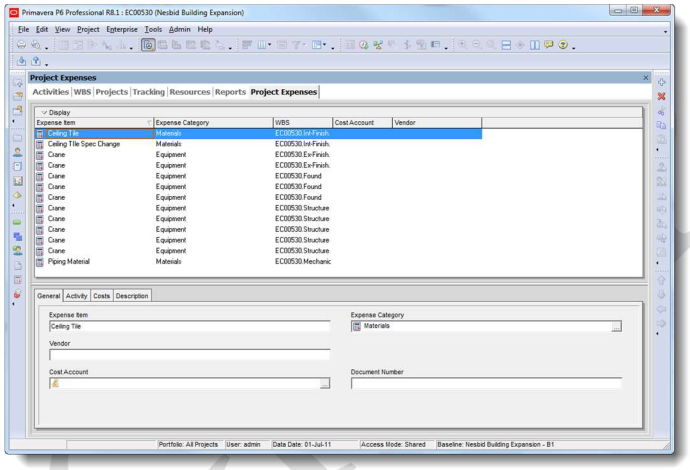
3. Enter the expense's information in the tabs of the Expense Details.

Viewing Expense Details

Select the Project menu, Expenses, and click the Display bar, Expense Details.

1. **General tab** – define general information regarding the expense item such as item name.
   a. Enter the item's Vendor, Cost Account, and Document Number (i.e. P.O. or invoice number)

   b. Enter the expense category to classify the type of cost. Expense categories are established under the Admin menu, Admin Categories, Expense Categories tab.



2. **Cost tab** – enter the number of units the assigned activity is expected use and the price per unit.
   a. Enter the Unit of Measure to quantify the expense. The Budgeted/Planned Cost of the selected expense is calculated as (budgeted [planned] units x price per unit).

   b. Select the Auto Compute Actuals checkbox to calculate an expense's actual cost based on the activity's completion percentage. Actual/remaining units are updated when project actuals are applied. (All work for the activity is proceeding as planned.)

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3. **Activity tab** – enter the accrual type to apply to the expense:
   a. *Start of Activity* – accrues the entire expense to the start date of the activity

   b. *End of Activity* – accrues the entire expense to the finish date of the activity

   c. *Uniform Over Activity* – evenly distributes the expense over the duration of the activity

   d. Select the activity to assign the expense. Activities can be grouped by WBS element, activity status, start and finish dates, and primary resource.

4. **Description tab** – enter a brief description of the expense.

Comparing Costs and Measuring Performance

Compare costs by opening the Project window and customize the columns to report earned value for all projects within the Enterprise. Use cost spreadsheets, profiles, and cost control reports to monitor spending. Measure Earned Value to determine performance using three indicators: planned value, earned value, and actual cost.

Use the Activity Usage spreadsheet and cumulative cost information to view cumulative remaining expense cost for each activity per month.

View activity profiles to see a graphical representation of cost flow for all / selected activities. Use a time-base graphic to gauge when and where costs are expended – is spending within the budget parameters.

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Budgets

Budgets are the total estimated effort (quantities) necessary, the cash flow required, and the change adjustments made to complete the project.

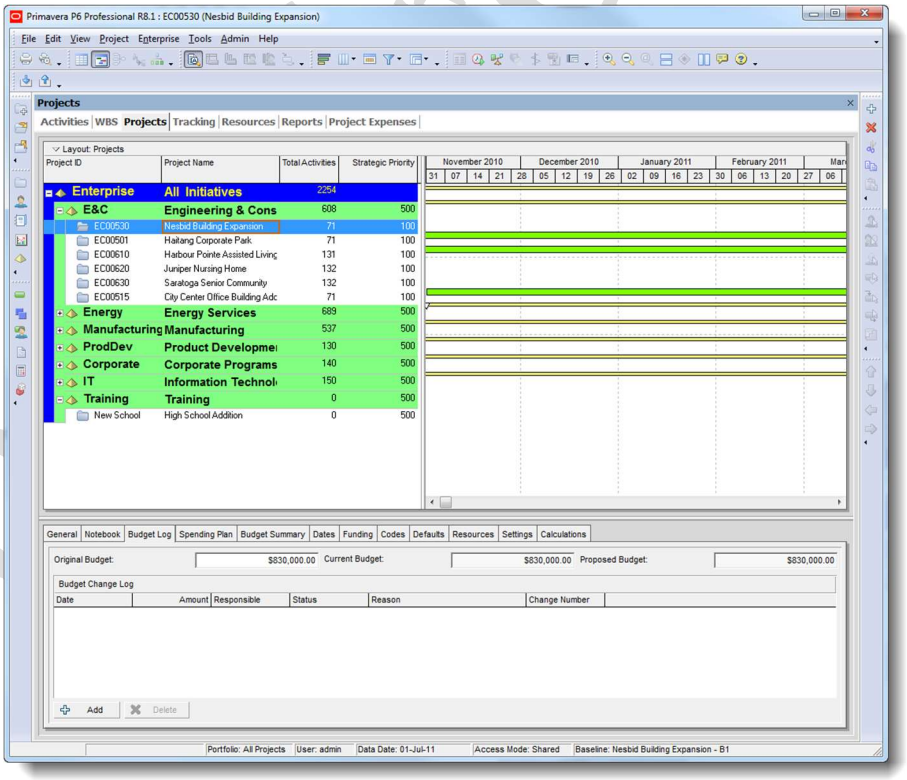
Budgets and the EPS

Top-Down approach to cost management – involves high-level budget estimates normally set at each EPS node by the resource or cost manager then distributed to each project maintained below the EPS node.

Creating a Budget Amount for the EPS Node

Set up the total budget for the EPS node before applying portions of the total budgeted amount to individual projects within the node using the Enterprise menu, Projects.

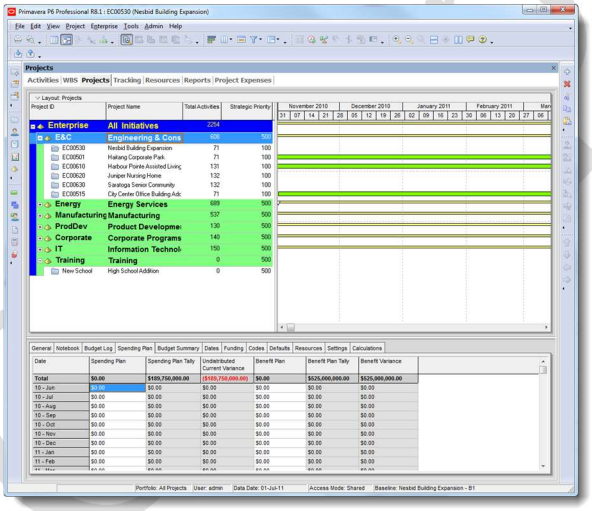
1. Select the Budget Log tab in the Details window (Display bar, Show on Bottom, Project Details).
2. Enter the total budgeted amount to complete the project in the Original Budget field.



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Creating a Monthly Spending Plan for the EPS Node

Create and compare a monthly spending plan of the EPS with those subprojects to track cash flow for each sub-node and/or project. Verify that the monthly spending does not exceed the original budget estimates. ***Negative Variance*** *indicates a need to reevaluate planned spending for the period in question.*



1. Select the Spending Plan tab to distribute the monthly budget and total the spending plan of lower levels of the EPS for comparison.

2. Enter total expenditures for each month in the Spending Plan column for the highlighted EPS node or project.

3. View the monthly spending plan of lower levels of the EPS node in the Spending Plan Tally column.

4. View the variance between the EPS node's monthly spending plan and those of its sub-nodes or projects tally in the Undistributed Current Variance column. *Red denotes over budgeted amounts or negative variance.*

5. Enter a benefit amount or profit return on monthly spending in the Benefit Plan column to track the financial benefit of performing the project on a monthly basis.

6. Benefit plans are automatically combined in the Benefit Plan Tally column of lower levels of the EPS.

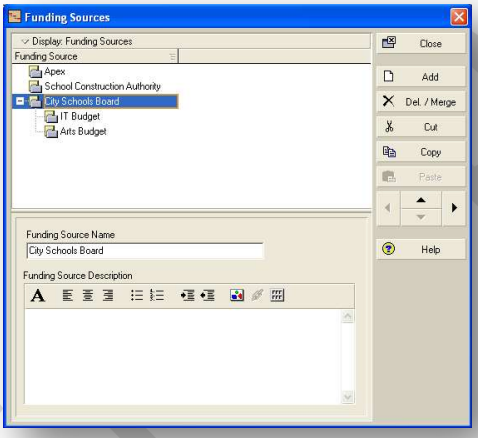
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Creating Funding Sources

Create a Funding Source dictionary to assign specific funding sources to budget items.

1. Create unlimited Funding Sources for projects such as government or capital projects normally funded by outside sources.
   a. Select the Enterprise menu, Funding Sources.

   b. Highlight an existing Funding Source node with the same level of the new Funding Source to be created and select Add.



   c. Using the arrow keys move the item to its correct location or placement within the Funding Source hierarchy. Indent (►) for a more detailed position or outdent (◄) for a more general position.

   d. Enter detailed information regarding the Funding Source in the Funding Source Description.

   e. View the Funding Source hierarchy in either the Table View or Chart View using the Display bar.

2. Assign funding sources in the Funding tab to different levels of the EPS.
   a. Amount – enter the dollar amount allocated by the funding source.

   b. Fund Share – enter the percentage of the total funded amount contributed by the funding source.
   COOL TOOL: View the sum of funding in the Project window by select the Display bar, Columns, Customize, Budget, Total Funding.

   c. The established Funding Sources hierarchy is available to any EPS node/project in the EPS hierarchy.

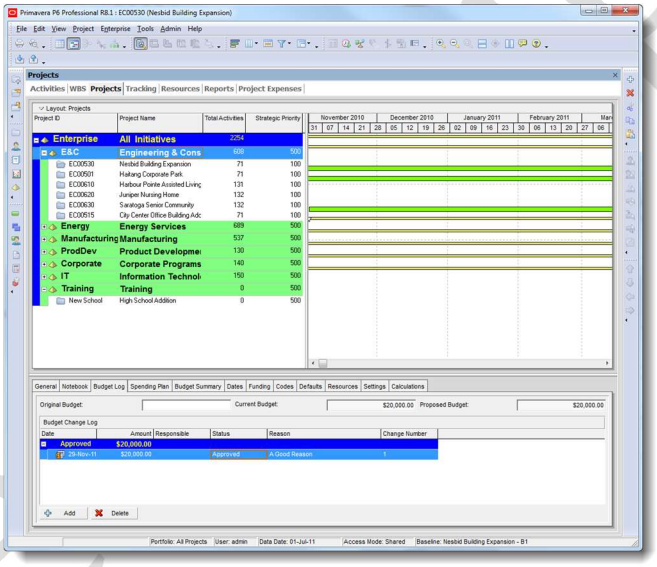
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Budgeting at the WBS Level

Set budget amounts and spending plans for the WBS levels once estimates and spending plans are firmly established using the Project menu, WBS.

Tracking Budget Changes

Track budget changes during the course of the project using change logs in order to calculate updated budgeted amounts.



1. Select the Budget Log to track modifications affecting the budget (i.e. changes in scope, resource reallocations, funding additions or withdrawals, etc.).

2. Enter the when, what, who, why, and status for each change.

3. Only changes with Approved status are calculated into the current budget.

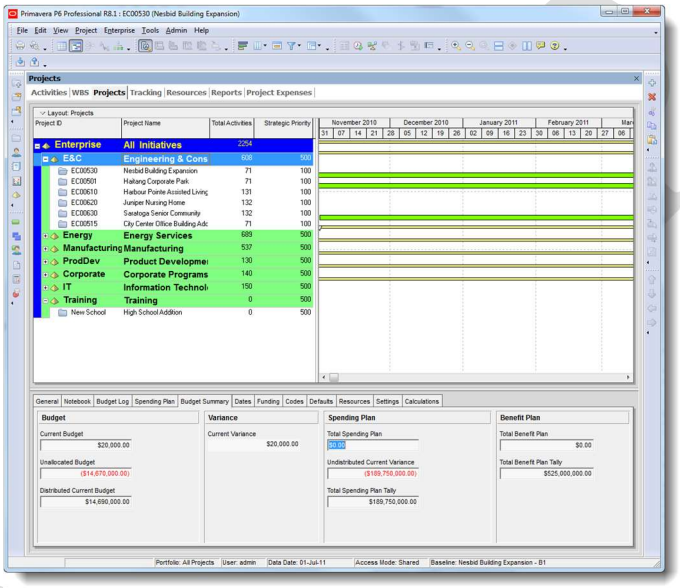
4. Pending status can only be changed to Approved or Not Approved by the authorized program or project manager to recalculate budgeted amounts.
   a. Proposed Budget = Original Budget + Approved Budget Changes + Pending Budget Changes

   b. Current Budget = Original Budget + Approved Budget Changes

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Monitoring Spending

Monitor monthly spending of budgeted funds and as actual costs are accrued track current and undistributed variance amounts.



1. Select the Budget Summary tab to track budget and spending totals, as well as changes, during the course of the project.
2. View the current budget, distributed current budget, and benefit plan totals to grade the project's success factor.
3. Current Variance = Current Budget – Total Spending Plan.
4. Undistributed Current Variance = Total Spending Plan – Total Spending Plan Tally.
5. Customize columns to produce resource and cost spreadsheets showing unit/cost distributions and variance of current versus original estimates by selecting the Display bar, Columns, Customize, Budget. *Negative variance amounts are shown in parentheses.*

Calculating Return on Investment

Calculate the return on investment (ROI) and the benefits for each completed project to assist in strategic planning for future projects.

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Chapter 11
Contract Changes

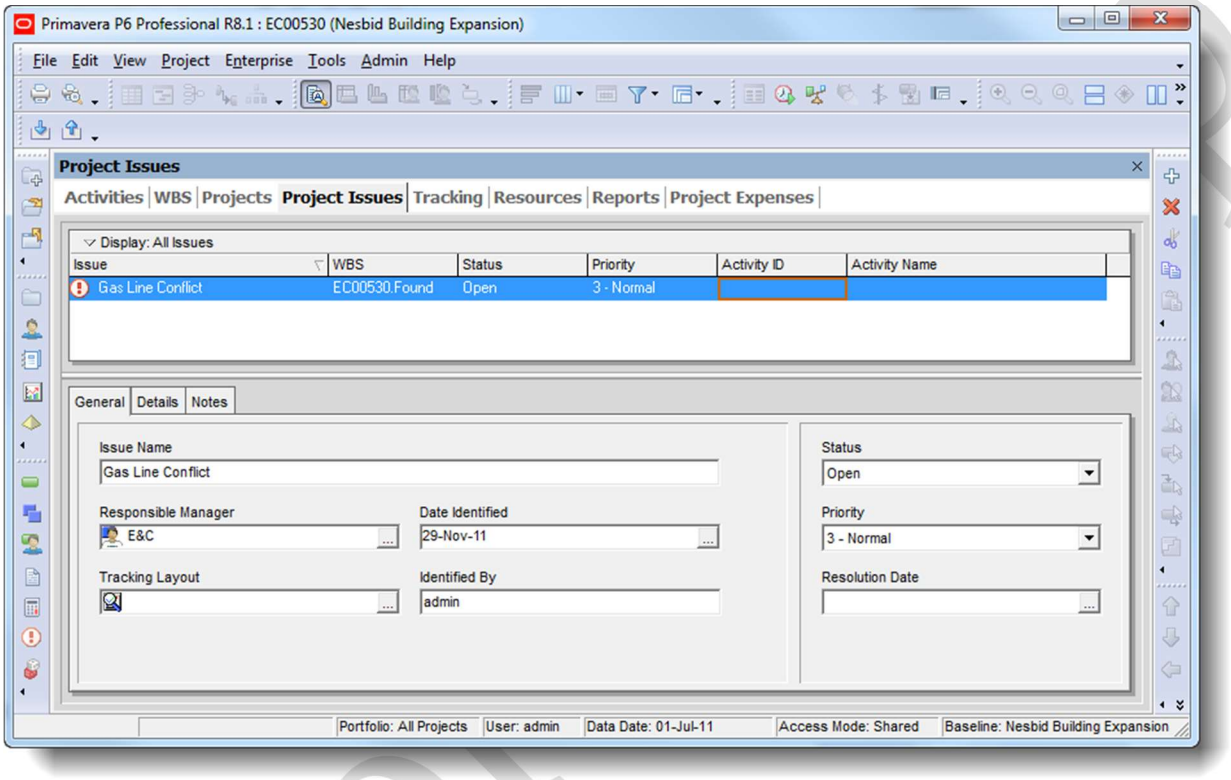
Track changes using Issues.



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Issues

Issues are potential or known problems that may require attention or corrective action. Create issues manually or define project thresholds to automatically generate issues. Issues can be associated with WBS elements, activities, or resources.



Assign priority levels, a tracking layout, and a responsible manager to an issue. Email the issue and its details (including comments) to other members in the project. View all information associated with an issue using the Issue Navigator.

Encor

Adding Issues

Add issues under the Project menu, Issues. Click the Display bar to show the Issue Details in the bottom layout. Select Add.

General tab – enter the Issue Name and select the Responsible Manager (defined in the OBS) for the issue. Select which layout to use in displaying the issue in the Tracking Layout field. The system date is automatically entered in the Date Identified field or browse to a different date. The user name is automatically entered in the Identified By field or type the name of the person who originally identified the issue. If the issue was automatically generated using the threshold monitor, Monitor is displayed. Enter the Resolution Date to track the duration it took in resolving the close d issue. Use the Status field to indicate whether the issue is Open, On Hold, or Closed and the Priority to categorize issues by urgency.

Details tab – enter the issue's Actual Value (numeric) in the Actual Value field. Associate the issue with a WBS element (other than a root WBS element) in the Applies to WBS field. Associate the issue with a specific resource or activity. These items are Read-Only if the issues was generated by a monitoring threshold. If the issue was generated by the threshold monitor, that parameter name is displayed in the Threshold Parameter field.

Notes tab – enter additional descriptive information about the issue.

Group and sort issues by project, WBS, EPS, etc. Select the Display bar, Group and Sort By.

View and add history to an issue by selecting Project, Issues. Highlight the issue and click Issue History. Enter any additional comments regarding the issue.

Send email about an issue using the Navigator dialog box under the Tools menu, Issue Navigator. Highlight the issue about which you want to send the email and select Notify. Enter the individuals to whom the email is to be sent and click Add. Select a recipient from a resource list or users. Then select a recipient from the Assign Recipients dialog box OR manually enter a Name and E-Mail Address. Select Assign. Or remove current recipients from the list by selecting Remove. Enter a subject for the email and any additional comments concerning the issue and select Send All.

Tracking Layouts for issues graphically display the information regarding the issue. Use the Issue Navigator (Tools menu, Issue Navigator) to open the layout in the Tracking Window. Assign these layouts in the General tab in the Issue Details window.

Use the Issue Navigator to maintain information regarding a specific issue. Jump from a highlighted issue to its associated tracking layout, or activity, WBS, or resource

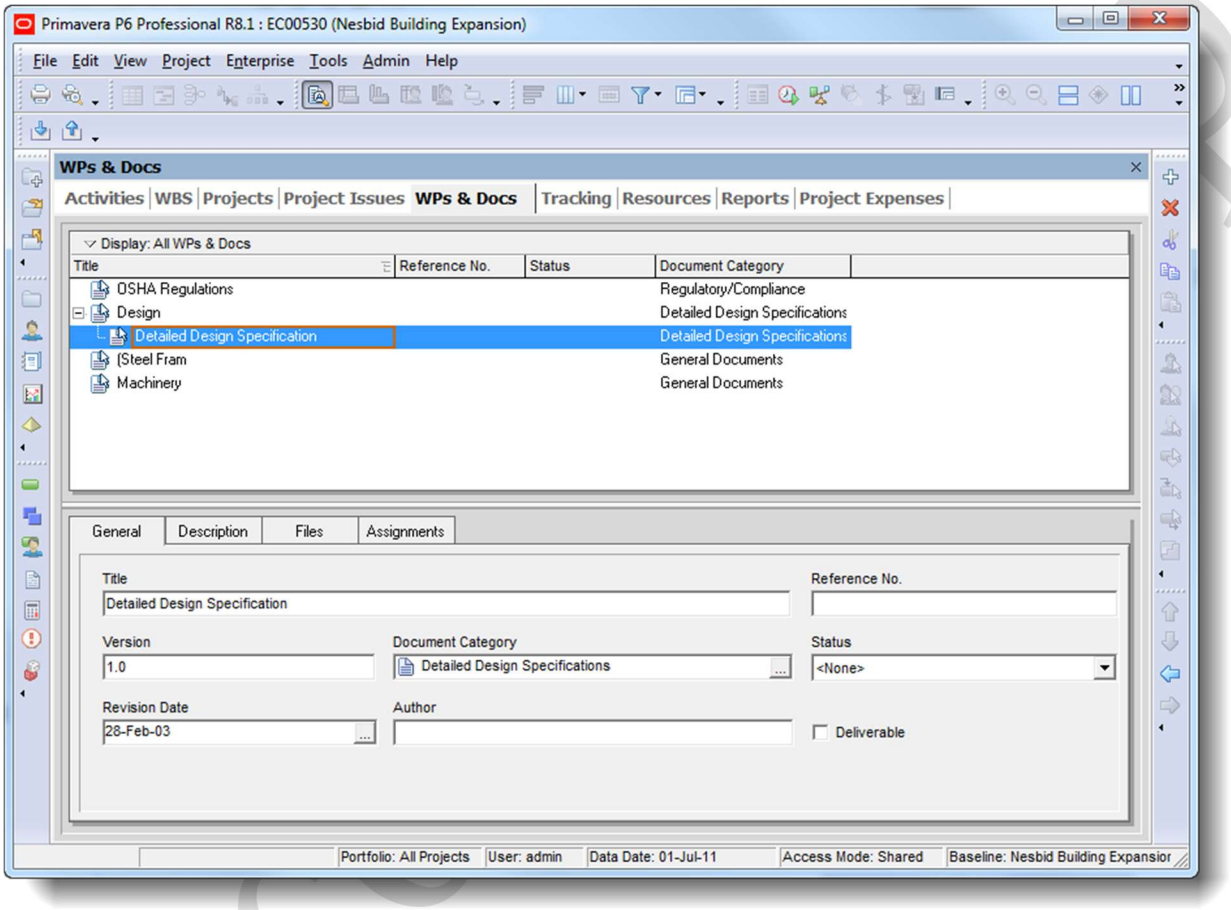
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assignment, or other details. Send e-mail to notify or share information with other users regarding the issue. If the item buttons are not available, then the issue is not associated with that topic. Set the Issue Navigator to open each time the project management application is opened by selecting the Edit menu, User Preferences, Application tab. Mark the Show the Issue Navigator Dialog at Startup checkbox.

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The Project's Document Library

Catalog and track all project-related documents and deliverables (i.e. guidelines, procedures, standards, plans, design templates, worksheets, etc.) using the Work Products and Document feature.



Maintain general information about project documents (i.e. revision dates, locations, and author information). Enter a reference document for an activity (i.e. standards and guidelines for performing an activity's work) or a document for the project (i.e. project standard). Documents can also be work products or activity output (i.e. testing plans and blueprints). Indicate whether documents are public or private.

Viewing the Library

View the library under the Project menu, Work Products and Documents.

Select the Display bar to view the WP & Doc Details.

Adding Documents to the EPS

Add a work product or document to an open project into the documents hierarchy. Select an existing document at the same level as the document to be created and select **Add**.

General tab – enter the Title of the document and a Reference No. or catalog number. Enter the document's Version and Revision Date (last update date). Select the Document Category to organize various types of documents. (Define categories under the Admin menu, Admin Categories, Document Categories.) Select the appropriate document Status and mark the Deliverable checkbox to indicate whether the document is a deliverable.

Description tab – enter a brief description of the document.

Files tab – specify two types of document location references, private and public locations. Users of the project management application can view Private Location references. All project participants can view Public Location references (most likely those documents that are stored in widely accessible network locations).

Use the Launch options to open/view the files.

Assignments tab – assign activities or WBS elements to the document. Hold the Ctrl key and select multiple documents in the Assignments tab, Assign.

Remove document assignments by selecting the item, Remove. Mark the Work Product box indicating the item is a work product.

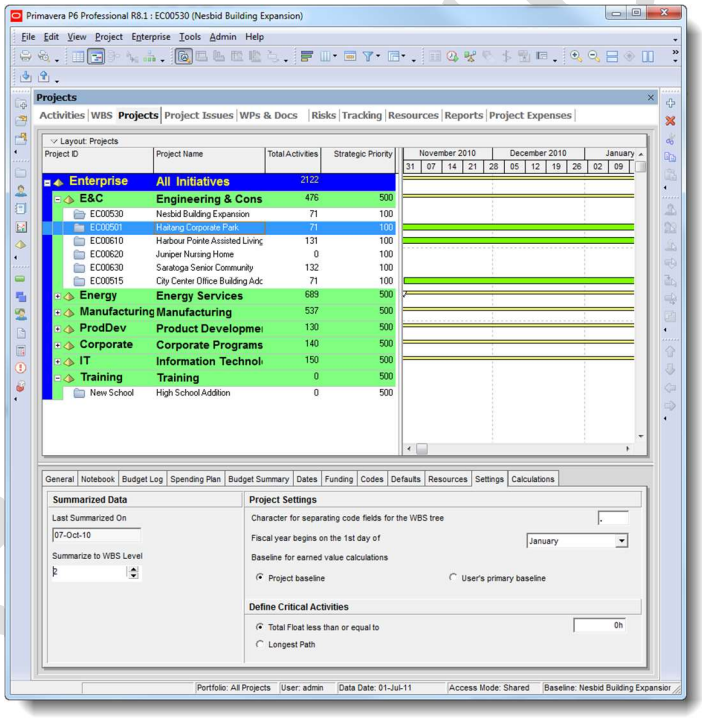
Deleting documents removes the document's link not the actual, physical file.

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Summarizing Projects

View the hierarchy of the EPS down to the individual nodes or WBS by specifying which projects should be summarized and the WBS level to which the data should be summarized.

Viewing Summarization



1. Select the Enterprise menu, Projects, and specify the project to be summarized.

2. In the Project Details window, select the Settings tab, Summarized Data section.

3. Enter the maximum WBS level to which the project can be summarized in the Summarize to WBS Level field. This number indicates to which level other users can view in the Progress Reporter and Portfolio Analyst applications.

4. Select the baseline to use when summarizing baseline project data in the Baseline for Summarization field.

5. View the date when project data was last summarized in the Last Summarized On field.

6. Use the Contains Summarized Data Only if the project data is to be linked in other applications.

"On-the-fly" Summarization

Summarize data "on-the-fly" under the Tools menu, Summarize. Select to summarize the open projects in the current view, all projects in the EPS (both open and closed), or summary-only projects (those projects for which the Contains Summarized Data Only checkbox is marked – Project Details window, Settings tab).

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Summarize a single project or a specific group of projects by selecting the projects to be summarized in the EPS, right-clicking and choosing Summarize Project.

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Using Job Services

Or using the Job Services option (Tools menu, Job Services) schedule a time when data is to be summarized taking a "snapshot" of calculated summary data to be viewed in the Portfolio Analyst. Make sure to select Summarize in the Service Type field. Specify the sequence in which the service is to be performed in the Job # field. Enter Enabled in the Status field to activate the summary service. Service can be suspended at any time by selecting Disabled in the Status field. Schedule when the service is to be run in the Run Jobs area (i.e. everyday, weekly, bi-weekly, or monthly).

Summary information is calculated and displayed for costs and quantities, dates, durations, float, progress, baselines, variance, resources, cost data, and custom user fields.

Cost and quantity data items (i.e. budgeted/planned costs and budgeted/planned quantities) are totaled by adding the values for each activity in the summary.

Dates (i.e. earliest early [actual] start, earliest early [actual] finish, as well as latest late dates and baseline dates) are displayed.

Durations for both original and remaining are summarized for activities with progress. For those activities with no progress, the original and remaining duration is calculated as the number from the earliest early start to the earliest early finish date. For activities with an actual start date, the original duration is calculated from the actual start to the latest early finish date. Remaining duration is calculated from the "internal" early start (original early start date retained before the actual start was applied) to the latest early finish date. If the activity is completed, the original duration is calculated as the actual duration and the remaining duration is zero.

Float (total float) of summarized data can be based on the start dates, finish dates, or most critical dates under the Tools menu, Schedule, Advanced, Compute Total Float as field.

Progress is the means by which percent complete is calculated using duration to compare the amount of time remaining to complete the activities to their original duration. [(Summary Current Original Duration – Summary Remaining Duration) / Summary Current Original Duration] x 100

Baseline data, comparison data, and variances are set by modifying date and duration summarization settings.

Custom user fields are summarized according to type. A numeric field is totaled for the summarized data. A text field is blank unless values are the same for all activities. Start dates display the earliest start date, and finish dates display the latest finish dates.

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Chapter 12
Admin and User Customizations

Establish preferences as the Administrator and as a User.



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Admin Customization

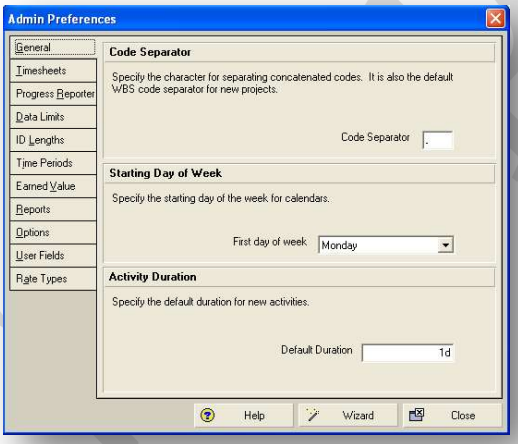
Setting Admin Preferences

Admin Preferences can be established by the Project Controls Coordinator by selecting Admin, Admin Preferences.

1. **General tab** default options can be modified.
   a. first day of week

   b. hierarchy level separator

   c. default duration for new activities)



2. **Data Limits tab** options establish hierarchical maximum levels as well as maximum number of baselines and activities per project. (Changing limits applies to future additions.) Maximum limits include:
   a. EPS/WBS tree – 1-50

   b. OBS tree – 1-25

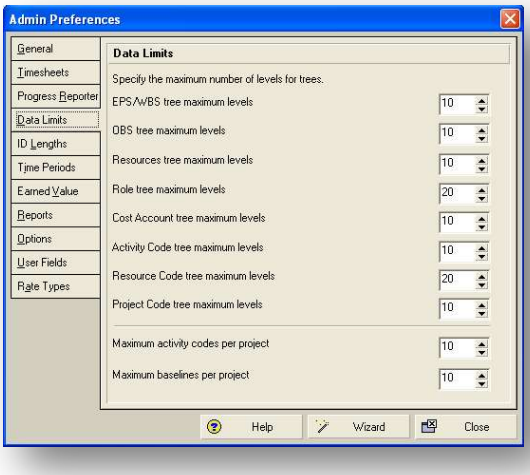
   c. Resources tree – 1-25

   d. Roles tree – 1-25

   e. Cost Accounts tree – 1-25

   f. Activity Code tree – 1-2

   g. Resource Code tree – 1-25
   h. Project Code tree – 1-25



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i. Activity Codes – 0-500

j. Baselines – 1-50

3. **ID Lengths tab** options establish maximum character length for IDs and codes. (Changing limits applies to future additions.) Maximum limits include:
   a. Project ID – 1-20

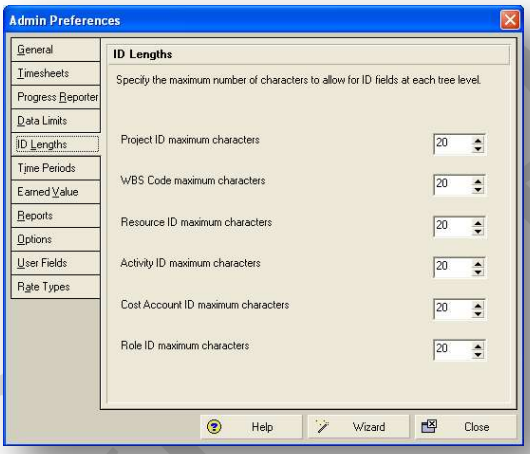
   b. WBS Code – 1-20

   c. Resource ID – 1-20

   d. Activity ID – 1-20

   e. Cost Account ID – 1-20

   f. Role ID – 1-20

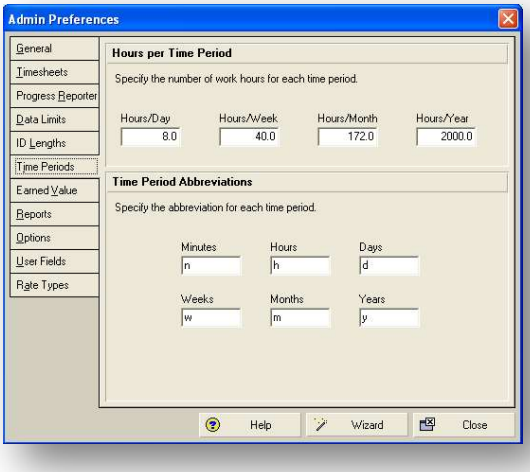


4. **Time Periods tab** options establish standard work hours per day, week, month, and year. Also set abbreviations (one character per field) for time periods. Hour entries set include:
   a. Hours/Day – 1.0-24.0

   b. Hours/Week – 1.0-168.0

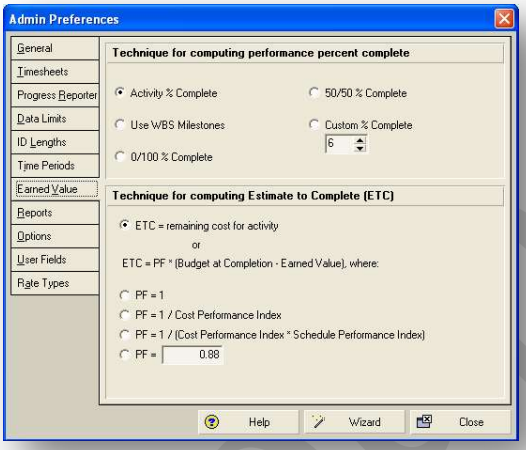
   c. Hours/Month – 1.0-744.0

   d. Hours/Year – 1.0-8784.0

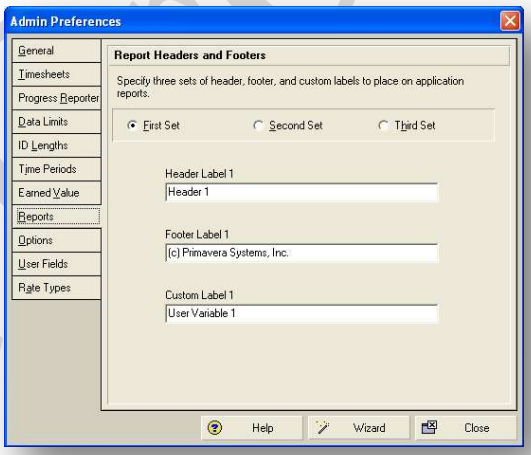


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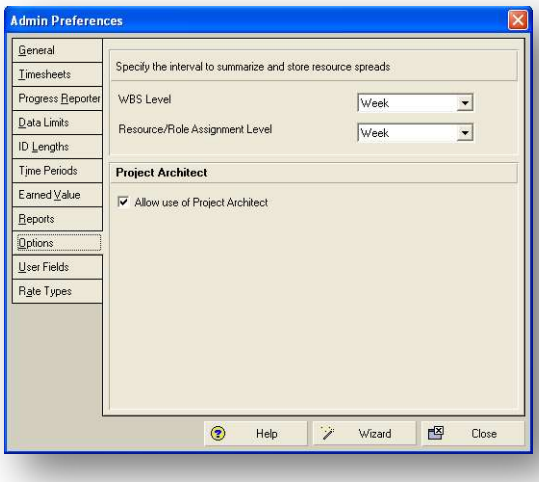
5. **Earned Value tab** options modify earned value calculations.



6. **Reports tab** options allow for three sets with each having its own headers, footers, and custom labels to be used in reports.

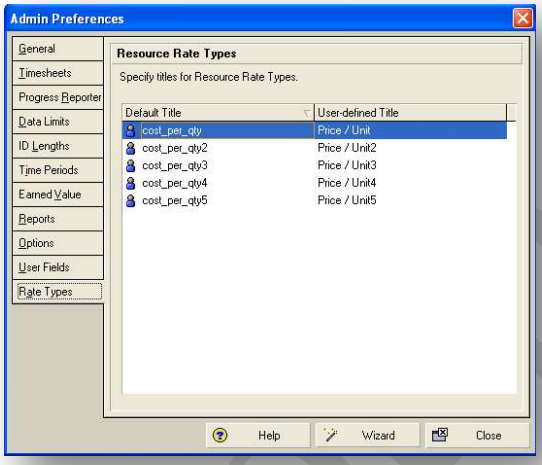


7. **Options tab** establish time intervals to which activity costs and quantities are to be calculated by both WBS Level and Resource/Role Assignment Level. Also give the security level of the Project Architect to access processes in adding activities or creating projects.



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8. **Rate Types tab** options establish titles for Resource Rate Types.

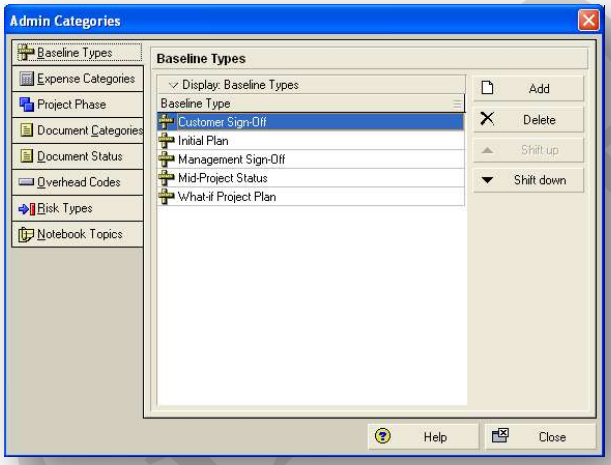


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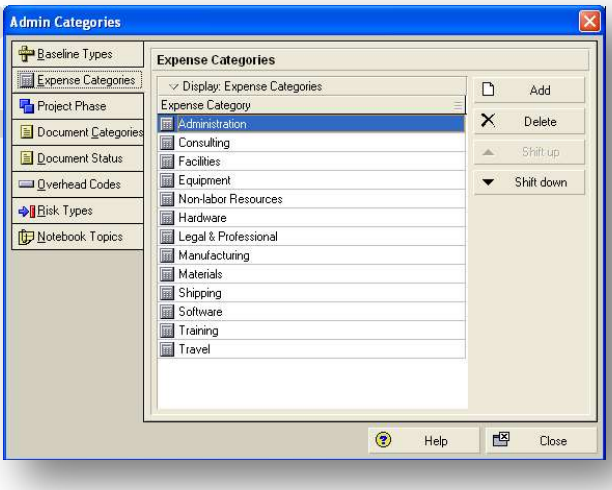
Setting Admin Categories

Admin Categories can be set through the Admin menu, Admin Categories.

1. **Baseline Types tab** options allow for addition, editing, or deletion of baselines across multiple projects. Items can reordered in the list view by selecting Shift up or Shift down. (Exception: when sorting alphabetically)

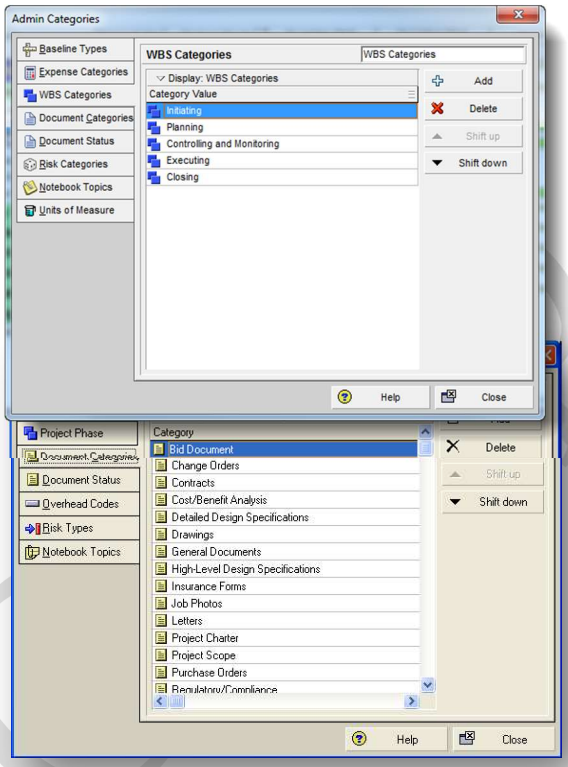


2. **Expense Categories tab** options allow for addition, editing, or deletion of categories of project expenses. Items can reordered in the list view by selecting Shift up or Shift down. (Exception: when sorting alphabetically)



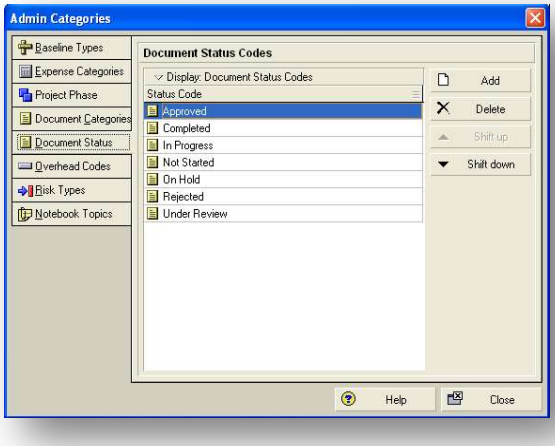
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3. **WBS Categories tab** allows for definition of custom WBS categories and values. Items can reordered in the list view by selecting Shift up or Shift down. (Exception: when sorting alphabetically) Values and assignments are *not* changed with category changes.



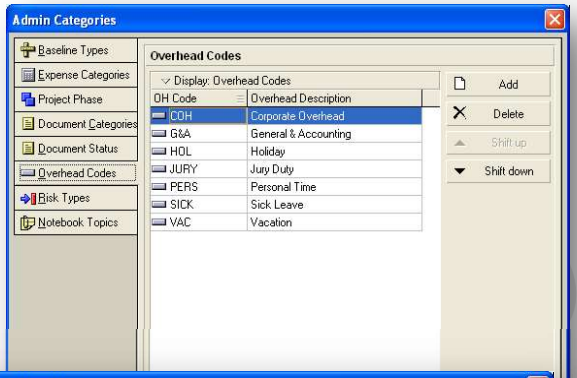
4. **Document Categories tab** options establish categories for project documents and work products. Items can reordered in the list view by selecting Shift up or Shift down. (Exception: when sorting alphabetically)

Document Status tab options allow for addition, editing, or deletion of status types of project documents and work products. Items can reordered in the list view by selecting Shift up or Shift down. (Exception: when sorting alphabetically)

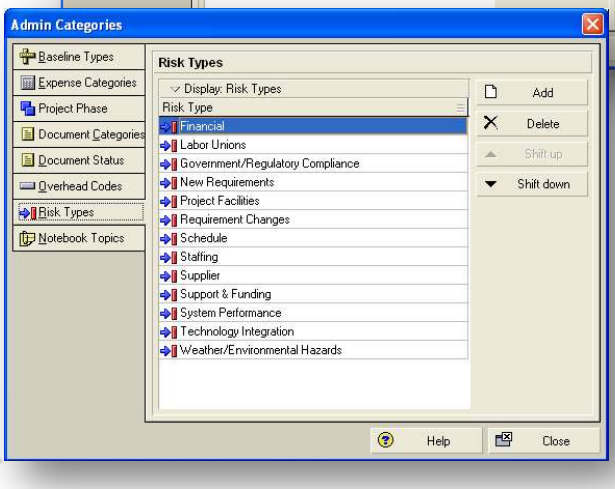


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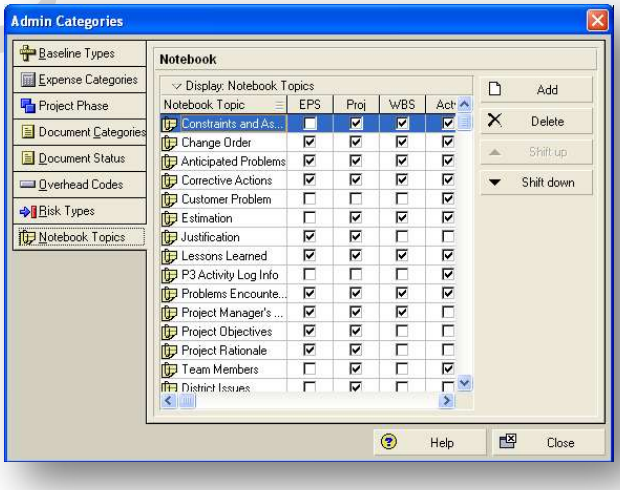
5. **Overhead Codes tab** options allow for addition, editing, or deletion of overhead codes used for timesheets and those items not associated with project activities. Items can reordered in the list view by selecting Shift up or Shift down. (Exception: when sorting alphabetically)



Risk Types tab options allow for addition, editing, or deletion of categories of risks or potential risk types. Items can reordered in the list view by selecting Shift up or Shift down. (Exception: when sorting alphabetically)



6. **Notebook Topics tab** allows for addition, editing, or deletion of notebook topics or instructions for completing an activity or task at the EPS, Project, WBS, or Activity levels. Items can reordered in the list view by selecting Shift up or Shift down. (Exception: when sorting alphabetically)



Setting Currency

Currency or monetary units is set by selecting the Admin menu, Currency. The options used apply to all projects in the database.

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1. Establish which currency is set as the base to set cost information. The base currency Exchange Rate should *always* be set to 1.0 to establish exchange rates with other currencies set. (Default = U.S. dollars)

2. Choose Appearance to tab to set the appearance of the currency selected.

3. Admin security level *only* has the access to modify the base currency.

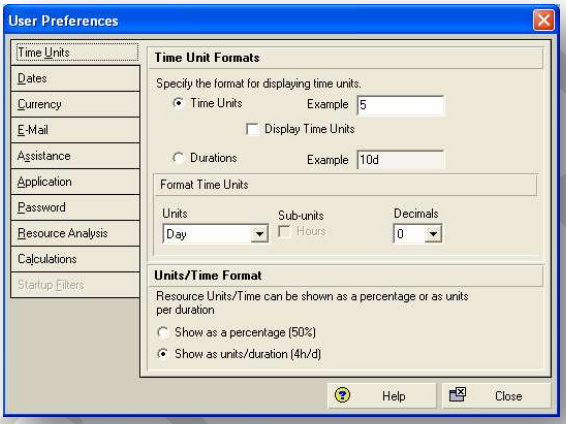
4. Users can select in which currency to view costs by selecting Edit, User Preferences, Currency.

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User Preferences

User Preferences can be customized by selecting Edit, User Preferences.

1. **Time Unit tab** options establish unit formatting for display in layouts, durations, resource prices, etc.
   a. Choose Time Units to display units and sub-units, if applicable OR select Durations to express sub-units in duration displays. (8h 30.00m OR 1.5d both equaling one-and-one-half days)

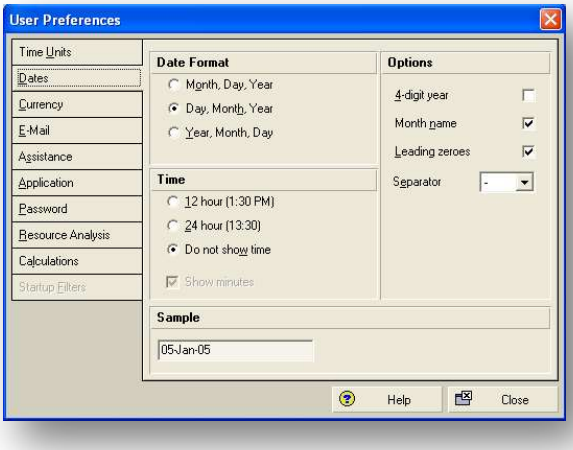


   b. The basic Units used as the default are also set and whether sub-units are to be used.

   c. Select Sub-units to display the minimal time unit for the Units selected. Establish the decimal places to be displayed.

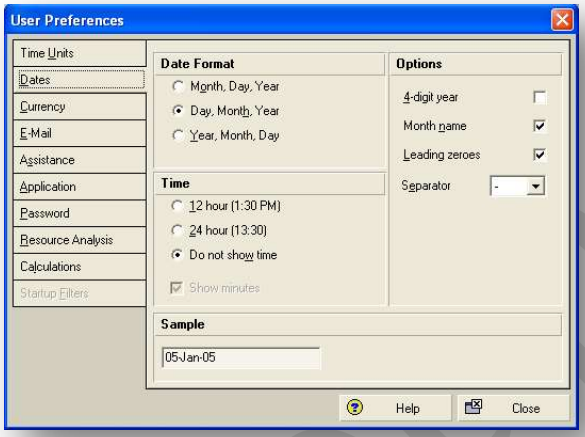
   d. Units/Time Format options determine resource units display preferences in either percentages or units per time/duration.

2. **Dates tab** options establish date formatting and display p references including date order, time display, and separating character.

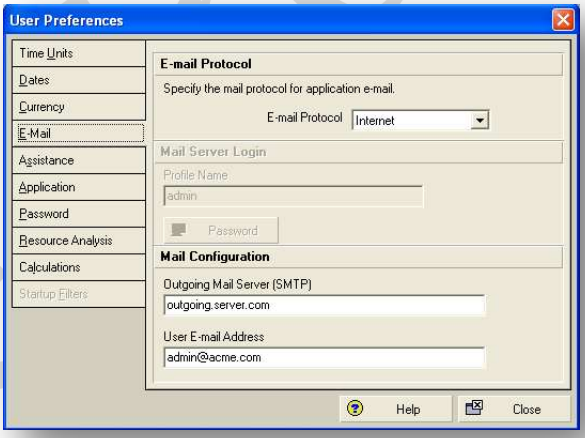


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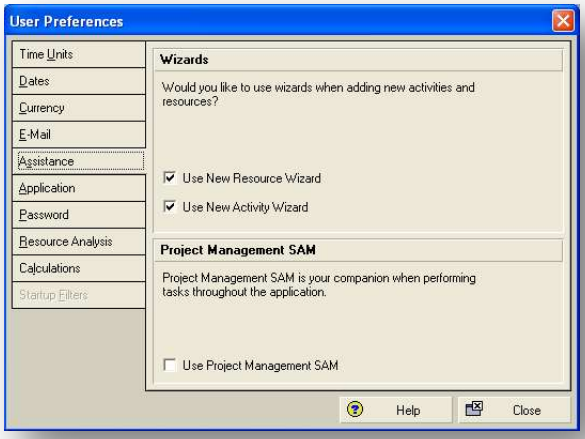
3. **Currency tab** options allow the user to specify currency displayed for cost data, whether to show currency symbol and/or decimal digits. (Currencies available are established by the Administrator.)



4. **E-Mail tab** options allow for transfer of information by establishing E-mail Protocol, either MAPI (Messaging Application Interface) and Internet (or SMTP).
   a. MAPI protocol requires a profile name in the Mail Server Login Name, Profile Name Field – established in Control Panel, Mail Settings.
   b. Outgoing Mail Server (SMTP) requires the Internet mail server domain name or its IP address.
   c. User E-Mail Address requires the individual's e-mail address sending the information (via Microsoft Exchange).



5. **Assistance tab** sets up Wizards to assist the user in adding resources and activities. The Use Project Management SAM option displays an animated character to help in the use of wizards. This character can be changed by selecting Edit, Project Management Sam.

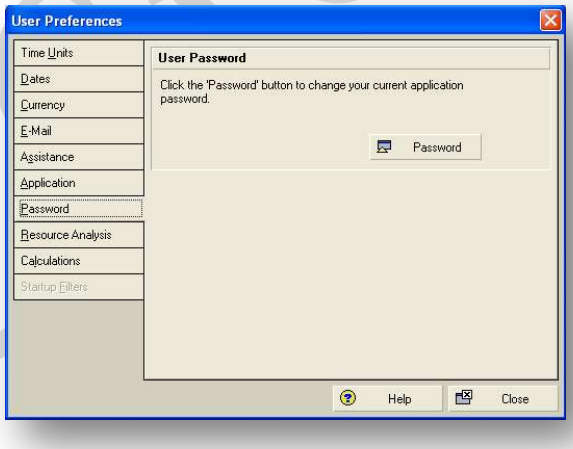


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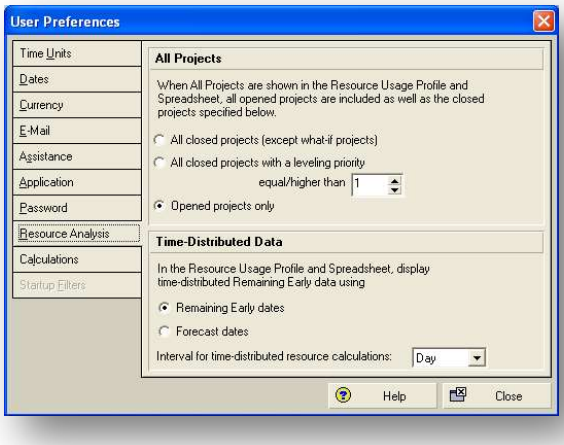
6. **Application tab** option modifies the start window that is displayed when Project Manager is opened. Actions performed can be recorded in a log file.



7. **Password tab** allows for a change to the user's password.



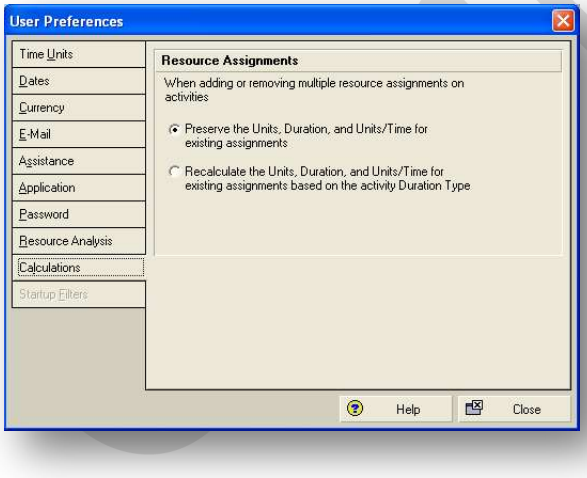
8. **Resource Analysis tab** options determine which projects are to be used in the Resource Usage Profile and Spreadsheet.
   a. Along with all open projects (which are displayed by default) the user can select:
      1. All closed projects (except what-if projects)
      2. All closed projects with a specific leveling priority
      3. Opened projects only



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b. Time-Distributed Data sets the starting point for calculating remaining units and costs for Resource Usage Profiles and resource Usage Spreadsheet displays.
   1. Remaining Early dates base remaining units and costs on the current remaining estimate.
   2. Forecast Dates base remaining units and costs on a forecast date set by moving the project bar forward in a "what-if" scenario.
   3. Select the Interval for time-distributed resource calculations.

9. **Calculations tab** options modifies Resource Assignments that will determine whether resource units, durations, and units/time are modified when adding or removing multiple resource assignments on activities.



10. **Startup Filters tab** establishes data filters applied to projects or all enterprise data when starting Project Manager. (Access rights determine the data displayed when selecting no filter.)

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Project Check in/Check out

Enable projects outside of the enterprise database to be tracked such as projects taken to a remote site for updating. The project cannot be modified in the project management application while checked out. After the update the project can then be checked back in to the enterprise database.

Checking Out Projects

Projects checked out using the proprietary format (XER) enable use of activity, cost, resource, and other types of project information with other installations of the project management application regardless of the database server.

See whether a project is checked out in under the Enterprise menu, Projects, Project Details, General tab, Check-out Status OR a checkmark symbol on the project symbol signifies a checked out project.

Open projects to be checked out and use the Check Out wizard to check out project(s) under the File menu, Check Out. Mark the checkboxes under the Check Out column to check out multiple projects in the XER format. Specify a file name and location to save the information. Multiple projects checked out to a single .XER format preserves interproject relationships.

Checking In Projects

Checked out projects can be checked back in to any installation of the project management application under the File menu, Check In option. Using the Check In wizard Browse to the location. If multiple projects are contained within the same XER file, the projects will be listed. For preexisting projects the Match checkbox is selected. Double-click the Import Action field next to each project to prevent the data from being overwritten.

Update Existing Project – the existing project is updated with any new/modified data in the XER file.

Replace Existing Project – the existing project is deleted and replace with the project from the XER file.

Ignore this Project – the project in the XER file is not checked in.

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When importing project data, specify which options are to be used in importing and matching data. These options or configurations can be created and saved to be used in later check in operations. (One configuration can be used to import at any one time.) Modify existing configurations using the Modify button. Mark the items in the Del. Unref. column to delete risks, issues, thresholds, or activities in the project to be updated but are not in the import file. To keep the data in the original project as is, clear the checkbox.

Keep Existing – does not overwrite existing project data with updated information; adds only new data if the records do not exist.

Update Existing – overwrites existing project data with updated information; adds new data if the records do not exist.

Insert New – retains existing project data while adding any new data items.

Do Not Import – retains existing project data and does not import any updated information.

Select the Currency type to use in the project before finishing the import.

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